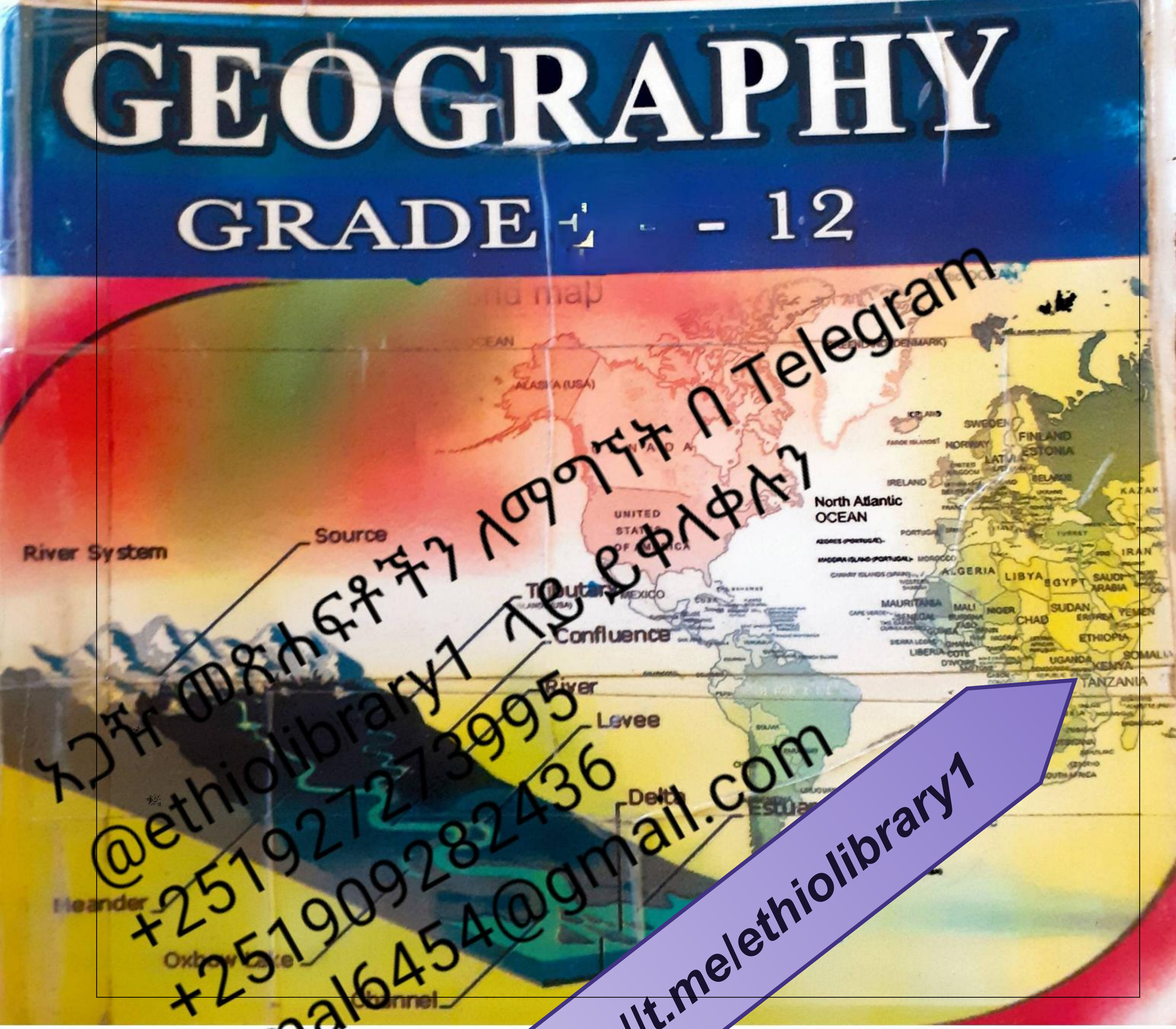
Extreme Series

GEOGRAPHY

GRADE 1 - 12



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UNIT FIVE

Basic Research Methodologies in Geography

5.1. DEFINITION AND CONCEPT

The term research came in to English from

- The old French root word "cerchier", which means to search or seek; and
- The prefix 're-' which means again.

- It means to seek again with a view to becoming certain. (Desta Hamito; 2000)
- It means to refined, re discover the problems and foreward possible solutions based on the organized facts.
- It is a processes of formulating questions, problems, or hypothesis, collecting data, relevant to the questions or problems; analyzing and interpreting the collected data (Nunan; 1992)
- It is an organized and systematic search for the answers to the questions we ask (Hatch and Lazaxation; 1991)
- It is also a systematic inquiry in to causes or the discovery of new facts through planned and organized effort that requires time, money and skill.

5.2. The Significance of Research

- Identifies the causes of problems, and the degree of these problems.
- Addis new knowledge
- Enables us to visualize or understand the gap between the real world and the formulated theory.

- Provides a basis for additional investigations that validate the research findings or that advance the study of the problem by acquiring more data.
- Enlarges one's Owen reading, writing and thinking skills.
- Strengthens our capacity for critical observation and enforces our power of prediction.
- Enables us to make rational decisions.
- Serves as an aid to formulating policy.

Illustrative Questions

1. Which of the following is NOT among the general uses of research in Geography? (UEE, 2004/2012)
   A. Being able to predict about the future.
   B. Solving social or environmental problems
   C. Narrowing down the broad area of knowledge
   D. Improving one's writing, reading and thinking skills.

Explanation: Among the general uses of research in geography is add to the body of knowledge about the problem by providing new information and knowledge.

Answer: C

2. Which of the following is true about the common characteristics of research
   A. Formulating questions
   B. Suggests some recommendation
   C. Identify the problem
   D. Problem oriented and solution looking

Explanation: The common characteristics of research is searching problems and solution looking.

Answer: D

3. Define the term research?

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Explanation: Different scholars explain in different ways but all of them argue that the research is an organized and systematic search for the answers to the question we ask.

4. What do you think that research enables us to make rational decisions.

Explanation: Research as a science or profession which follow synthific and systematic approach. This situation make clear and fertile ground in any economic, social and political aspects. In other words, the synthefic approach of research create right decision to right time. All the situation occurred based on facts.

5. Mention the significance of research?

Explanation: The solving naturally caused problems of research work includes identifying causes of problems, solving naturally caused problems of the society, setting new knowledge improving one's writing reading and thinking skills and being able to predict about the future.

5.3. Approaches of Research

- There are two basic approaches to research.
- Their differences are mainly in the attributes to be measured and the techniques to be used for collecting and analyzing data, they includes the quantitative approach and qualitative approach.

A. Quantitative Approach

→ It verify a given geographical theory

→ It can be measured with stastical techniques.

→ It is advantageous for the quantitative technique employed by one researcher can be used by another researcher for different objectives.

→ **Examples**:- the decrease in land – lease prices as one travels from the center of acity to its suburbs;

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→ The decrease in vegetation diversity as one travels from the equator to pole direction.

→ The decrease in temperature towards the poles from the equator.

- In the above examples, distance from a center is the determining variable. It determines land – lease prices, vegetation and temperature.
- The quantitative approach can be subdivided.
  - → Inferential quantitative approach = the target – study uses an existing data base and infers characteristics or relationships from it.
  - → Experimental quantitative approach the research work manipulates variables to see their effects on other variables.
  - → Simulation (model) approach = this approach involves constructing on artificial environment (model) to represent the actual environment we are studying.
- The quantitative approach and hypotheses regarding an existing theory are established and tested.
- It is also mathematical analysis is frequently used

B. Qualitative approach

→ In this approach, data are used to explain a new theory.

→ No previously exsiting theory or hypothesis is tested.

→ Quantitative techniques (mathematical analysis) are not employed.

→ The study cannot be repeated by other researchers.

Common techniques for gathering data are

→ Group interviews

→ Telephone interviews

→ In – person interviews (face – to face)

→ Questionnaires

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→ Personal observations

→ Examples – differences between urban and rural populations of Ethiopia.

→ Effects of harsh climate on human activity.

Illustrative Questions

6. Which one of the following statement about a research paper is NOT correct? (EUEE, 2003/2011)
   A. It is based on the writer's knowledge and experience
   B. It has much in common with other types of writing
   C. It could be based on primary or secondary data
   D. It relies on various sources of information

Explanation: The research paper is based on the writer's knowledge and experience's it could be based on primary or secondary sources and it relies on various sources of information.

Answer: B

7. Which of the following is not a common techniques for gathering data in the qualitative approach
   A. Group interviews
   B. Inferential approach
   C. questionnaires
   D. personal observation

Explanation: Inferential approach is part of quantitative approach that the target – study uses an existing data base and infers characteristics or relationships from it.

8. What makes quantitative research approach different from the qualitative research approach?

Explanation: The difference is in the case of quantitative approach

- The research work done on existing theory
- It use statical techniques
- The quantitative technique employed by one researcher can be used by another researcher on the other hand

- Qualitative approach

- Data are used to explain a new theory
- Quantitative techniques are not employed and
- The study cannot be repeated by other researchers.

9. Describe the concept of simulation (model) approach?

Explanation: This approach involves constructing an artificial environment (model) to represent the actual environment we are studying.

10. Explain briefly the determining factor that chosen quantitative or qualitative research approach?

Explanation: Most scholars argue that the nature of the title and the researchers skills determine the selection of the above two research approaches.

5.4. The nature of Geographic Research

- The geographic research distinct from other disciplines is its concern for studying the distribution and organization of phenomena and activities over the surface of the earth. Hence, its nature is largely the concern of spatial interaction.
- This history of geographical research work is classified in to

a) Traditional or early geographical researches

b) Modern geographical researches

a) Traditional Geographical research

→ It is related to the locations of places and people.

→ They were simply enquiries made unscientifically.

→ They were largely based on human's curiosity to know

→ The questions asked were either left un answered, or where explained unscientifically.

b) Modern geographical researches

→ It coincided in the 16th century with; the age of exploration and discovery and the period of the Renaissance.

→ In this time the traditional views started to die out being substituted by reasons.

→ GIS produced a powerful investigative tool for geographic research.

The concerns of geography and geographers are mainly to

- Assess the spatial distribution of the population
- Determine the forces that governed the distribution.

→ Is the distribution due to physical factors? Or

→ Is it due to economic factors?

- Examine patterns of population distribution (is it sparsely or densely populated, is there uniformity or not:)
- Question whether there is a relationship between the distribution and the factor observed.
- To investigate possible relationships between population distribution and other factors.
- Look at the impact of populations on the surrounding natural resources.
- Forward views on the sustainability of the area or locality for the future.

Geographic Information system (GIS)

- It is a computer system that records, stores and analyzes information about features of the earth's surface.
- It provided is its ability to generate two – dimensional and three dimensional images of an area.
- It can receive geographical data from maps, satellites, photographs and printed texts and books.
- It allows geographers to conduct research on environmental changes
- It is an information – acquisition and interpretation device, has enhanced geographer's ability to perform accurate high – level research.

Illustrative Questions

11. Which of the following brought a major change to geographic research as of the 1960's? (EUEE, 2005/2013)
    A. The application of spatial analysis as a new technique in geographic studies
    B. Advancements made in exploration and discovery
    C. Advancements made in the field of Astronomy
    D. The use of maps as sources of data

Explanation: In the 1950s and 1960s, the adoption of quantitative techniques or application of spatial analysis has a major change to geographic research.

Answer: A

12. Which of the following is NOT an expected outcome of modern geographic research? (EHEECE, 2002/2010)
    A. Challenging the laws of nature
    B. Enhancing knowledge
    C. Predicting about the future
    D. Finding solutions to certain problems

Explanation: Modern geographic research focus on enhancing knowledge, predict about the future, finding solutions to certain problems

Answer: A

13. The major focus of research in geography is explaining the (EHEECE 2002/2010)
    A. History of nations and nationalities
    B. Variable character of the earth's surface
    C. Elements that constitute the core of the earth
    D. elements that constitute the upper part of the atmosphere

Explanation: The major focus of research in geography is studying the distribution and organization of phenomena and activities over the surface of the earth

Answer: B

14. A system that use sophisticated computer hard ware and software's to collect, store, manipulate and process spatial data is
    A. The geographic information system
    B. Arial photo interpretation system
    C. Map reading system
    D. Cartographic system

Explanation: A system that use sophisticated computer hardward and softwares to collect, store, manipulate and process spatial data and also analyzes information about features of the earth's surface is geographic information system (GIS)

Answer: A

15. Describe briefly the nature of traditional and modern geographical researches.

Explanation: The nature of traditional geographic research focus on location of places and people and also if enquiries made unscientifically on the contrarily modern research its focus on reasoning view and discovery of things like GIS.

5.5. Basic Research Methodology in Geography

- The current trends of modern geographic studies are being highly assisted by scientific research activities, since it is believed that it solves social and economic problems of a given society.
- Scientifically based geographical research works demand time, experience, finance and other supporting things.
- Research, research method and research methodology are related to one another, but vary in scope.

→ A research project is an inquiry into a problem. It is also the foundation of research methods and methodology.

→ A research method or technique is as kill that uses different steps or elements to solve the identified problem and arrive at a possible solution.

→ A research methodology is much broader than research and research method.

Basic Elements of Research

After considering basic questions of the research methodology for their projects, geographers conduct their research by performing the following tasks.

A. Identifying or Formulating the Research problem

It is a step performed after identifying the problem to be studies. Here, the researcher visualizes the nature, depth and effect of the problem under study.

B. Defining the significance of the study for solving the problem

- Its significance is its importance at a local, regional, national or global level.
- A study must benefit a society or community by providing new discoveries, improving existing situations demanding new decisions and enriching or refuting already existing theories.

C. Studying the problem Area

- When we study the problem area that a research project addresses, we create a detailed bird's eye – view for ourselves and others who will access the research. As we study the problem area, we identify problems and eliminate ambiguity

D. Defining the objectives of the study

- This step is actually concerned with what the researcher intends to achieve finally after studying the problem.
- In this step, general and specific objectives are set. The general objectives provide short statements of the goal, while the specific objectives provide detailed inter-connected statements.

E. Defining the scope and Delimitation of the study

Whenever a problem is identified for a research work, care should be taken to limit its extent in reference, material

consumed and the current capacity of the researcher. Thus, this is known as scope or delimitation of the study.

F. Limitation of the study

While the research is on, the researcher may come across certain shortages, such as lack of proper or adequate information, time constraint, limited finance, etc. thus, the research may have some sort of bias at the end. Therefore, such a phenomena is known as limitation in a research work.

G. Reviewing the Literature

- It means identifying, reading and analyzing documentation related to the research you plan to perform.
- For example; books, academic journals, conference proceedings, government reports and published and published and unpublished documents related to the problem.
- For quantitative research, the literature review can help identify the variables and their relationships.
- For qualitative research, the literature review helps to expand your knowledge of the research subject.

H. Developing or Formulating a working Hypothesis

- It is a preliminary assumption or tentative explanation that accounts for a set of facts, taken to be true for the purpose of investigation and testing a theory.
- It refers to the formulation of a potential, or attentive solution to the problem identified.
- It should be tested before the final conclusion is made.
- It can be formulated based on the ideas gather from inducting discussions with experts, inferring from the work of other on similar studies and personal investigations of conducting origin field interviews with interested parties and individuals.
- A hypothesis must have the following characteristics includes it should be clear and precise, capable of being tested or

verified, state relationships between the variables determined and explain the facts that give rise to the explanation.

I. Preparing the Research Design

- The research design is the conceptual frame work with in which the research could be conducted.
- The preparation of an appropriate research design for a particular problem involves the following considerations.
  - → **The means of obtaining the information**
  - → **The competence of the researcher**
  - → **The time available for the research**
  - → The money available for the research

J. Defining the sample Design

- If a population is very large, and we try to address the entire population, out project might be too large to be practical. In this case choosing the population sample is called sampling. The resulting study is called a "sample study".
- Sampling is one of the statically methodologies used in geographic research to make statically valid inferences. Thus, sampling is important to obtain a representative specimen of a total population. The specimen so taken to be valid and dependable is called sample.

Types of Sampling

1) Random Sampling (Probability Sampling)

- It is the most accurate method where every member of the sample has equal chances to be selected.
- It avoids bias.
- It is called chance or lottery sampling.

2) Systematic Sampling

- Unlike random sampling a sort or personal interference is involved and thus some bias is un avoidable. In this technique, the researcher reorganizes the available data

depending on the nature of the data. Thus, in systematic sampling, the data are collected in a given regular way.

3) Stratified Sampling

- This technique of sampling involves the division or segmentation, or stratification of clearly observable disparities in character or nature, or some other elements of the study area related to the problem.

4) Clustered Sampling

- This is relatively a complex technique involving the above three methods of sampling together

K. Collecting and organizing the Data

- Data are varieties of information (evidence) gathered for a purpose.
- There are two types of data collection: primary and secondary.

i) Primary Data Collection

→ It is original in character

→ The research produces the data, rather than relying on existing data

Ways of Collecting Primary data

→ The researcher's personal observations

→ Interviews conducted through telephone conversations and in person.

→ Questionnaires

→ Discussion with groups of people

ii) Secondary Data Collection

→ It investigation data that has already been collected, organized and arranged by someone else.

→ It may be either published or unpublished

→ Published data include

→ Publications by governments

→ Publication by NGOs
→ Journals
→ Magazines and newspapers
→ Reports, by universities, scholars, etc.
→ Historical records, etc.

→ Unpublished data include.
→ Diaries
→ Letters
→ Unpublished biographies, etc.

L. Analysis of the Data

- It involves the use of stastical techniques to measures relationships of the geographic variables defined in the hypothesis.
- The important issue in this step is the selection of appropriate measuring techniques.
- Example, simple ratios and percentages, pie charts, compound graphs measures of averages and raw – data variances.

M. Preparing the Research Report, Based on the Findings

- The report must be clear and accurate.
- A good report is the product of concentration, accuracy and the logical arrangement of facts and ideas.

N. Compiling the Bibliography/References

- It is the final step of your research project
- It is a list of written materials relevant to the research study performed.
- It includes all books, documents etc.

Illustrative Questions

16. Which of the following indicates the correct order of conducting geographic research? (EHEECE, 2002/2010)
    A. Identifying the research problem, data analysis and interpretation, data collection.

B. Data collection, data analysis and interpretation, identifying the research problem.
C. Identifying the research problem, data collection, data analysis and interpretation.
D. Data collection, identifying the research problem, data analysis and interpretation.

Explanation: The correct order of conducting geographic research is identifying the research problem, data collection, data analysis and interpretation

Answer: C

17. Which of the following is considered to be the least biased in spatial sampling?
    A. Random sampling
    B. Systematic sampling
    C. stratified sampling
    D. cluster sampling

Explanation: In sampling technique, random sampling method where every member of the sample, has equal chances to be selected. Thus, this avoids bias.

Answer: A

18. Which one of the following is the initial stage in modern geographic research? (EHEECE,2002/2010)
    A. Preparing maps and charts
    B. Identifying the research problem
    C. Conducting field survey
    D. Defining the research objectives

Explanation: Identifying the research problems is the initial stage in modern geographic research.

Answer: B

19. A tentative answer to a research problem or questions is referred as _____ (EHEECE, 2003/2010)
    A. Research methodology
    B. Hypothesis
    C. Data gathering
    D. Research objective

Explanation: A tentative answer to a research problem or questions is referred as hypothesis.

Answer: B

20. In Geographicresearch, if is not always easy to take the entire population as a study unit. What technique do you use to decide your study unit? (UEE – 2004/2012)
    A. Sampling techniques
    B. Use of qualitative method
    C. Changing the research topic
    D. Dividing the study unit among many researchers.

Explanation: If a population is very large, and we try to address the entire population we changing the population sampling.

Answer: A

5.6. Conducing Action Research

- It is an ongoing research process by which a particular real problem is identified, conformation/ data are gathered, practical solutions are tested, conclusions and recommendations are reached and finally improvements are made.
- The basic features of action research includes
  - → It is based on reality
  - → It enables researchers to put ideas in to action
  - → There is rapid feedback
  - → It aims at improving situations, not at producing new knowledge
  - → It encourages reflections and further developments.
- The following steps are important in conducting action research
  1. Selecting the problem
  2. Objectives of the study
  3. Making Review of literature
  4. Establishing working Hypothesis
  5. Determining the sample size
  6. Collecting the Required Data

7. Analysis of the data
8. Generalization and Interpretation
9. Preparation of the report

Illustrative Questions

. Which of the following statement is NOT correct about the features of action research?

A. There is rapid feedback
B. It encourages reflections
C. It is based on assumption
D. It is not producing new knowledge

xplanation: Action research is an ongoing research processes based on reality not based on assumption.

Answer: C

. What are the important aspects or consideration in the processes of selecting a research problem?

xplanation: Identify, problem is the first necessary step in conducting action research. In this step the researcher should considered the problem not be overdone, not be adebatable issue, simple and clear and it does not dis coverage the participants.

. What is the main function of Ahypothesis?

xplanation: The main function of ahypothesis is to put forward an assumption that can be verified by the research.

. Write the basic elements of action research?

xplanation: Basic elements of research includes identifying research problem, significance of the study, defining the objectives, defining the scope and limitations, reviewing the literature etc.

. Describe the special characteristics of action research?

xplanation: Like the other type of research action research also to fulfill all steps starting to problem is identified up to

recommendation, however, it is special for finally assessing improvements are made or not.

26. Which of the following is **NOT** among the main uses of geographic research?
    A. Getting new knowledge
    B. Gaining ability to predict about the future
    C. Finding answers to the question we ask
    D. Using old skills to solve new problems

Explanation:
As dynamic science Geography does not use old skills to solve new problems

Answer: D

27. Which of the following comes first in a standard research proposal?
    A. Statement of the problem
    B. Research objectives
    C. Research Methodology
    D. Hypothesis

Explanation:
Stating the problem is the primary task in a standard research proposal

Answer: A

28. Which of the following is a secondary source of data?
    A. Questionnaires
    B. Interviews
    C. Newspaper
    D. Personal observations

Explanation:
Secondary or documentary data sources are
- News paper • Journals • Historical records
- Publications by NGOs and governments…

Answer: C

Review Questions on Unit Five

Part I Write "A" if the statement is true and "B" if it is false

1. Modern geographic research focus on the location of places.
2. Research is a time and money consuming activity.
3. The significance of a research project is similar at everywhere.

4. Data analysis involves the use of stastical techniques to measure relationships of the geographical variables.
5. Inferential quantitative approach manipulates variables to see their effects on other variables.

Part II. Match items in column "A" with that of column "B"

A	B
6. Research problem	A. Existing theory
7. Ahypotheiss	B. Delimitation
8. Non probability sampling	C. Assumption or tentative explanation
9. Probability sampling	D. Action Research
10. Reviewing the literature	E. Interview
11. Bibliography	F. Deliberate sampling
12. Analyzing the data	G. Second stage of research
13. Scope of the study	H. New theory
14. Objectives of the study	I. Traditional Geographic Research
15. GIS	J. Modern Geographic Research
16. Research Design	K. Initial stage of research
17. Primary data	L. General and specific
18. Secondary data	M. Interpretation
19. Qualitative approach	N. Analyzing documentation
20. Quantitative approach	O. Journals
	P. Random sampling
	Q. References
	R. conceptual framework of Research

Part III. Completion questions

Fill in the blank space with appropriate words or phrases.

21. _____ is a skill that uses different steps or elements to solve the identified problem and arrive at a possible solution.
22. ___ means identifying, reading and analyzing documentation related to the research plan.
23. _____ is the conceptual frame work with in which the research could be conducted.
24. In the quantitative approach, hypothesis are established and _____

25. A systematic inquiry in to causes or the discovery of new facts through planned and organized effort is _____
26. _____ is often called chance or lottery sampling
27. _____ is a type of research that put ideas in to action.
28. A list of written materials relevant to the research study is called _____
29. In qualitative approach group interviews done through _____ and _____
30. _____ is technique of sampling involves the division or segmentation or stratification of clearly observable disparties in character, or nature

Answer Key on Unit Five

Part I.

1.B 2.A 3.B 4.A 5.B

Part II.

6.K 7. C 8.P 9. F 10.N 11.Q 12.M 13.B 14.L 15.J
16.R 17.E 18.Q 19.H 20.A

Part III.

21. Research method
22. Review literature
23. Research design
24. Tested
25. Research
26. Probability or random sampling
27. Action Research
28. Bibliography References
29. Telephone and in-person interviews
30. Stratified sampling

UNIT SIX

Map use and Map Work

6.1. THE STUDY OF TOPOGRAPHIC MAPS

6.1.1.. Meaning and uses of Topographic Maps

Meaning of Topographic Maps

- Topographic is a Greek word made up of "topos" and "graphy". "Topos" means places, and "graphy" means writing or description. Therefore topography means the description or explanation of places.
- They are often known as topo sheets, scenery and reference maps.
- They are large scale maps that give detail information of a given place or area often with a scale of 1:250,000 to 1 = 25,000.
- It provides a great variety of both physical and human made features.

Nature of Topographic Maps

A topographic Map is

- A map that portrays rivers, streams, drainage, vegetation, roads, buildings, land uses, etc.
- Shows multiple natural and anthropogenic features;
- A large scale map drawn to show a relatively small area.
- A map that uses definite symbols, known as conventional symbols.

Uses of Topographic Maps

Topographic maps are multipurpose maps and are useful for various study as well as research works some of the major uses are stated below.

1. They portray a number of features both physical and cultural, they enable us to recognize the relationship between people and their environment.

2. Show the existing land forms and help us to understand the geological forces responsible for the formation of these visible landforms.
3. Topographic maps are very much useful for economic planning purpose.
4. Topographic maps are also very useful in identifying the location of the enemy and other military targets.
5. Show the various natural and cultural landscapes.
6. Help engineers to select the right engineering design viewing the portrayed features.
7. Act as documents for tourists
8. Provide information about the landscape

Illustrative Questions

1. Define the term topographic map?

Explanation: Topographic map is a map which is consisting of two Greek word topos and graphy. Topos means places and graphy means description. Therefore topography means explanation or description of places.

2. List the nature or unique features of topographic maps?

Explanation: In general topographic maps provide a variety of information i.e. both natural and human made features.

3. Write the major uses of topographic maps?

Explanation: The major uses of topographic maps includes they portray a number of features both physical and cultural, useful for economic planning purpose, useful in identifying the location of the enemy etc.

6.1.2. Conventional signs and Symbols on Topographic Maps

- The signs and symbols on topographic maps are listed and explained in the map's key or legend.
- A map symbol could be expressed in various ways eg. Diagrams, letters, numbers, or abbreviations etc.
- The legend contains the symbols most commonly used in particular series or on that specific topographic map sheet.
- They are vary in amount and kind from country to country.

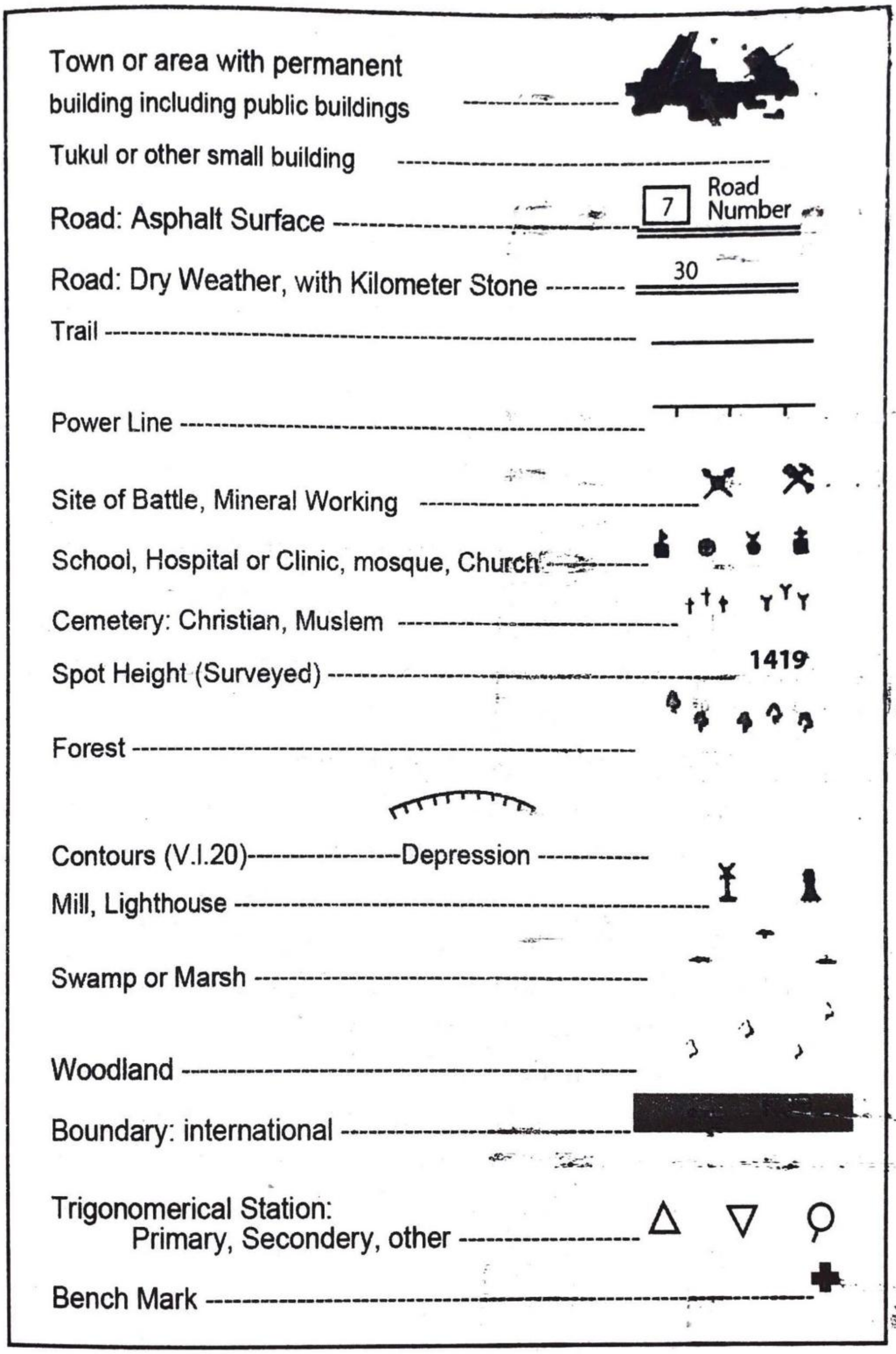


Figure 6.1. Conventional Symbols used in the topographic map of Ethiopia

Illustrative Questions

4. Write the importance of conventional map symbols?

Explanation: conventional map symbols serve as a key to help the map user to read a map effectively.

5. What are conventional signs and symbols?

Explanation: one vital principle of maps is symbolization i.e. maps use their own languages in order to communicate with users of the information found in the map

6. Explain the term key related to conventional signs and symbol?

Explanation: Key is one of the main types of marginal information. The meaning of map signs and symbols is given in key, also called legend or index or Reference.

7. Mention the common characteristics of conventional signs and symbols?

Explanation: Common characteristics such as standard colors that are internationally and standard symbols, that represent physical or human made feature.

8. List the universally accepted conventional map colours of relief, drainage and communications?

Explanation: Relief represented by brown colour, drainage blue color and communication represented by black, red or brown colors.

6.1.3. The study of Distribution Maps

- Distribution maps are maps that show the spatial distribution of geographic features.
- Different kinds of distribution maps are used world-wide.
- Based on their purpose maps classify in to.
  - → Reference or general – purpose maps
  - → Special purpose or thematic maps
  - → Charts

Reference or General Purpose Maps

→ They are topographic maps that portray different features as the same time.

→ They are made at a larger scale covering smaller areas to show details.

Thematic or special – purpose maps

→ They are emphasize the distribution of a single geographical – element or character.

→ They are drawn of small scale

→ Example soil map of Ethiopia
→ They are classified in to two sub groups. They differ from each other on grounds of the nature of information they convey.
  → Qualitative maps
  → Quantitative maps

A. Qualitative Distribution maps

→ They show the areal spread of geographic elements without taking in to consideration their quantity.
→ They are also called non – quantitative areal distribution maps.
→ They are purely descriptive maps containing no measurable facts.
→ They are sub-divided in to the following two types.
  i. Chorochromatic maps and
  ii. choroschematic maps

i) Chorochromatic Maps

→ They show the spread of various geographic features means of different colors, tints, dots, or line symb without any conditional of values.

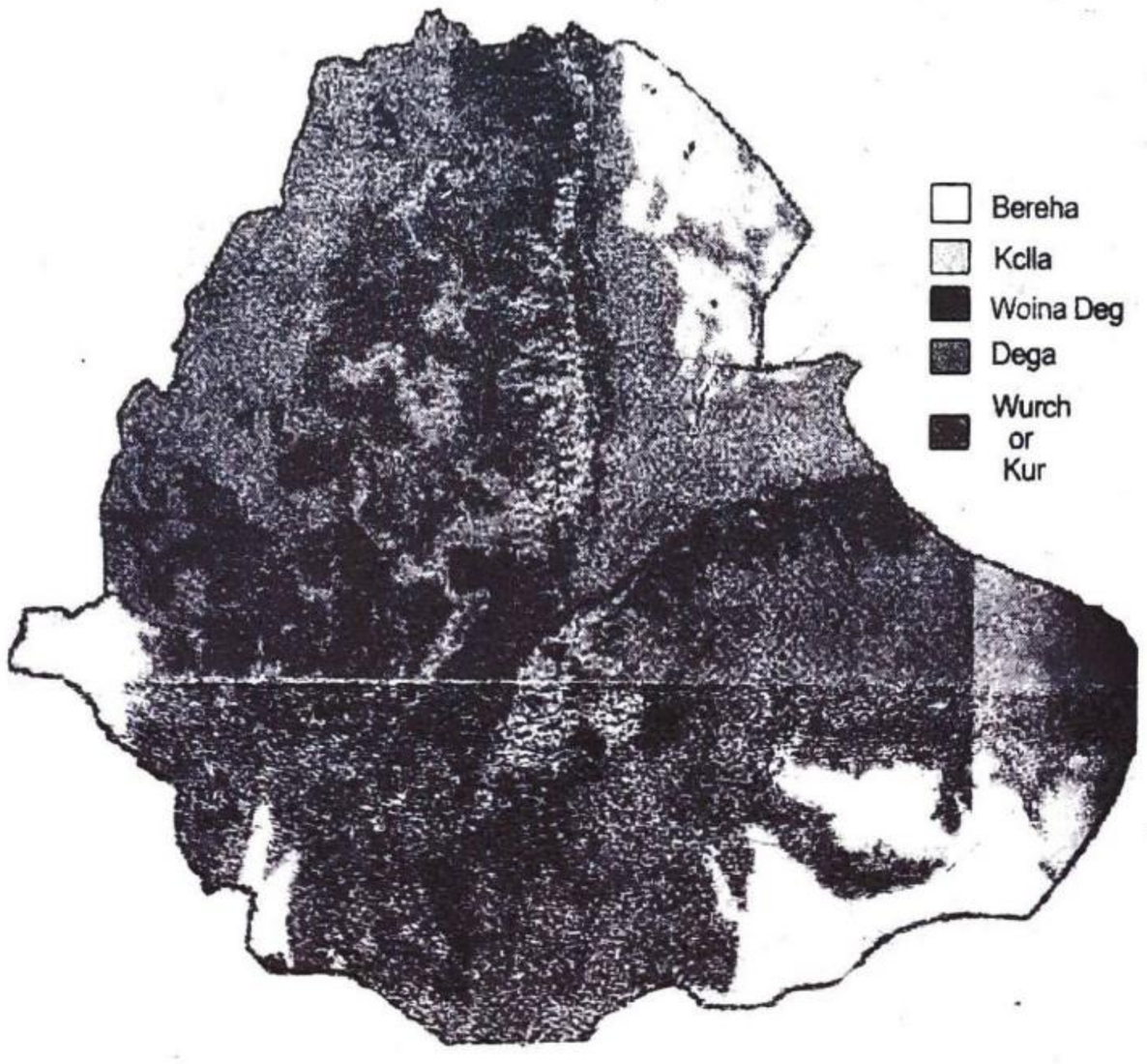


Figure 6.2. Agro climatic Zones of Ethiopia

ii) Chroschematic Maps

→ They use pictures and letters.

→ They are relatively simple to understand

→ Example, pictures of trees for forests, pictures of huts and houses for human settlements etc.

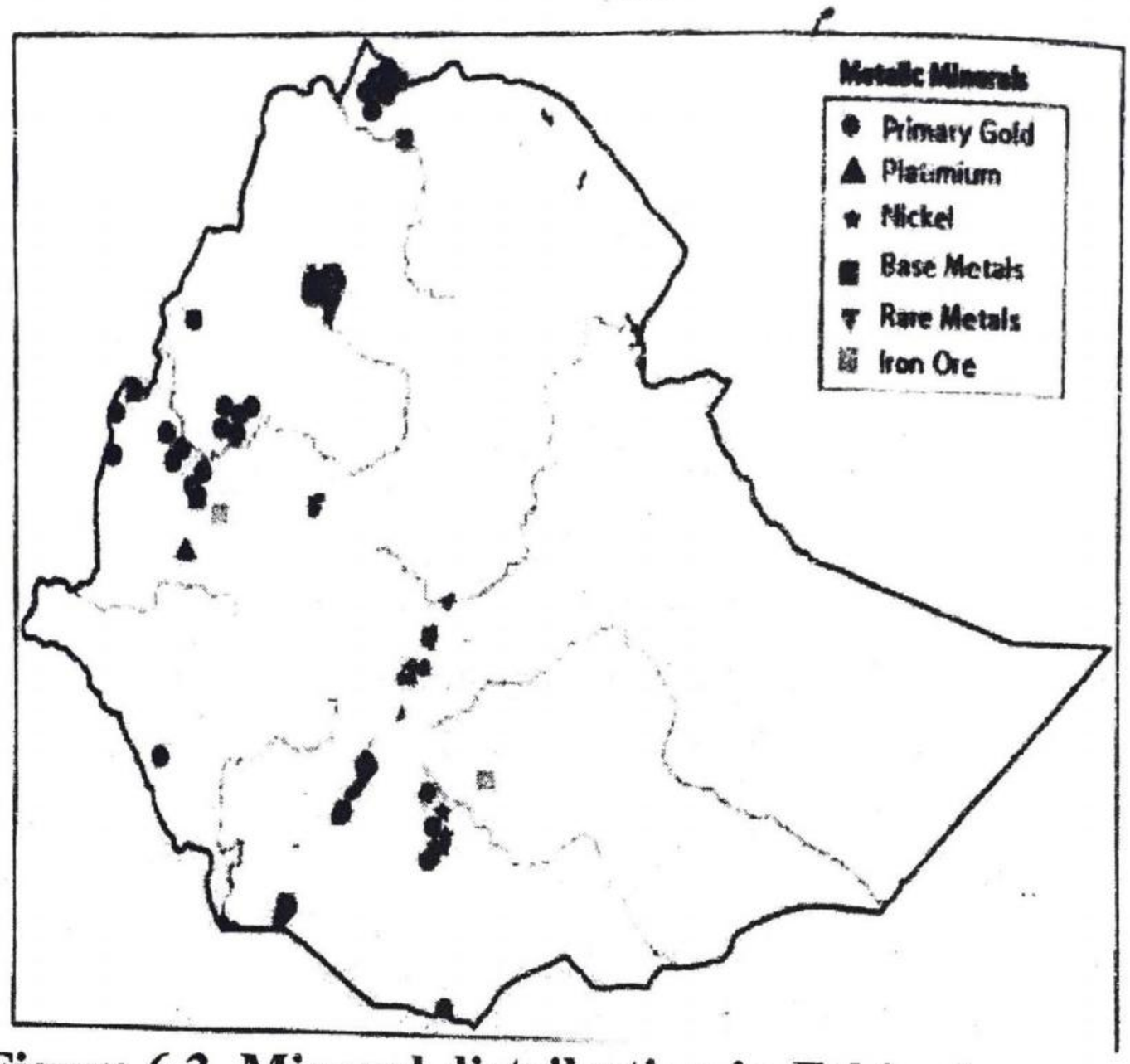


Figure 6.3. Mineral distribution in Ethiopia

B. Quantitative Distribution Maps

→ They display spatial distribution of geographical elements, representing quantitative values.

→ The most widely used quantitative distribution maps include:

i. Dot maps
ii. choropleth maps
iii. Isopleth maps
iv. Graphs and diagrammatic maps

i. Dot Maps

→ They show the areal spread of geographic features by means of dots of specific value.

→ They show the distribution of economic products, human and other animal populations.

→ In dot quantitative maps, each dot represents a given quantity or value.

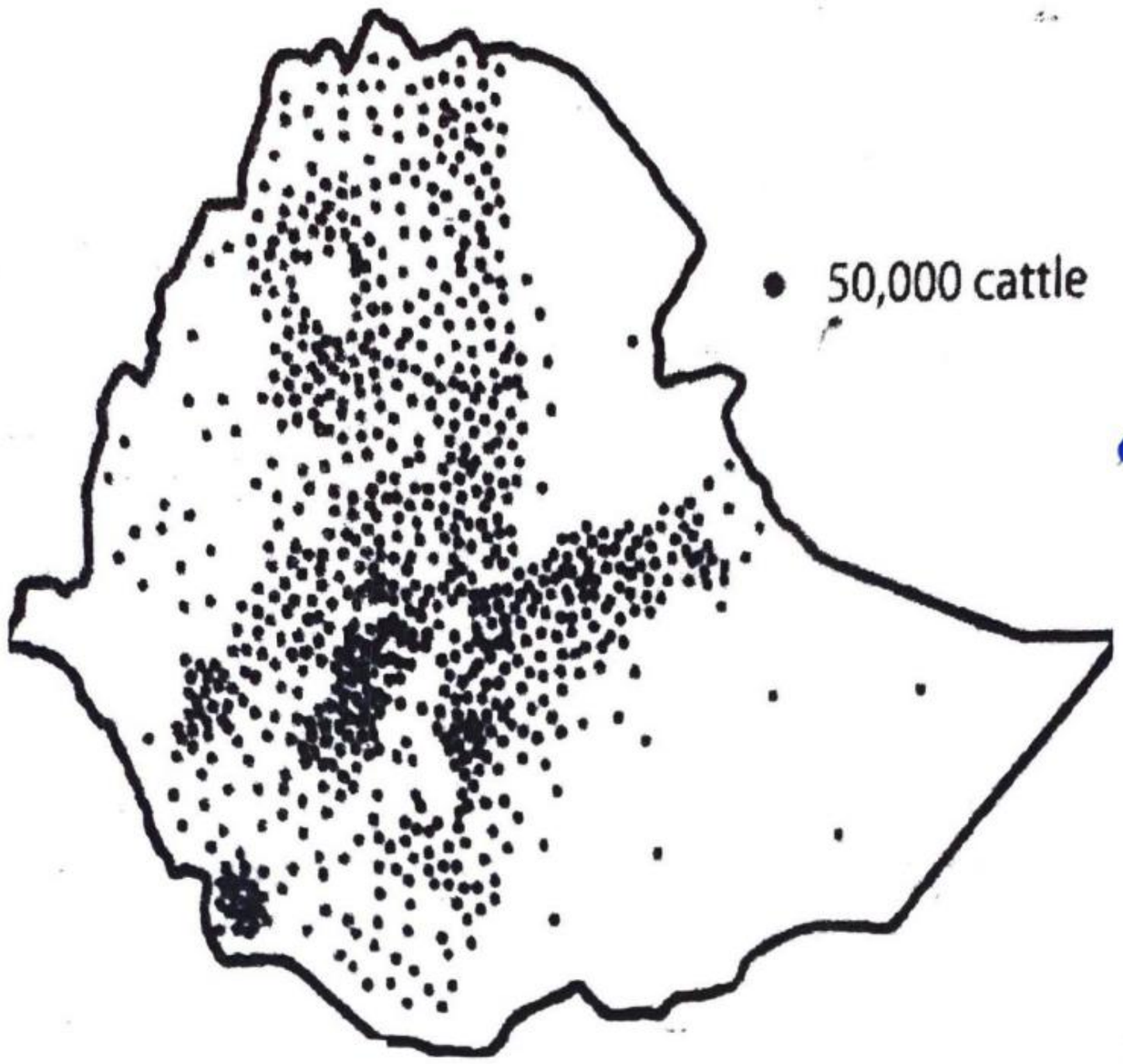


Figure. 6.4. Cattle distribution in Ethiopia

ii. Choroplethic Maps

→ It referring to average values using a varying intensity of shading.

→ They are called shaded maps.

→ The display quantifiable features

→ They show the distribution of average values such as population density, cropping intensity, etc.

→ The varying intensity of shading shows a progressive increase or decrease of the value.

→ Lighter shades give lower density, while dark shades suggest higher density. This kind of shading is called graded shading.

iii. Isoplethic Maps

→ They are lines of equal value or magnitude

→ It is equivalent to isolines

→ They are most commonly used for representing climatic elements, transportation costs, etc.

- Isolines are those imaginary lines that are connecting places of equal value of measurable geographical data. Some of the element of isolines are:

→ Isotherm maps are ones that show equal temperature points.
→ Isobar maps are those that show equal pressure distribution
→ Isodapane maps are those that show equal transportation cost distribution.
→ Isoneph maps are those that show equal cloud cover distribution
→ Isohypose are lines that connect places having the same altitude (height);
→ Isobaths are lines that connect places having the same ocean depth.
→ Isohyet maps;are those that show equal rainfull distribution.

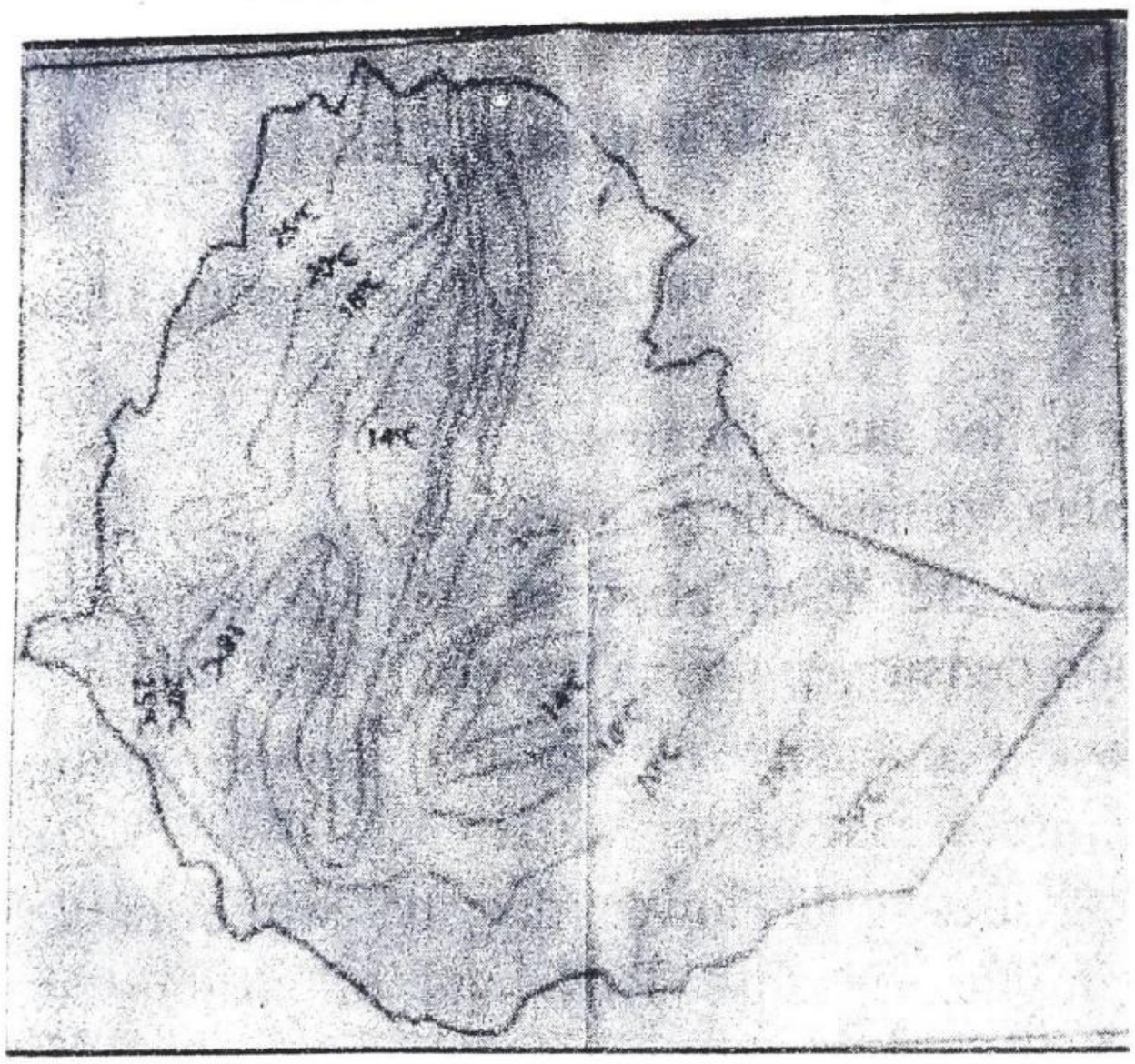


Figure 6.5. Temperature distribution in Ethiopia

iv. Graphs and Diagrammatic Maps These are often, used to show economic data, population data and other quantifiable distributions.

Graphs:

- The simplest and most widely used graphs and line graphs and bar graphs.
- They are used to show the relationship between two geographic variables in time and space series.

- Both line graphs and bar graphs can use a single or double format.
- Bar graph representing climatic data especially temperature and rainfall.
- There are different types of bar graphs includes simple, multiple, compound and percentage bar graph.

Diagrammatic maps

- They showing quantitative distribution involves the use of geometric figures and diagrams representing varying values.
- They are more explanatory than graph maps since they are two dimensional or areal.
- It is expressed by means of square, proportional circles, proportional cubes and proportional arrows.

Examples Using proportional squares

- It is a method of representing quantity by using square
- The area of each square is made proportional to the quantity it represents
- To show distributions using proportional squares, follow this procedure:

  Step 1: Calculate the square – roots of the values given

  Step 2: Let the least square root computed represent a side of the square that measures $1cm \times 1cm = 1cm^2$.

  Step 3: Compute the side of the values in an ascending order and draw the squares, using the computed square roots.

Example: Major rivers of Ethiopia and their catchment area (km^2)

1. Wabishebelle – 205, $407km^2$
2. Baro – 75, 781
3. Ghenalle – $168,141km^2$

Solution:

The square root for =

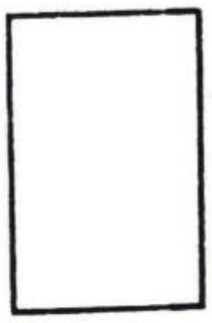
1. Wabishebelle $\sqrt{205,407} = 453.21$
2. Baro $\sqrt{75,781} = 275.25$
3. Ghenalle $\sqrt{168,141} = 410.04$

Let, the square root of the quantity of the lowest value, be represented by a square of 1cm side. Then,

275 = 1cm

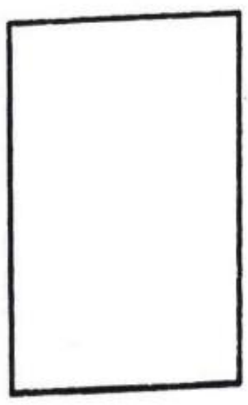
410 = 1.5cm

453 = 1.6cm



1cm × 1cm

Baro River



1.5cm × 1.5cm

Ghenalle River

1.6cm × 1.6cm

Wabishebele River

Proportional Circles

- They are also known as a real diagrams or graduated circles.
- Each areal diagrams has an area which is proportional to the quantity it represents.
- They are used for comparing quantities.
- They are also used as proportional symbols to make comparison between the quantities and whose locations are marked on the maps.
- For drawing proportional circles, a suitable value (i.e. the least value being represented by I centimeter) is selected and radic of circles for representing different values or magnitudes are calculated.
- The area of a circle is made proportional to the quantity of represents.

Example: Population of a given hypothetical towns A,B,C and D in a given year

Town	Population
A	14,400
B	32,400
C	46,225
D	62,500

Solution: Suppose 14,000 persons are represented by a circle of 1.0cm radius

$$\text{Town A} = 1.0 \times \sqrt{\frac{14400}{14,000}}$$

$$= 1.0 \times \sqrt{1.03}$$

$$= 1.0 \times 1.01$$

$$= 1.01\text{cm}$$

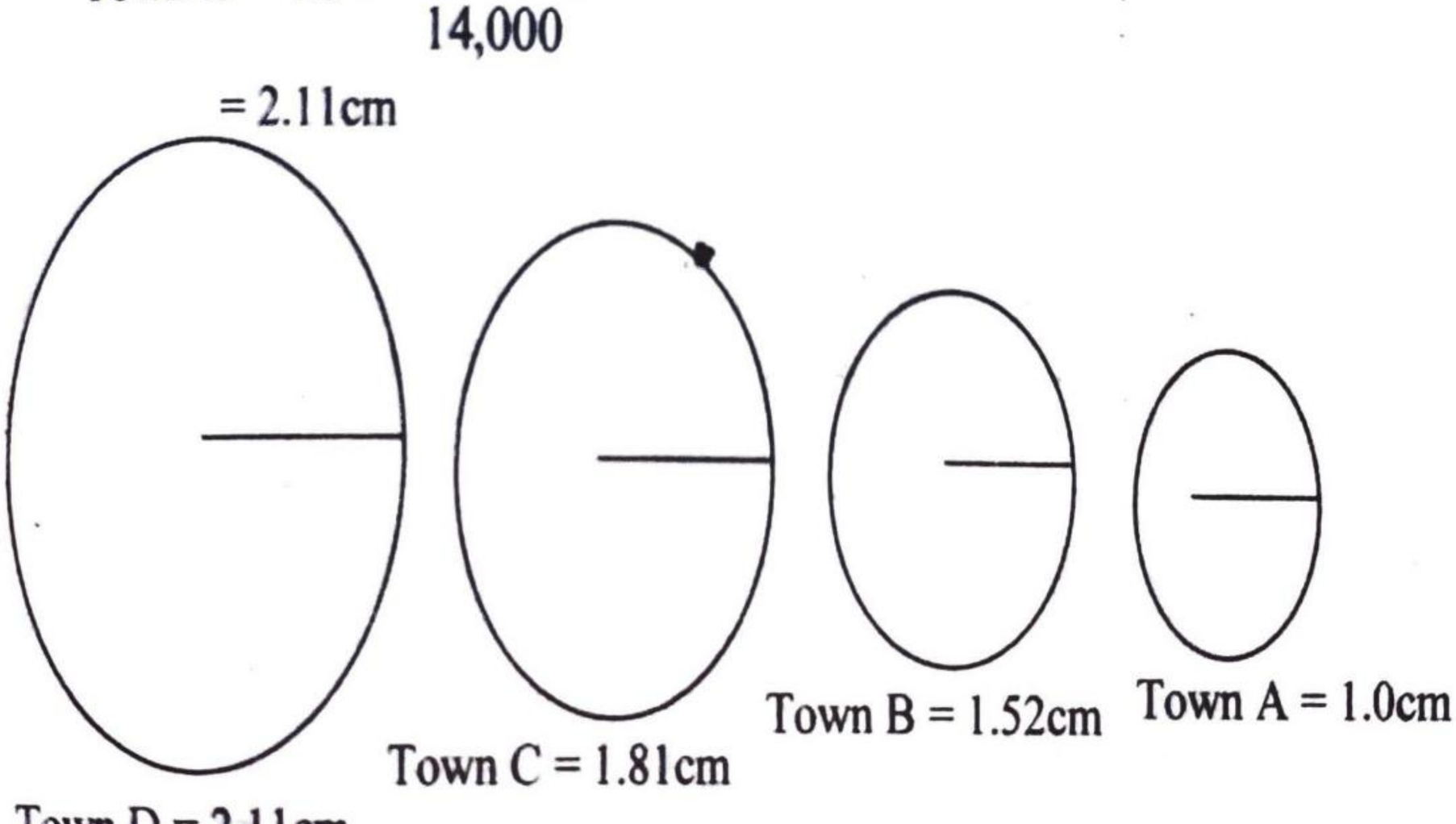
$$\text{Town B} = 1.0 \times \sqrt{\frac{32,400}{14,000}}$$

$$= 1.52\text{cm}$$

$$\text{Town C} = 1.0 \times \frac{\sqrt{46,225}}{14,000}$$

$$\text{Town D} = 1.0 \times \frac{\sqrt{62,500}}{14,000}$$

$$= 2.11\text{cm}$$



Town D = 2.11cm

Proportional arrows or flow map

- Flow maps using different arrows of varying thickness represent the movement of goods in varying quantities from one region to another.
- They are varying thickness can be sued to show different magnitudes or flows.

- They are used to portray degree or volume of movements of goods, traffic flow, migration of people, direction of exports and imports, etc.

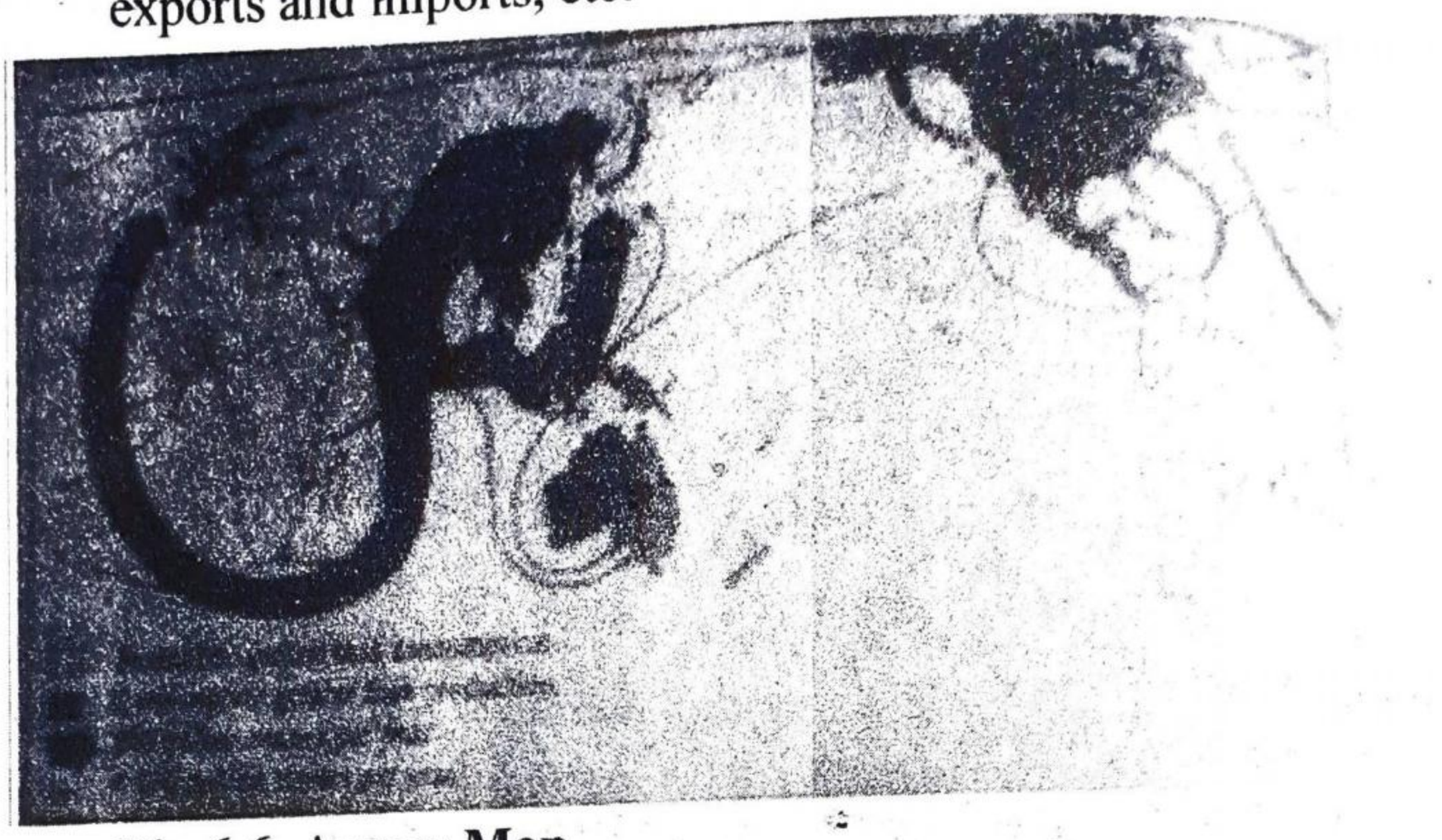


Fig 6.6. Arrow Map

Illustrative Questions

9. One of the following methods of presentation of data is more used in geography than in other disciplines
   A. Bar graphs
   B. Line graphs
   C. Maps
   D. Tables

Explanation: In geography bar graphy is more important because it representing climatic data especially temperature and rainfall.

Answer: A

10. Which of the following types of maps is different from the others?
    A. Dot maps
    B. Chorochromatic
    C. choroplethic
    D. Isoplethic maps

Explanation: All are grouped under thematic quantitative distribution maps where as chorochromatic is qualitative distribution maps.

Answer: B

11. _____ Maps are those that show equal rainfall distribution
    A. Isotherm
    B. Isobar
    C. Isohyte
    D. Isodopane

Explanation: Isohyte maps are those that show equal rainfall distribution.

Answer: C

12. Which of the following statements is correct about reference maps?
    A. Emphasize an a single geographical element
    B. Show different features as the same time
    C. Drawn at small scale
    D. A and B

Explanation: General reference maps that portray different features as the same time.

Answer: B

13. Proportional arrows shows that there is
    A. Two – dimensional or areal
    B. Two geographic variables in time and space series
    C. Used for making comparison
    D. Used to portray the degree of migratic of people

Explanation: Proportional arrows or flow map are used to portray the degree or volume of movements of goods, traffic flow, migration of people direction of exports and imports, etc.

Answer: D

14. Maps showing non – quantitative area distribution by picture, symbols, names, letter or numbers are called
    A. Chorochromatic maps
    B. Isopleth maps
    C. choropleth maps
    D. chroschematic maps

Explanation: Maps showing non – quantitative area distribution by picture, symbols, names, letter or numbers are called choroschematic maps.

Answer: D

6.2. Globe and Map

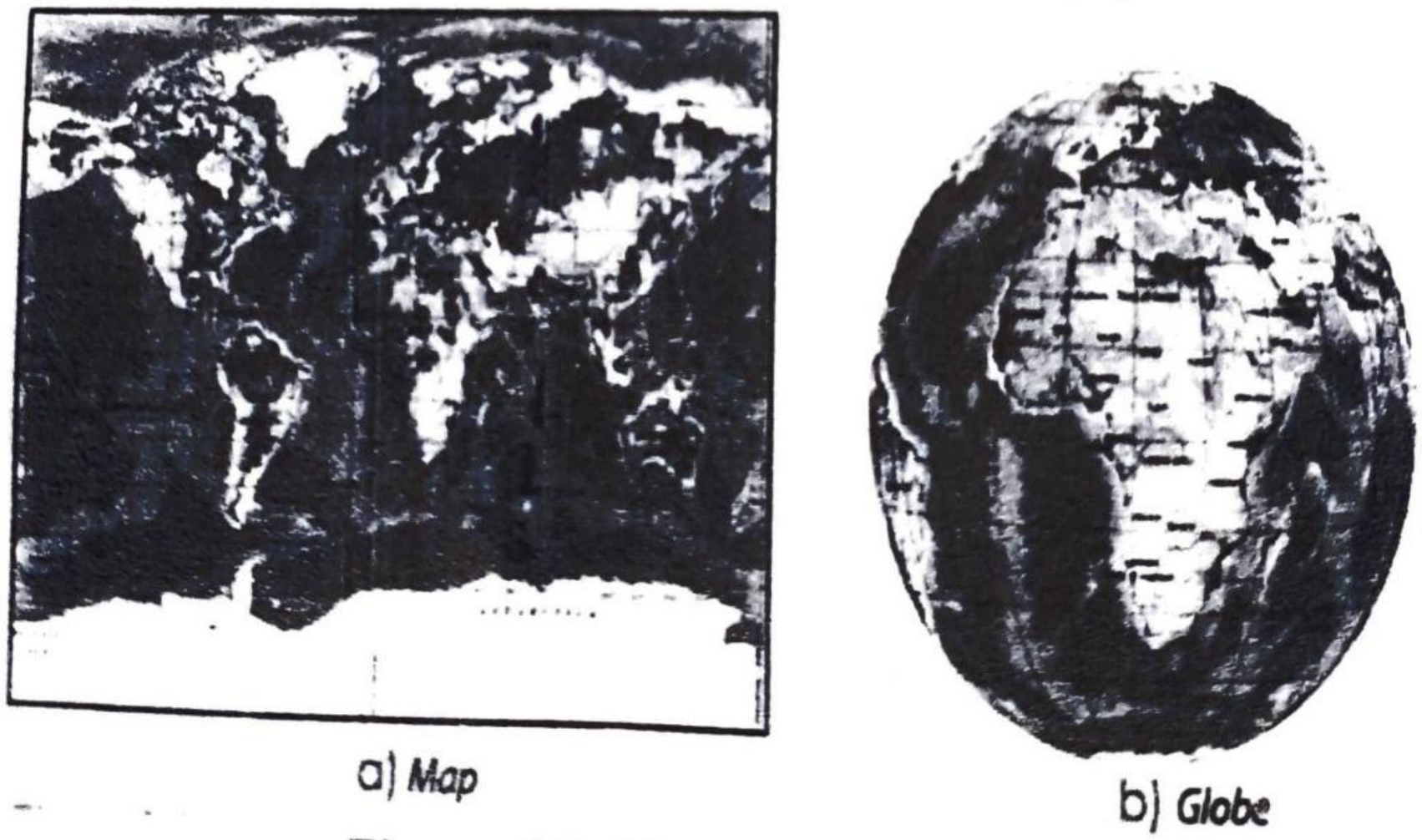
- Maps and globes are essential tools that geographers use in order to show geographic facts.
- A map is a representation of the earth an reduced scale.
- Globe is a spherical representation of the earth surface.

Properties of a Globe

- Globes are the most appropriate models for representing the surface of the earth because they are spherical.
- All meridians converge at the poles and are true north – south lines
- All lines of latitude (parallels) are parallel to the equator and to each other.
- Parallels decrease in length as you draw nearer to the poles;
- Meridians and parallels intersect at right angles.
- The scales on the surface of the globe is everywhere the same in every direction.

Advantages of a map over a globe

- Globes are expensive to reproduce and update, but maps are not
- Globes are not easy to fold and handle, but maps are.
- Globes must be rotated to show the entire surface of the earth, and they are not convenient for showing locations and distributions over very large areas. With a map, you can immediately see the whole of any area shown on it.
- Most globes are less than one meter in diameter and therefore are too small to provide detailed information.
- Globes are not visually convenient, but maps are.



a) Map

b) Globe

Figure 6.7. Map and Global

Illustrative Questions

15. Enumerate the difference between map and globes?

Explanation: Both are representation of the earth surface the difference is maps are a flat model representation a flats sheet of paper on the other hand, globes is spherical representation of earth surface.

16. Why do you think that maps are more advantageous than a globe?

Explanation: Because maps are visually convenient, relatively cheap and we can see all information at all corner etc.

17. What are the particular properties of a globe?

Explanation: Globes are spherical representation, All meridians converge at the poles, all meridians and parallels a globe intersect at right angles its are the particular properties of a globe.

6.3. Map Projection

6.3.1. Meaning and significance of Map projection

- It is a technique that cartographers use to transfer information from a globe to a flat surface in order to create a map.
- It is the technique or system of showing the curved surface of the earth on a map (flat surface) by transferring parallels, meridians coastlines and major features of the earth (on the globe) or a map either geometrically or mathematically with scale alternation.
- A map projection is not free from drawbacks and can be sued as a complete substitute for a globe. Because basic features of our earth such as shape area and direction which are correctly shown on globe, are not maintained on maps.
- Graficule is the network of parallels (latitudes) and meridians (longitudes) formed on a certain map projection.

The significance of map projection

The significance of a map projection is to eliminate all difficulties encountered when using a globe include

→ We cannot see all geographical phenomena on a globe at a place because we can only see half the globe at a time.

→ It is difficult to measure distances on a globe due to the spherical nature of its surface.
→ It is difficult to construct a large – sized globe
→ It is cumbersome to carry a globe from one place to another.
→ It is not possible to trace maps from a globe accurately because a tracing paper coming in contact with the globe develops many creases.
→ As globes are always made on small scales, they compress a lot of geographic information in to assumingly very small place.
→ As globes are costly, they are difficult to be reproduced or up dated.

6.3.2. Properties of Map Projection

- Map projections are not free from some sort of distortions.
- Map users need to know in what respects a particular map correctly reproduces, or in what respects it distorts, earth features.
- The four main properties of map projection include.
  - → area
  - → shape
  - distance and
  - direction (angle)
- These properties are distorted in different ways and to different degrees by various projections.
- A surface, such as that of the earth, that cannot be converted in to a flat surface without distortion is called an undevelopable surface.
- Based on the properties and distortions, that most commonly used projection are
  - → Equal area (Homolagraphic projection)
  - → Equidistant projection (True Distance)
  - → Projection maintain shape (orthomorphic)
  - → Projection maintain direction (Azimuthal)

6.3.3. Geometrical Map projection

- In a perspective map projection or geometrical map projection, the parallels and meridians of the globe are represented on as surface geometrically from apoint.

- Based on the shape of the developable surface, the most commonly used projections are
  - → Cylindrical
  - → Conic (Cone)
  - → Planar (plane) – also known as azimuthal

I. Cylindrical Projections

- They are best suited for the projection of tropical regions since deformities increase pole wards.
- These type of projections are developed geometrically from a cylinder, wrapped around the globe cylindrical projections are the result of transferring from the globe with out distortion.

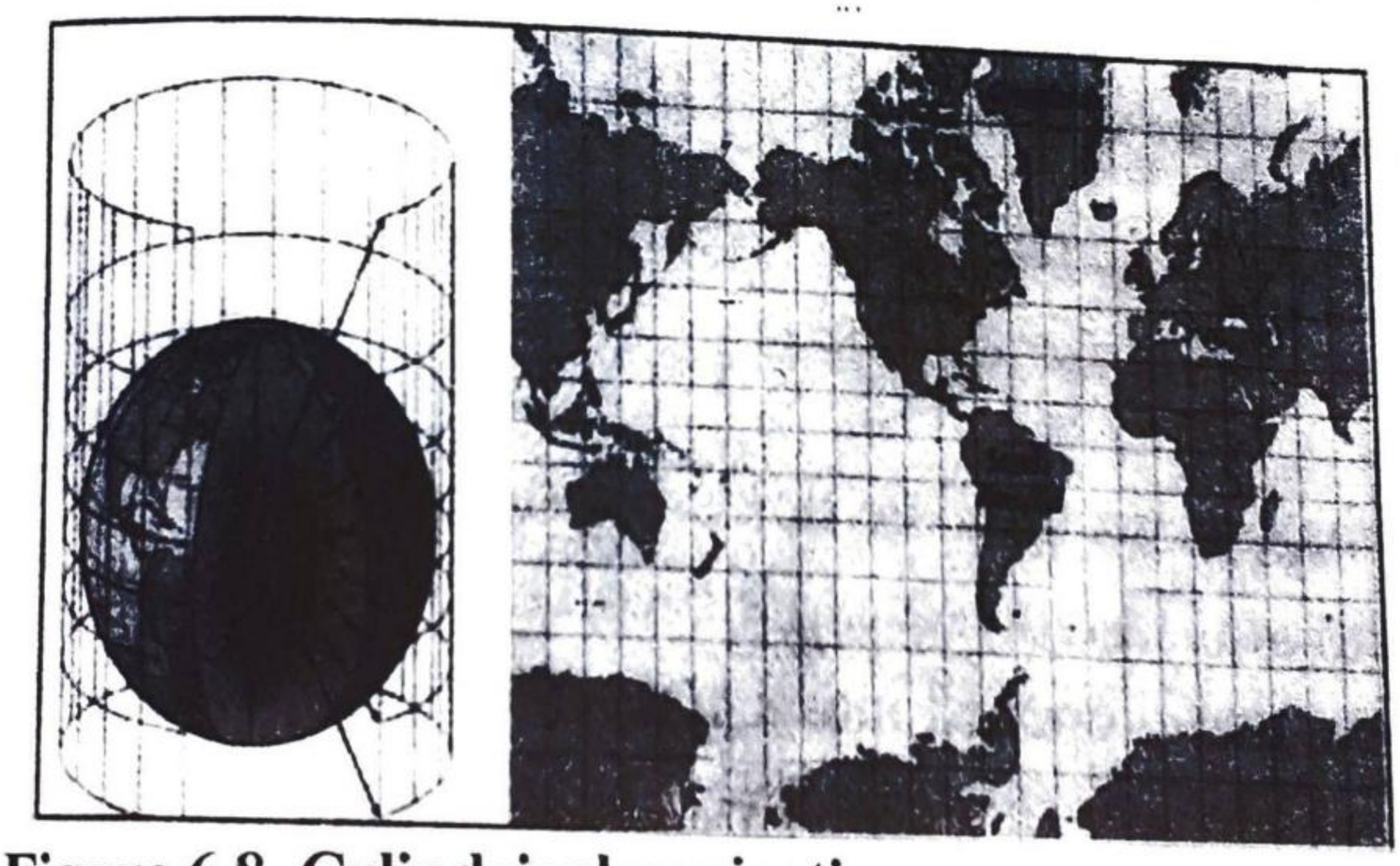


Figure 6.8. Cylindrical projections

Characteristics of cylindrical projection

- → Parallels and meridians in such a projection are straight lines interesting at right angles.
- → The lines tangent to the developable surface are true to scale.
- → All parallels are equal in length to the equator.
- → Distortions increase poles wards from the equator

II. Conic Projection

- It is derived from geometric projection. It is often employed to depict hemisphere or smaller parts of the earth.
- It obtained by covering the globe with a cone – shape developable surface.

- The cone is placed tangent to the globe a long one, two or more parallels, with the apex of the cone located above a pole.

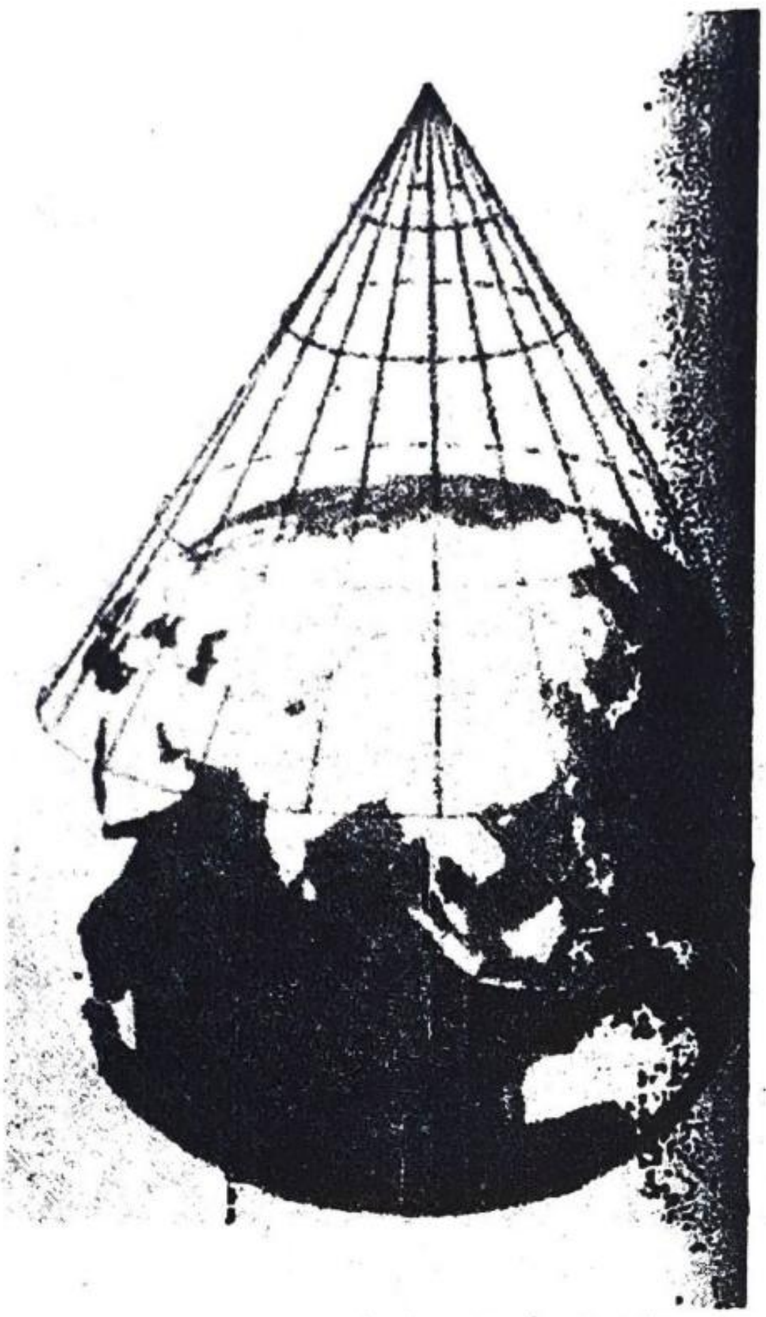


Figure 6.9. Conic Projection

Characteristics of conic projection

→ Parallels make arcs of concentric circles.

→ Meridians are straight lines radiating from the pole

→ The parallel tangent to the cone is true to scale. It is known as the standard parallel (sp.)

→ Distortions increase as parallels move away from the standard parallel.

→ Conic projection cannot be used to show worldwide distributions

→ Linear features, such as transcontinental railways and forests, for example, coniferous forest regions, can be depicted accurately.

III. Zenithal Projection

- These types of projection sometimes called Azimuthal projection

- They are constructed by placing a plane tangent to the reference globe at a single point. In order to visualize the pole (south or north)the cartographer concerned the plane as he/she/ wishes.
- It refers to over head position that is perpendicular.
- True distances and direction in such projections are retained by placing a developable surface at one of the poles.

Figure 6.10: Zenithal project

Characteristics of Zenithal Azimuthal/ projections

→ Parallels are concentric circles.

→ Meridians are straight lines radiating from the poles.

→ Planar projections produce circular maps.

→ Planar projections are very much suited to map of polar land masses (high latitude regions).

→ Parallels and meridians intersect at right angles.

Illustrative Questions

18. One of the following may not explain best the characteristics of globe
    A. They are too small
    B. Globes are expensive to make

C. They do not show all the surface of the earth at a time
D. Easy to carry.

Explanation: One of the disadvantage of globe are not as convenient to use as a flat map is easy to carry.

Answer: D

19. Identify the correct combination
A. Azimuthal – planar
B. Homolagraphic – shape
C. Orthomorphic – Area
D. A and B

Explanation: The other name of developable surface of Azimuthal is planar.

Answer: A

20. Which of the following statements best represents the characteristics of conic projection.
A. All parallels are equal in length to the equator
B. Parallels make arcs of concentric circles
C. Distortions increase pole wards from the equator
D. Meridians are straight lines radiating from the poles

Explanation: The best characteristics of conic projection is parallels make arcs of concentric circles.

Answer: B

21. One of the following is a Zenithal perspective map projection.
A. Gnomonic projection C. General purpose projection
B. Orthomorphic projection D. equal – area projection

Explanation: The Azimuthal equal – area projection is commonly used for continents world maps interred on the North pole.

Answer: D

22. Describe the significance of map projection?

Explanation: The Main significance of map projection is to eliminate the draw backs of the globe.

6.4. Drawing Sketch Map

6.4.1. The Meaning and Purpose of Sketch Maps

- They are simplest types of maps drawn manually.
- They are free hand drawings.
- They are rough drawings prepared to show distribution or arrangements based on observations.

Characteristics of Sketch maps

→ It can be drawn quickly
→ It is a guideline
→ It is easily recognizable
→ It provides skills for roughly showing or recording landscapes
→ They are made by using maps, photographs or field observation.

- In the study of geography, sketch maps play the role of conventional maps and photographs but not exactly the same as the conventional maps and photographs.

Basic differences between sketch maps and conventional maps

Sketch	Conventional Maps
1. Are simple and hand made with less skilled required	Are not mainly drawn and require professional skill
2. Are not true to scale and accurate	Are true to scale and accurate
3. Much more selective and are often helpful to very smaller areas	Are also selective but comprise larger areas
4. Are the effect of human observation	Are beyond human's eye horizon
5. Are not expensive	Are expensive in relation to drawing sketch maps
6. Are quickest methods or not time taking	Are time as consuming

6.4.2. Basic Guidelines for Making Good sketch Maps

The following points help us to make a good and meaningful sketch maps

→ Decide the areal extent to be sketched and the most important elements to be included.
→ Establish mental map or set up things to be done in your mind.
→ Create the map's boundaries

→ Begin from background details and apply measurements to fox the elements to be sketched in their proper places.
→ Reduce the elements to be sketched in to simple lines or symbols in order to avoid clumsiness and congestion of objects.
→ Use a well sharpened pencil, ruler and soft rubber eraser.
→ Try to sketch out the very important elements or objects with neatness and clarity.
→ Annotate your sketch map i.e give it title, set the key, nothing and other essential marginal information.

6.4.3. Producing sketch Maps

Types of Sketch Maps

Like other maps, there are different kinds of sketch maps that are useful for the daily life of human beings. Among these

A. Location sketch Maps

- They are widely used in geography to show the location of things.
- They built up areas, vegetation, human – made features, etc.

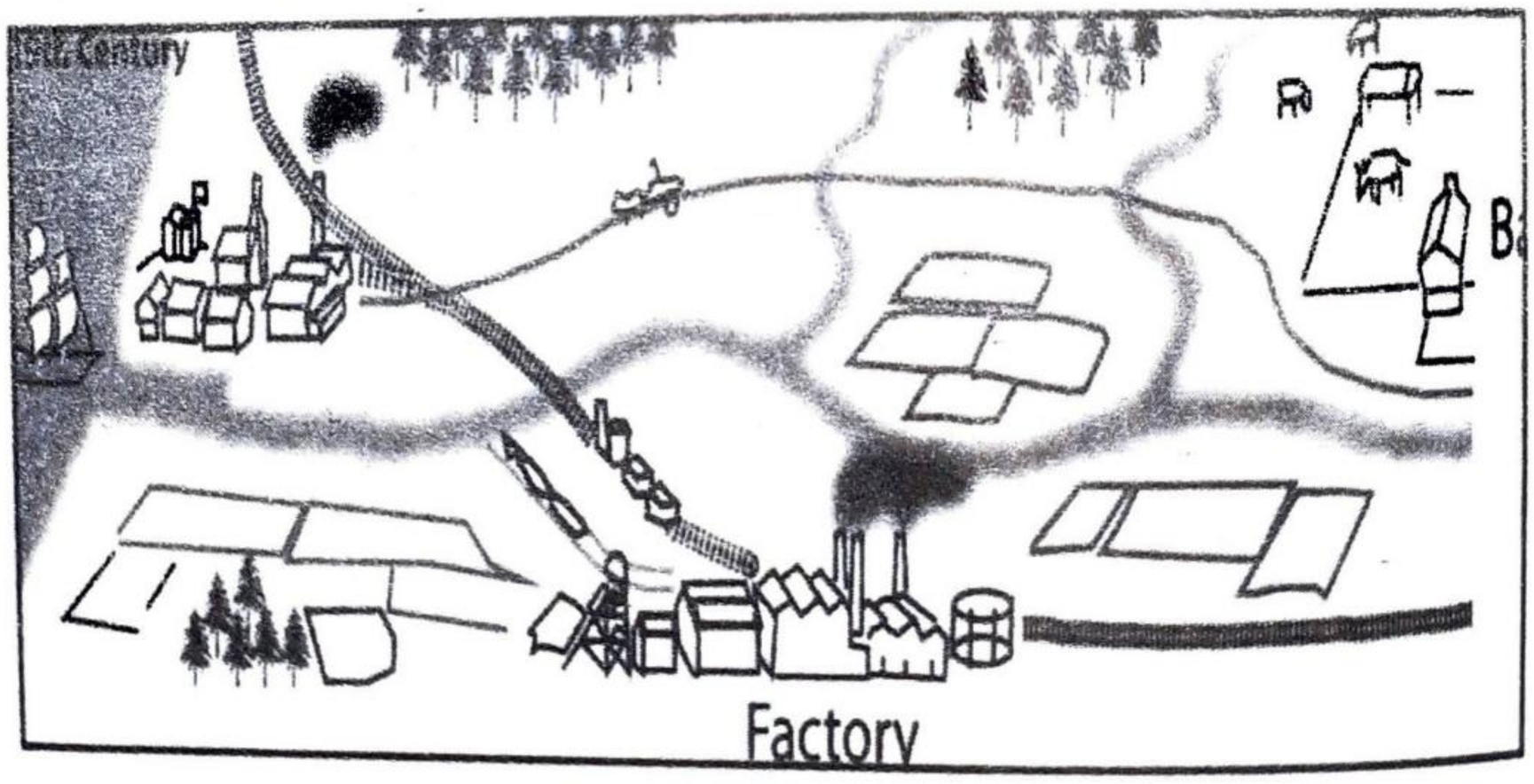


Figure 6.11 A village

B. Route Sketch Map

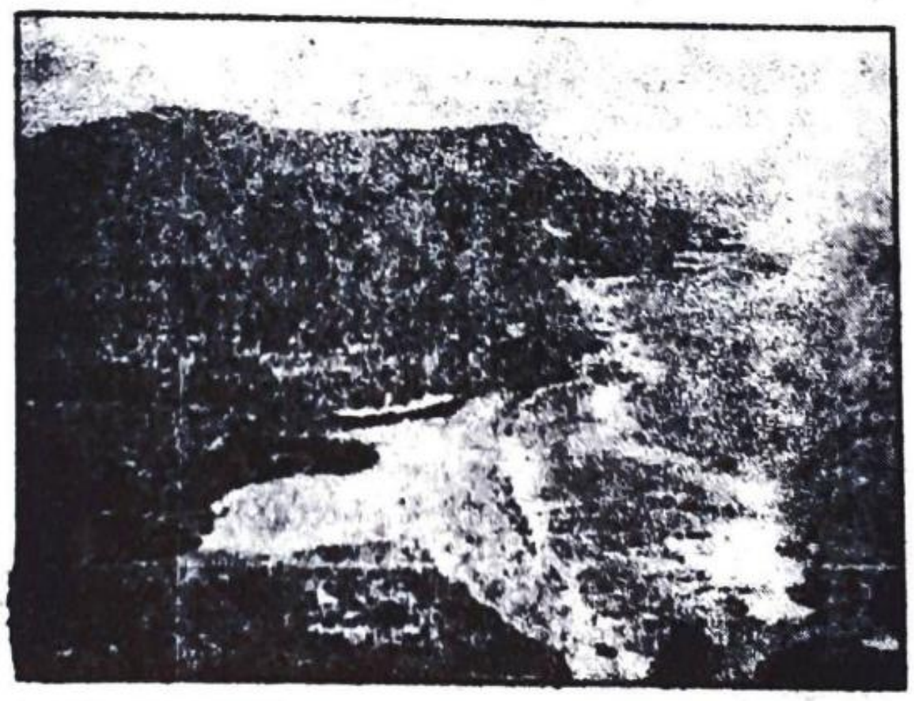
- They are essential and are frequently used naps in our everyday life.

- They show things like the way to specific place, such as home, school, slope, etc with in a village, town or city; the inter journeys (path ways) from one place to another, along high ways, rail ways, air ways, water ways, foot paths; traffic flows, taxi ways, city bus service lines, trade and caravan routes.

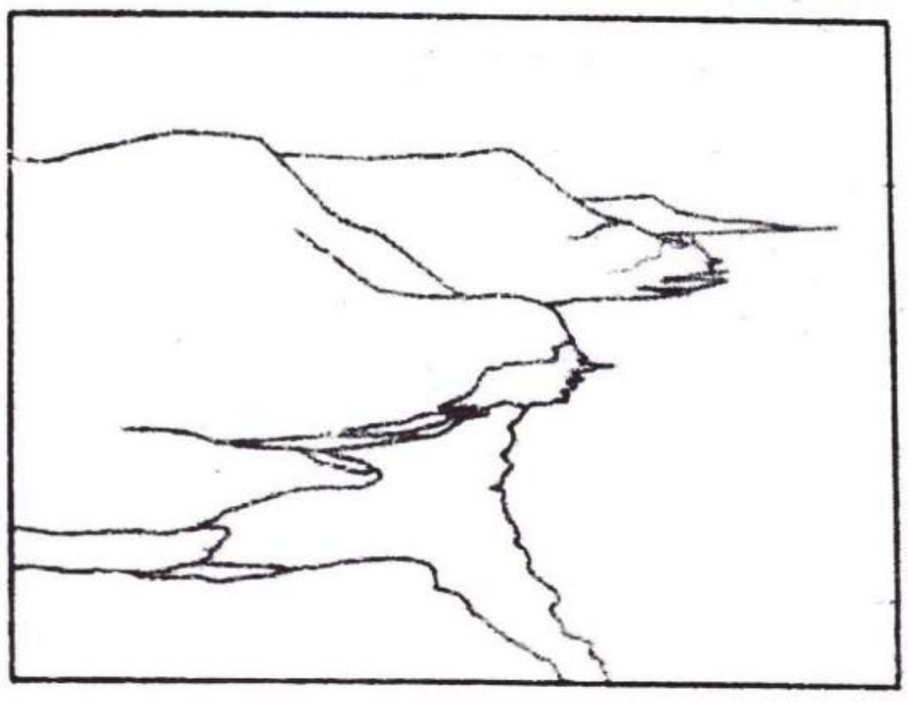
C. Landscape Sketch Maps

- They are field sketches used to show surrounding areas.
- They could be drawn from topographic maps, aerial photographs or from direct field observation.
- They are maps useful to represent land features such as drawing, land from, roads and farm lands.

Example:



a) *Aerial photograph*



b) *Sketch map*

Figure 6.12

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Illustrative Questions

23. One of the following stamens is true about sketch map?
    A. Sketch maps is not crude
    B. It have scale
    C. It can be drawn quickly
    D. It is hardly recognizable

Explanation: One of the characteristics of sketch map is drawn quickly

Answer: C

24. Which one is frequently used map in our everyday life?
    A. Route sketch map
    B. Location sketch map
    C. Field sketch
    D. outer space map

Explanation: Route sketch map are essential and frequently used maps in our everyday life.

Answer: A

25. Which steps first in the processes of sketch map drawing?
    A. Create mental picture
    B. Give it title
    C. Decide the areal extent
    D. Apply measurements

Explanation: In the processes of sketch map drawing decide the areal extent is the first and necessary step.

Answer: C

26. What is sketch map?

Explanation: Sketch maps are the simplest type of maps drawn manually with no true scale and also it is a free hand drawings.

27. List the distinguish feature of sketches and maps?

Explanation: Sketch maps are less skilled required, not true scales, not expensive etc where as conventional maps are require professional skill, true scale, time consuming etc.

28. Mention the disadvantages (draw backs) of sketch maps?

Explanation: The draw backs of sketch maps include it is crude, does not have scale and can be both misleading and clumsy.

29. The Process of plotting or locating boundary lines on maps or aerial photographs by cartographer is known as _____.
    A. Boundary allocation
    B. Boundary definition
    C. Boundary demarcation
    D. Boundary delimitation

Explanation:

Boundary definition ⇒ Boundary Delimitation ⇒ boundary allocation

⇒ Boundary Demarcation

Boundary allocation is the technical task of the cartographer in which plotting of boundary lines map or aerial photograph

Answer: A

30. _____ are lines on maps that show the distribution of equal transportation cost

A. Isohyets
B. Isodapanes
C. Isobar
D. Isotransport

Explanation:

Isoplettic maps are quantitative distribution maps that use lines. Isodapanemaps show equal – transport ation costs

Answer: B

31. The type of map projection which is particularly suited to the polar region is

A. Zenithal projection
B. Conic projection
C. rectangular project
D. cylindrical projection

Explanation:

Zenithal/ planar type of map projection use **Plane as developable surface** best suited to the polar region

Answer: A

32. The type of map projection which is particularly suited to the polar region is

C. Zenithal projection
D. Conic projection
C. rectangular project
D. cylindrical projection

Explanation:

A. Zenithal/ planar type of map projection use **Plane as developable surface** best suited to the polar region

Answer: A

33. ____ is a type of drainage pattern where rivers flow from surrounding high grounds towards a central basin

A. Triangular drainage pattern
B. Rectangular drainage pattern
C. Circular drainage pattern
D. Centripetal drainage pattern

Explanation:

Centripetal drainage patterns are found in areas, where rover flow **Drom surrounding high grounds** in to **one common cente like lake**

Answer: D

Review Questions on Unit Six

Part I. True or False questions

Write "A" if the statement is true and "B" if it is false

1. Graphys are used to show the relationship between two geographic variables in time and space.
2. Zenithal projection refers to overhead, that is perpendicular.
3. Conventional signs and symbols are the same in amount and kind.
4. Topographic maps are often known as topical map.
5. Cartographers developed map projection as they tried to solve the problems that globes presented.
6. Isopleths are lines of equal value or magnitude.
7. Choroschematic map are qualitative maps that use colour tint.
8. Location sketch map are used to show the relative location built up and human made areas.
9. Globes are not visually convenient.
10. Maps are a true representation of the earth surface.

Part II. Matching Question

Match items in column "A" with that of column "B"

A	B
11. Topographic maps	A. Topical
12. Thematic maps	B. Natural and Human made features
13. Chorochromatic maps	C. Picture
14. Choroschematic maps	D. Shaded maps
15. Choroplethic maps	E. Tourists, travelers
16. Isoplethic maps	F. Freehand drawing
17. Location sketch Map	G. Topo sheets
18. Route sketch map	H. Field sketches
19. Landscape sketch map	I. climatic elements
20. Sketch map	J. Symbol
	K. Pictures and letters
	L. Colour, dots, symbol

Part III. Completion questions

Fill in the blank space with appropriate words or phrases

21. The other name of Zenithal projection is _____
22. A surface that of the earth cannot be converted in to a flat surface without distortion is called _____
23. _____ is a technique that cartographers use to transfer information from a globe to a flat surface in order to create a map.
24. The simplest and most widely used graphys are _____ and _____
25. Isotherm maps are an example of ____ quantitative distribution map.
26. _____ are maps used to show the distribution of average values such as population density, cropping intensity, etc.
27. _____ maps show non – quantitative attributes.
28. Thematic maps are classified in to _____ and _____
29. The most common properties of map projection are _____, _____ ____ and _____.
30. _____ maps are maps that show both human made and natural features.

Answer Key on Unit Six

Part I.

1.A 2.A 3.B 4.B 5.A 6.A 7.B 8.A 9.A 10.B

Part II

11.G 12.A 13.L 14.K 15.D 16.I 17.B 18.E 19.H 20.F

Part III

21.Azimuthal
22. Undevelopable
23. Map projection
24. Line graphes and bar graphs
25. Isoplethic
26. Choroplethic map
27. Qualitative distribution
28. Qualitative and quantitative
29. Area, shape, distance and direction
30. Topographic

UNIT SEVEN

Physical Geography of Ethiopia and the Horn

7.1. LOCATION OF THE HORN OF AFRICA AND SIZES OF MEMBER COUNTRIES

7.1.1.. Concept of Location and Location of the Horn.

- The Horn of Africa, a part of the continent is a narrow tip that protrudes in to the northern Indian ocean, separating it (from the Gulf of Aden to its surroundings. This important term expressed in the form of.
- In spatial distribution every place has its own particular location in relation
  - → Absolute Location
  - → Relative Location

Absolute Location

- It is expressed as a geographical extent, in terms of latitudes and longitudes.
- It is also called Astronomical location.
- The absolute location of the Horn of Africa is bounded within, 18^0N – 1^0S latitudes and 33^0E – 51 24'E longitudes.
- As the result absolute location, countries of the Horn have tropical characters and lie in the GMT + 3 time zone.

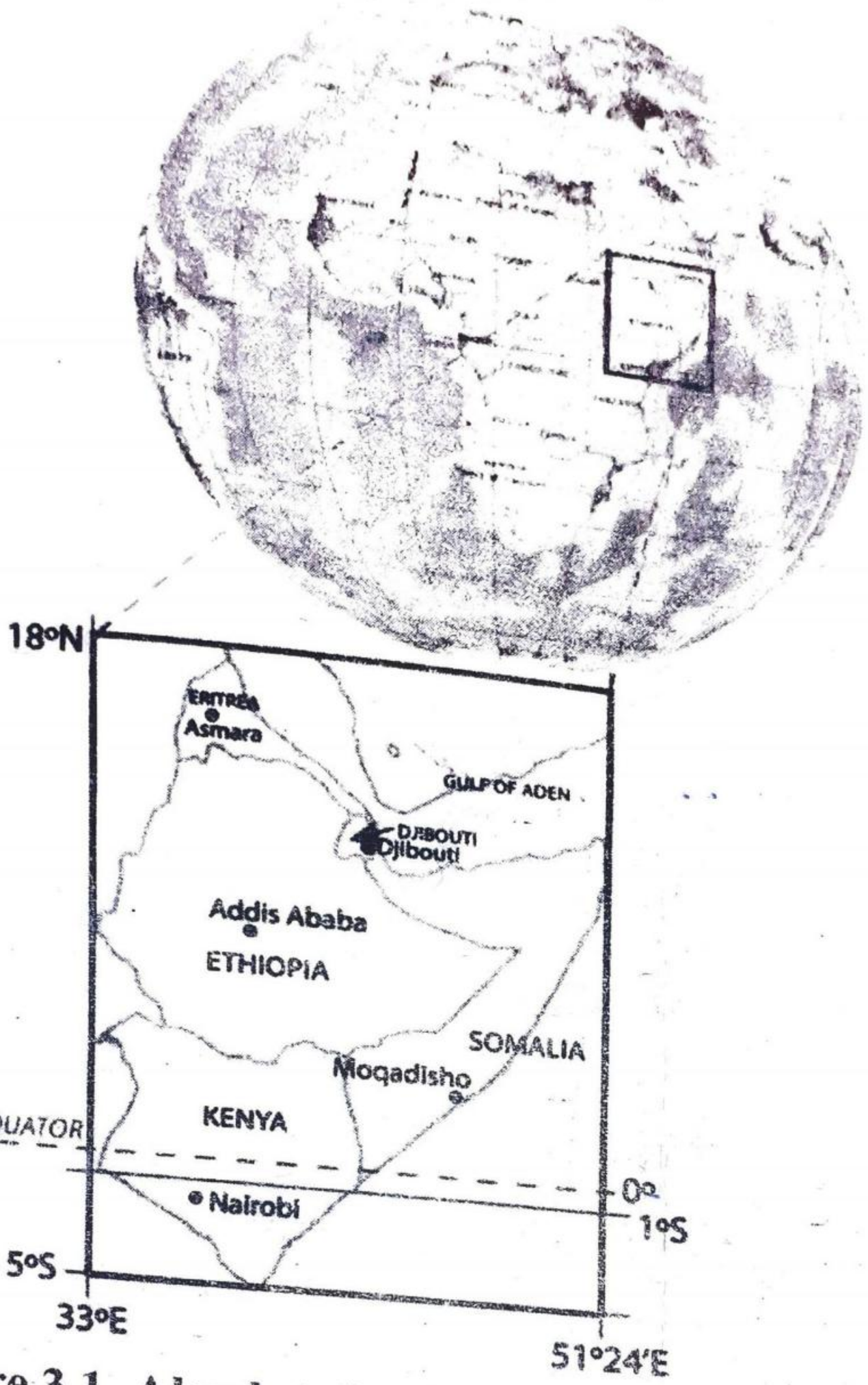

Figure 3.1. Absolute location of the Horn.

elative Location

- It refers to the position of a place in relation to the location of other geographic features.
- It can be described as a region or subregion bounded by
  - → The Indian ocean in the south east.
  - → The Red sea and Gulf of Aden in the north east,
  - → Sudan, in the north and west, and
  - → Kenya in the south
- It can be expressed in the following two ways; namely; vicinal location and geological location (natural).

- Vicinal location shows the location of a country in relation to its neighboring counties, while the geological location describes a country's location in reference to big land masses or water bodies

7.1.2. Sizes of Countries of the Horn

- Size refers to area coverage of a country
- The countries of the Horn vary in size. This variation has both advantages and disadvantages for individual countries
- In geography, relative size implies relative possibilities for possessing diversities in resources, culture, agro-ecological zones.
- The Horn of Africa consists of four countries. Djibouti, Ethiopia, Eretria and Somalia.
- It covers an area of 1.88 million square kilometers.
- Iterms of size comparatively speaking Ethiopia is the biggest, Djibouti is the smallest and Eretria and Somalia are less than Ethiopia by almost ten times and twice respectively.

Table 7.1. Comparison of size among countries of the Horn of Africa

Countries of the Horn of Africa	Total Area (in km^2)
Djibouti	23,200
Eritrea	118,000
Ethiopia	1,106,000
Somalia	637,661
Total	1,884861

Illustrative Questions

1. Explanation the difference between vicinal and geological location

Explanation: Vicinal location indicate the location of a country in relation to its neighboring countries, while the geographical location describes a county's location in reference to big land masses or water bodies. .

2. What do you think the disadvantage of small area size?

Explanation: Small area size implies small possibilities for possessing diversities in resources, culture, agro ecological zone.

3. Where is the Horn of Africa generally found?

Explanation: The Horn of Africa, a part of the continent, is a narrow tip that protrudes in to the northern Indian ocean, separating it from the gulf of Aden.

4. What is the implication of absolute location of the horn of Africa?

Explanation: As a result of absolute location, countries of the Horn have tropical characters and lie in the GMT + 3 time zone.

5. List member countries of the horn of Africa and write the biggest and smallest countries in terms of size.

Explanation: The Horn of Africa consists of four countries Djibouti, Ethiopia, Eritrea and Somalia. In terms of size Ethiopia is the biggest while Djibouti is the smallest.

7.2. Location, Size and Shape of Ethiopia

7.2.1. Location of Ethiopia

- It is the identification of a place, a country or a region in the world.
- It can be expressed in two ways relative location and absolute location.

Relative Location of Ethiopia

Relative location can be expressed in vicinal and geological terms.

I. Relative (Vicinal) location of Ethiopia

→ Ethiopia is a land locked country that is surrounded by five neighboring countries.

→ Each country shares different lengths of Ethiopia's borderlines.

→ The total length of Ethiopia's boundary line is 5260km.

→ Sudan Shares the longest length of boundary line (1750km), followed by Somalia (1600km).

→ The Republic of Djibouti shares the smallest boundary line length (1600km)

S/N	Bordering countries	Shared boundary line length in kms
1	Djibouti	310
2	Eritrea	840
3	Kenya	760
4	Somalia	1600
5	Sudan	1750
	Total	5260

Table 7.2. Ethiopia's boundary line length, as shared with neighboring counties

II. Strategic (Global, Geological) Relative Location of Ethiopia

It is found.

→ To the southwest of the Asian continent
→ To the south Europe,
→ To the northwest of the Indian ocean,
→ In the Nile Basin, and
→ In north eastern Africa.

Absolute (Astronomical) Location of Ethiopia

- The astronomical location of Ethiopia lies between 3^0N – 15^0N latitude and 33^0E – 48^0E longitude
- The extreme points of the country lie at

→ Badime me in the north (Tigray) – about 15^0N
→ Moyalle in the south (Borena) – at $3^0$30'N
→ Akobo in the west (Gambella) – at 33^0E.
→ The tip of Ogaden in the east (Ogaden) – at 48^0E.

Significance of Ethiopia's Location

Ethiopia's relative location – especially geologic location affects or influences its.

Climate: Although altitude plays a significant role in modifying the temperature of Ethiopia, its relative location in the tropical zone and its proximity to the two big oceans, (the Atlantic and India oceans) have influenced her to experience.

→ High temperature in most of the year
→ Relatively lesser annual range of temperatures
→ Heavy summer rains and moderate spring rains

Social-cultural aspect: Ethiopia's proximity to the middle east and the Mediterranean Europe has influenced her to have diversified religions, cultures, languages, ethnic group, etc.

Geopolitics: the poltical history of Ethiopia has been considerably influenced by its location around the Red Sea (suezcanal).

Time Zone: because of its longitudinal location ($33^0E - 48^0E$), Ethiopia belongs to GMT + 3 time zone. That means, the local time in Ethiopia is 3 hours a head of the Green which standard time.

7.2.2. Size of Ethiopia

- Ethiopia is the tenth largest country in Africa, with a total area of 1,106,000 square kilometers.
- It contain about 0.7 percent of the world's land area and about 3.6 percent of Africa's landmass.
- Ethiopia is essentially a country of a fairy larger areal size.

Table 7.3. Comparison of Ethiopia's areal size with its Neighbors

Country	Total area extent (km^2)	Rank	Size in comparison of Ethiopia
Ethiopia	1,106,000	2	-
Djibouti	23,200	6	Almost 50 times smaller
Eritrea	118,000	5	9.42 times smaller
Kenya	580,00	4	1.91 times smaller
Somalia	637,661	3	1.73 times smaller
Sudan	2,505,813	1	2.3 times bitter

Advantages Ethiopia's large size lets it

- Possess diverse agro-ecological zones, resulting in a wide variety of fauna and flora.
- Possess a large amount of arable land
- Have a great variety of mineral resources.
- Be the home of diverse ethnic groups
- Requires expensive defense and administrative expenditures
- Faces expensive expenditures for the establishment of infrastructure.

- Has the opportunity for greater defense in depth from temporary attack, i.e. deeper penetration is necessary from the periphery to the central area.

Disadvantages Its large size compels Ethiopia to

- Require great financial power to construct infrastructural facilities,
- Have a large army to protect its sovereignty
- Preserve an efficient and popular government to administer its vast territory.

7.2.3. Shape of Ethiopia

- Countries vary not only in location and size but also in shape.
- Some countries have nearly circular (compact) shapes, others have elongated (linear) shapes, and still others have truncated (shortened) shapes.
- These shapes have implications on the administrative, defense and economic integration within the country.
- Ethiopia has more or less compact (circular) shape.
- Compact shape has militaristic advantage. This can be well explained by considering border line and area ratio.

$$\frac{B(Borderline)}{A(Area)} = \frac{5260}{1106{,}000} = 1 = 210$$

- There are three theoretical indicators of the compactness of an area.
  - → The boundary – circumference ratio (B/C)
  - → The area – boundary ratio (A/B)
  - → The actual area – area of the inscribing circle (A/A')
- Index of compactness explained by.

$$\frac{\text{Boundaru length}}{\text{circumference of the inscribing circle}}$$

Illustrative Questions

6. Which of the following is correct about the location of Ethiopia?
   A. The country shares the longest border with the Sudan
   B. The northern point lies at 18 degrees North of the equator

C. The country shares longer border with Kenya than with Eritrea
D. The southern point of the country lies at about five degrees south of the equator

Explanation: Compared with other neighboring counties Ethiopia has longest boundary with Sudan more than 1000 kms.

Answer: A

7. Which one of the following is most likely outcome of the location of a country?
   A. Its geographic size
   B. Its geopolitical character
   C. Its geographic shape
   D. A and C correct answers

Explanation: The geopolitical character of an area related to the location of a country.

Answer: B

8. The location of Ethiopia between 33^0E and 45^0E has a marked influence mainly on
   A. Time zone the country finds herself in
   B. Level of economic development
   C. Ethnic diversity
   D. Climatic diversity

Explanation: The longitudinal location of Ethiopia (33^0E – 48^0E), influence mainly on 3 hours a head of the green which standard time (EHEECE, 2002/2010)

Answer: A

9. Which of the following statements about Ethiopia is NOT correct?
   A. The E → W distance of Ethiopia is shorter than its N→S extent.
   B. The geographic centre of Ethiopia is not very far from Addis Ababa.
   C. Ethiopia has a surface area of more than I million square kilometers.
   D. Ethiopia is relatively more compact in shape than most of its neighbors.

Explanation: The geographic centre of Ethiopia is not very far from Addis Ababa, the surface area of north than one million square kilometers and its is relatively more compact in

shape than most of its neighbors while the E→W distance of Ethiopia is larger than its N→S extent.

Answer: A

10. The astronomical location of a country is expressed (EHEECE 2003/2011)
    A. Using lines of latitude and longitude
    B. In relation to the location of water bodies
    C. With reference to the location of other landmasses
    D. In relation to the location of other countries

Explanation: The astronomical location of the country is expressed using lines of latitude and longitude.

Answer: A

11. Which one of the following is NOT strongly linked to the large size of Ethiopia? (EHEECE, 2003/2011)
    A. It underdeveloped economy
    B. Its large arable land
    C. Its great ethnic and cultural diversity
    D. Its climatic diversity

Explanation: The size of Ethiopia linked with its arable land, ethnic and cultural diversity and climatic diversity

Answer: A

12. Which of the following is NOT directly associated with the large territorial size of Ethiopia? (EUEE – 2005/2013)
    A. Its diverse climates
    B. Its large population size
    C. Its great variety of natural resources
    D. Its high rates of natural increase

Explanation: All are advantages of Ethiopia territorial size except natural increase. High rates of natural increases related to population.

Answer: D

7.3. Geological Structure and Relif of the Horn of Africa

7.3.1. Geological History of the Horn of Africa.

The geological history of the Horn is not something extra ordinary. It is part of the geological processes that acted within and upon the earth's surface for millions of years.

The geological history of the Horn shows four major geological eras.

- Each era is divided in two periods, and each period is subdivided in to epochs.
- Each geological era is distinguished from the other's, based on
  - The relative positions of the continents
  - The character of the prevailing climate
  - The predominant life form

The Precambrian Era (from 4.5 Billion to 600 Million years ago)

- It is the oldest and longest geological era, covering about 5/6 of the earth's geological time.
- Frequent organic movements
- Intensive volcanic activities
- Denudation during the later periods
- Formation of folded mountain ranges in a NNE – SSW direction
- The first forms of life emerged. such as amoeba, and jellyfish.
- The oldest rock formed – the old crystalline basement. This rock underlies all other rocks.
- In some parts of Ethiopia, the basement complex rocks are exposed to the surface due to severe river erosion and lack of over lain recent rock formation.
- The Precambrian racks are exposed in the
  - → Northern Ethiopia - in northern and central Tigray
  - → Southern Ethiopia – in Abbay and Baro river valleys, Asosa and methekel.
  - → Eastern Ethiopia - in wabishebelle river valley and south of. Harar town.
  - → In central. Western and Northern Eritrea.

The Paleozoic Era (from 600 Million – 250 Million years ago)

- It was an era of more denudation and peneplanation.
- It is known for the predominance of invertebrates.
- It is the era with very little rock formation in Ethiopia
- There was no organdy volcanism in the Horn of Africa

Mesozoic Era (from 250 Million – 70 Million years ago) an Era of Reptiles.

- The major geological events in this era in the Horn were the sinking and uplift of the landmass. In geological terms, this processes is called Epeirogenesis.
- In Ethiopia, the land was generally titled east wards, hence lower in the south east and higher in the north west.
- It is also known for the predominance of reptiles. Huge reptiles, such as dinosaurs, were dominant.
- The three periods of the Mesozoic era experienced Epeirogenesis in one form or another.

Triassic Period (250m)

- Land subsidence (sinking) began in the south eastern part of Ethiopia and progressed towards the North western part of the country due to internal forces.
- Sinking was followed by transgression of a nearby sea into the mainland of today's Somalia, and South eastern Ethiopia.
- Transgression of the sea resulted in the deposition (sedimentation) of the first sedimentary rock in Ethiopia which is now called Adigrat (lower) sandstone.
- The Adigrat sandstone is younger in the north west and older in the south east.

Jurassic period (142m)

- This period manifested the continued sinking of the land and deepening of the sea in Ethiopia.
- This event deposited another sedimentary rock known as Hintalo limestone.
- The Hintalo limestone in the north west is older while it is younger in the south east.

Cretaceous period (70m)

- During this period great up lifting began.
- The repression was followed by the deposition of the third layer, called upper sandstone.
- Upper sandstone is the youngest sedimentary rock, and therefore overlies the rest.

- The upper sandstone is thicker and older in the northwest and younger and thinner in the south east.

The Cenozoic Era (from 70 Million to Recent years)

- It is the most recent geological era.
- Most of the existing land forms of Ethiopia were formed. Many of these events occurred during the tertiary period.
- The Cenozoic era has two periods, namely tertiary and quaternary period

Geologic Events of the Tertiary period – (70 million – 2 million)

- The uplifting that began in the cretaceous period of the Mesozoic Era continued and reached its maximum height.
- In Ethiopia and the Horn it formed huge blocks of dome over greater part of the region. As the uplifting continued through time, great cracks opened in the crust and resulted in the pouring out of extensive basaltic lava (known as the Trappean lava settles.) the lava resulted in the formation of.
  - → The Afar, Horst that extends into Djibouti
  - → The active volcano of Ertalle in Afar
  - → The dormant volcanic mountain of Fentalle in Eastern Oromia.
  - → The extensive lava field of Methara.
  - → Lava ridge.
  - → Thermal springs and fumaroles (in the rift valley)
- The occurrence of marked climatic variations in the quaternary period resulted in quaternary sediments or recent deposits.

  These climatic variations or changes included
  - Heavy rains often known as 'pluvial' rains occurred.
  - Lakes and marshy areas became numerous and deep.
  - The excessive run off increased the volume of river which had high erosion power. Thus, the river carried a lot of sediments which were deposited on the lower areas, marshy areas, lake shores and beds of river basin,
- The climate changed in to warm and dry conditions during the inter pluvial and post pluvial periods.

- Pluvial period is characterized by different types of depositions, namely:
  - → Lacustrine deposits, deposits on the former lake beds and swamps
  - → Fluvial deposits, deposited on the banks of rivers, flood plains etc,
  - → Aeolian deposits: wind deposits on low lying areas
  - → Glacial deposits: ice deposits
  - → Marine deposits: deposits on once sea invaded area
- Generally, the Cenozoic era in its quaternary, period is assumed to be the period in which modern man evolved.

A simplified 'Rock' profile of Ethiopia

Quaternary (Recent) lava deposited	Ingeous Rock → Cenozoic era
Tertiary lava deposited	
Upper sandstone	Old marine sedimentary Rock →Mesopoic Era
Hintailo limestone	
Adigrat sandstone	
Old basement complex, (crystalline basement)	Metamorphic Rock → Precambrian era

Illustrative Examples

13. Transgression of the sea form the south eastern parts of Ethiopia and the deposition of shales and gypsum characterized the (EHEECE 2001/2009)
    A. Pleistocene epoch
    B. Precambrian era
    C. Triassic epoch
    D. Pliocene epoch

Explanation: Transgression of the sea from the south eastern parts of Ethiopia and deposition of shales and gypsum is the characteristics epoch

Answer: C

14. In which part of Ethiopia do you expect to find the thickest layer of the Adigrat sandstone? (EHEECE, 2002 2010)

A. In the western most areas of the Tigray National regional state
B. In the eastern most parts of the Somali national regional state
C. In the northern most parts of the Afar National, Regional state
D. In the western most part of the Gambela national regional state.

xplanation: As its name indicate the Adigrat sandstone located in the north west part of Ethiopia in the Tigray National state.

Answer: A

5. The Hintalo lime stone of Ethiopia was deposited during the _____ period (EHEECE 2002/2010)

A. Tertiary
B. Jurassic
C. Triassic
D. quaternary

xplanation: The Hintalo limestone of Ethiopia was deposited during the Jurassic period of the Mesozoic era.

Answer: B

6. In which part of Ethiopia is the upper sandstone the youngest? (UEE 2004/2012)

A. In the Abay valley
B. In the south eastern Lowlands
C. In the North western Highlands
D. In the south western lowlands

Explanation: The upper sandstone is thicker and older in the North west and younger and thinner in the south east.

Answer: C

7. In which parts of Ethiopia is quaternary lava most extensively found? (UEE 2004/2012)

A. In the Baro river valley
B. In the Ethiopia rift valley
C. In the wabi shebelle river valley
D. In the North Western lowlands

Explanation: Emission of Aden series lava also took place in the rift valley and south of lake Tana.

Answer: B

18. To which geologic area in Ethiopia do dinosaurs belong?

A. Early Cenozoic era
C. Precambrian era

B. End of Mesozoic era D. Early Mesozoic era

Explanation: The major geological event in the Mesozoic era was the appearance and disappearance of the dinosaurs.

Answer: D

19. Where do you find Quaternary extensive lava field in Ethiopia? (UEE 2004/2012)
    A. Ogaden C. elkere area
    B. Methara D. Borena area

Explanation: Quaternary extensive lava field in Ethiopia found in methara.

Answer: B

20. The pluvial period in Ethiopia has resulted in the (EUEE 2005/2013)
    A. Formation of potash C. emergence of crater lakes
    B. Lacustruine deposits D. formation of conical hills

Explanation: The pluvial period in Ethiopia has resulted in the lacustrine deposits.

Answer: B

21. The transgression of sea water from south – east to north Ethiopia took place during _____ era. (EUEE 2005/2013)
    A. Mesozoic C. Paleozoic
    B. Precambrian D. Cenozoic

Explanation: The transgression of sea water from south – east to north – west Ethiopia took place during Mesozoic era.

Answer: A

22. In which of the following parts of Ethiopia is the basement complex exposed? (EUEE 2005/2013)
    A. The Baro river valley
    B. The upper Awash river valley
    C. The upper Ghibe river valley
    D. The lower Wabe Shebelle River valley

Explanation: The basement complex rocks are exposed on the surface at different locations among these western Ethiopia in Abbay and Bar river valleys Asosa and Metekel.

Answer: A

7.3.2. Landforms of Ethiopia and the Horn

- The existing landforms of Ethiopia and the Horn are the work of endogentic force that from inside the earth and exogenic force that originates at the surface.

- The land forms of Ethiopia are largely the result of the Cenozoic era tectonic and volcanic activities.
- The land forms of Ethiopia and the Horn are made up of river – cut gorges, valleys, plateaus, mountains and rolling plains.
- Altitude varies form, about 116 meters below sea level at the Dallol depression (kobarsink) to 4620 meters above sea level at Ras Dashen in Semin.
- Based on the relief formation Ethiopia is described as the Roof of East Africa or water Tower of East Africa.
- 56 percent of the country is occupied by highlands and 44 percent by lowlands.
- 1000meters altitude is taken as a demarcating contour line between highlands and lowlands.
- The relief of Ethiopia and the Horn can be divided in to three main physiographic division.
  - → Highlands
  - → The rift valley
  - → The lowlands

The Highlands of the Horn

- Highlands are lands with altitudes of over 1000 meters above sea level (masl).

The Horn's highlands are

1. The North western Highlands
2. The south eastern highlands

1. The North Western Highlands of the Horn

- It stretch from Ras Kassar in Eritrea to the highlands of Gamo Goffa in South western Ethiopia.
- They are separated from the south eastern highlands by the rift valley, which is part of the Great East African rift Valley.
- It is the biggest physiographic region. It makes up 44% of the area of the country.
- They are the sources of large number of rivers and streams
- They consists of
  - → The plateau of Eritrea

→ The plateau of Tigray
→ The north central Massifs
→ The plateau of shewa
→ The south western Highlands

The plateau of Eritrea

- It is located between the course of the Barka River and the coastal plain of Eritrea.
- It is bisected by the upper basin of the Mereb River.
- The plateau is capped by basaltic rocks.

The plateau of Tigray

- It is the most northerly group of highlands bounded in the north by the Mereb river and in the south by the Tekkezze river
- It is separated from the Eritrean plateau by the Mereb River.
- It is the most physically degraded part of the Ethiopian highlands due mainly to long history of settlement and growing population pressure.
- The average altitude is about 2000m.
- Known mountain peaks on this subdivision include:
  - → Mount Tsibet – 3988M.a.s. l
  - → Mount Ambalage – 3291 m.a.s. l
  - → Mount Assimba – 3248 m.a.s. l
  - → Mount Sedie – 3988 m.a.s. l
  - → Mount Alequa – 3391 m.a.s. l

The North Central Massifs

- They are found between the Abay Gorge in the south and Tekkezze Gorge in the north
- They are the most rugged and dissected plateaus of Ethiopia.
- They are capped by basaltic and surrounded by deep gorges with in each of the plateaus are small arable lands known as ambas. The ambas are isolated from one another by gorges.
- This massif contains the highlands of Wello, Gonder, and Gojjam.
- Rivers Abay and Tekkezze along with their tributaries have formed a number of gorges and steep sided river valleys.

- They are known for the production of cereals, such as teff, pulses and oil seeds.
- Tekezze River has divided the North central massif area in to western and eastern massifs, which are connected by the yeggu– Wadla – dilanta plateau.

e western group of highlands

- It includes higher areas in Gondar and Gojjam.
- Sumits like Mt. Ras Dejin (4620m) in the semen mountains and Mt. Guna (4231m) the Guna massif are significant heights in the Gondar.
- The Gojjam highlands are located to the south and contain Mt. Berhan (4100m) in the choke mountains is the highest.
- These massifs form watershed between the drainage basins of lake Tana, Abbay and Tekeze/ At bara.

ıe eastern group of highlands

- It make up the Lasta and Wollo massifs. South of there is found the Gojjam Massif.
- Highest peaks in the eastern highlands include Mt. kolo (4300m), Mt Lalibella (4276m) and Mt. Abune Yoseph (4190m).
- Mount kolo in the Lasta Massif is the highest peak.

he plateau of shewa

- It is a dome – shaped plateau that serves as a watershed between the Awash and Abbay River basins.
- They lie south of the Abbay trough which is greatly eroded due to torrential rain that pours down on in the area for almost all of the year.
- It is the wettest region of the country with a totl average annual rainfall of above 1500mm.
- The most important rivers include Baro, Akkobo, Ghibe, Didessa and Dabus.
- The South western highlands are the most physically dissected uplands in Ethiopia which include.
  - → Gamo Konso Highlands (in Gamo Goffa)
  - → Gamo – Konma Highlands (in Kefa zone)
  - → Kulo – Konta Highlands (Kefa zone)

→ The highlands of Jimma
→ Benishangul mountain (in Benshangul Gumuz)
→ The shewa – Ghimira and Guraferda mountain
→ Tullu wallel (in west Wellega)

- Except Mt. Ghughe (4200 mas l), the highest peak in the subdivision, the south western highlands are lower than 2500 meters above sea level.
- The region from economic point of view, is the most dominant coffee producing area of Ethiopia and has relatively higher percentage of forest cover.
- In socio – economic terms, it is the most ethnically linguistically diverse region in Ethiopia.

The surrounding lowlands (Western lowlands)

- It extend from north to south bordering the Ethio – Eritrean and the Ethio Sudanese boarders with a general elevation of 500 - 1000 meters above sea level
- They are characterized by arid or Semi arid condition.
- These lowlands are sub – divided into smaller lowlands as follows
  → The Tekezze – Anghereb lowlands
  → The Abbay – Dinder lowlands
  → The baro – Akobo lowlands
  → The omo – Gibe lowlands
- The Tekezze – Anghereb lowlands are the most northerly lowlands, drained by Tekeze and Anghereb.
- The Abay – Dinder lowlands extend from south of Metema town – up to the southern part of the Abay river.
- The Baro – Akobo lowlandsm are found in the western part of Gambella. These lowlands are drained by rivers Baro and Akobo.
- In most parts of the western lowlands, communities practice nomadic and semi – nomadic way of life.
- The Baro – Akobo lowland is the wettest lowland in Ethiopia. That is why the rivers draining the region (particularly the Baro river) have almost a regular flow of water for much of the year.

- The main agricultural products are sesame, maize, sorghum, (Zengada and Mashilla and Millet (dagussa).
- The important towns on the western lowlands are Hummera, Metema, Omedla, Asosa and Gambella.

The South eastern Highlands of the Horn

- It is located to the south east of the rift valley. The region being the second largest covers 37% of the country's area.
- It is bounded by the eastern escarpment of the western highlands in the west and north the Ethio – Somalia and Ethio – Kenya borders in the north east, east and south east.
- They are bounded.
  - → In the west, by the fault line of the Rift valley
  - → In the east, by the Ogaden lowlands
  - → In the south by the Elkerie and Borena Lowlands
- They are capped by basaltic rock.
- They are the main sources of the wabishebelle and Genalle rivers.
- The south western Highlands of the Horn include the plateaus of
  - → Hararghe
  - → Sidama
  - → Arsi
  - Bale
  - Somalia highlands

The plateau of Hararghe

- It rises sharply from the fault line of the Rift valley and extends gently to the east up to Jijiga.
- It is also drained by the left bank tributaries of wabishebelle river.
- The basaltic rocks have been worn away, exposing limestone and earlier sedimentary rocks.
- It is significant in producing coffee, chat, sorghum and millet.
- The prominent mountain peaks include Mount Gara Muleta (3381 masl) and Mount Jebel Tita (3122masl).

The plateau of Arsi

- They are generally made up of flat rolling up lands, dissected mountain ranges and graded extent volcanoes.

- It consists of the Gugu and chillalo Massifs
- These featurs are due to the fact that erosion on the Arsi plateau has been comparatively low.
- It is known for its wheat.
- The highest summits in the Arsi highlands include Mount Chillalo (4136 masl), Mount Bada (413 masl), Mount kaka (4180 mas l) and Mount kara (4340 masl).

The Bale Massif

- It is separated from the Arsi highlands by the head streams of the Wabishebelle River (popularly known as the wabe).
- It consists of a plat form like basaltic plateau in the north central part and huge mountain massifs to the south.
- It is known for its barely.
- They are significantly dominated by the highest peaks are Mount – Tulu Dimtu (4377 masl) and Mount Batu (4307 masl).

The plateau of sidama

- The area constitutes the south west extension of the south eastern highlands.
- They are separated from the Bale heights by the Genale river.
- The plateau slopes away gently to the south and is drained by the Dawa river and its tributaries.
- The Jemjem is the dominant part of the plateau/ they are surrounded by the massive Borena lowland. These plateaus are known for coffee production and gold mines of the Kibre Mengest – Shakiso areas for long.
- The newly discovered Legedembi site is also another addition of the mineral wealth of the region.
- The highest mountain peaks are Dello mt. (3600m) and Guramba mt. (3400m).

The Somalia Highlands

- The highlands are the extension of the south eastern highlands of Ethiopia
- The average altitude doesn't exceed 1500 masl.

- They rise gently in the West and decend sharply to the Indian ocean coasts.

The south eastern lowlands

- These lowlands extend from the foot hills of the south eastern highlands along the Ethio – Somalia and Ethio – Kenya borders.
- They begin in Djiboutic and run all the way to Somalia in the south east, bordering the Indian ocean.
- They are also called the shebelle – Ghenale plains.
- These lowlands are highly extensive lowlands.
- They are characterized by low annual rainfall – often below 500mm and their general elevation lie between 500 and 1000 meters above sea level.
- It covers an area of 247,000 km^2, they make 54% of the south eastern highlands and lowlands and nearly one fifth of that of the country.
- They are covered by sandstones and recent marine deposits.
- The people practice a pastoralist way of life because of the harsh climate.
- They have high potential for cattle breeding and for the exploitation of fossil fuel energy resources.
- They consists of
  - → The Red sea coastal plains (in Eritrea)
  - → The Afar, plains (in Afar) which are included in the Afar Triangle.
  - → The Ogaden plains (in Somalia region)
  - → The Elkeri plains (in Bale)
  - → The Borena plains (in Oromiya)
  - → The Benadir plains (in Somalia)
  - → Djibouti

The Rift Valley System

- The Ethiopian Rift Valley system is part of the Horn's Rift Valley system, which is part of the Great East African Rift system.

- It is a set of fractures in the earth's crust that extend from the Dead sea in the north, through the Red sea, and then across East and central Africa to Mozambique in the south.
- It formed during the tertiary period and also it extends from the Ethio – Eritrea border in the north east up to Lake Turkana in the south along the Ethiopia – Kenya border.
- It is the most unstable physiographic division of an area.
- The floor is covered with active volcanic activity, fluvial and lucastrine deposits.
- It is a tectonically formed structural depression which divides Ethiopian highlands in to two broad groups.
- It covers a total length of 1700 kilometers in Ethiopia and Eritrea.
- The Afar triangle is becoming increasingly significant for its great potential in the extraction of salt, geothermal energy and irrigable agriculture.

Subdivisions of the Ethiopian rift Valley system

Because of internal variation, the rift valley is further sub – divided in to three physiographic regions as follows

→ The Afar Triangle (northern)

→ The lakes 'Region (the main Ethiopia, Rift or central) and

→ The Chew – Bahir Rift (southern)

The Afar Triangle (Northern)

- It is the largest and widest part of the system extending 200 – 300 kilometers
- It is also known as the Dankil plain or depression.
- Its altitude is generally low, ranging from 116 meters below sea level at the Kobarsink to about 900 meters above sea level at Awash.
- It is characterized by faulted depressions (the Dallol Depression) and grabens (also called troughs) and volcanic mountains.
- Alarge part of the area is covered by extensive salt plains and lakes (lake. Assale (116m b.s.l) and Lake Afrera (80m. b.s. l).

- The floor in many places in dotted by cinder cones and volcanic mountain the highest ones includes Mt. Fentalle, Mt. Bosseti and Mt. Aleltu.
- Its southern part is drained by the Awash river.
- It is inhabited by nomadic population.
- The area has economic advantages which include.
  - → Large seale extraction of salt
  - → Geothermal energy and
  - → The possibility of irrigation based agriculture for growing cotton and other tropical crops.
- It consists of fossil rich sediments and rich archeological sites which have shown us that the, area was the home of the ancient ancestors of primates and hominids.

'he main Ethiopian Rift (Lake Region or central part)

- It is found in the middle part of the rift Valley.
- It covers the area from the lower awash basin up to Lake Chamo.
- It is characterized by the following conditions it is
  - → The most elevated part of the rift valley region
  - → The wettest (most relate[redacted]section of the rift valley region
  - → The most densely vegetated and populated part of the region.
  - → The narrowest part of the Rift Valley.
  - → Sedentary farming is practiced.
  - → Endowed with numerous lakes whose economic and aesthetic values are enormous.

The Chew – Bahir Rift (Southern Subdivision)

- It is also known as the Omo – Ghibe trough.
- It is the smallest section of the Ethiopian Rift separated from the lake's region to the north by the konso highlands and its surroundings
- It consists of an extensive shallow marshy area covered by tall grasses.
- The characteristics feature of the area is the chew Bahir which was formerly called lake stephani.

- Rivers sagan and Woito flow in to lake chew Bahir.
- It is occupied by nomadic population
- The region is moderately hot and dry.

Illustrative Questions

23. Compared to the lowlands, the highlands of Ethiopia are characterized by (EHEECE 2001/2009)
    A. Hotter climatic conditions
    B. Poorer biotic life
    C. Denser human population
    D. Harsher ecological conditions

Explanation: Because of favorable climatic condition the highlands of Ethiopia denser human population than lowlands.

Answer: C

24. Most parts of the central highlands of Ethiopia are covered with (EHEECE 2001/2009)
    A. Trappean lava
    B. Marine sediments
    C. Quaternary lava
    D. crystalline rocks

Explanation: Most parts of the central highlands of Ethiopia are covered with trappean lava. **Answer: A**

25. Which one of the following is not a result of the formation of the Great Rift valley?
    A. Formation of Red Sea
    B. Division of Ethiopian highlands in to two
    C. Formation of lake Tana and the Blue Nile gorge
    D. Separation of Arabian peninsula from the Horn Africa

Explanation: As a result of the formation of the Great Rift Valley did not cause the formation of Lake Tana and Blue Nile gorge. **Answer: C**

26. Which lowlands are wetter than the lowlands found in other part of Ethiopia?
    A. The Baro – Akobo lowlands
    B. The Elkere or Bale plains
    C. The Genale or Borena Plains
    D. The Afar depression (triangle)

Explanation: The Baro – Akobo lowlands are wetter than the Eleker or Bale plains, Genal or Borena plains and Afar depression.

Answer: A

27. Which of the following mountains is different from the others in its location? (EHEECE 2002/2010)
    A. Mt chilola
    B. Mt. Batu
    C. Mt. Tkulu Dimtu
    D. Mt. Gughe

Explanation: Mt. Chilalo Mt. Batu and Mt. Tulu dimtu are belonged to the Arsi Bale massive where as Mt Gughe located in the northern massive.

Answer: D

28. Which of the following is NOT an outcome of the formation of the Great Rift Valley?
    A. The creation of lake Tana
    B. The creation of the Red sea
    C. The creation of Lake Abe
    D. The creation of Lake Abaya

Explanation: The outcome of the formation of the great Rift valley cause the creation of Red Sea, and lakes in the Rift – Valley.

Answer: C

29. What proportion of the land features of Ethiopia would the highland comprise if the 1000 meters contour line is used to separate highlands from lowlands.
    A. About two – thirds
    B. Approximately one–third
    C. Slightly more the one – half
    D. Slightly low than three-quarters

Explanation: Slightly more than one half proportion of the land features of Ethiopia would the highland comprise if the 1000 meters contour line is used to separate high land from lowlands.

Answer: C

30. Which one of the following may not be affected by the relief of a given geographical unit?
    A. Rainfall
    B. Population settlement
    C. Length of day and night
    D. Visibility between two points within the area

Explanation: Relief of a given geographical unit may not be affected the length of day and night it is related to revolution of the earth surface.

Answer: C

31. Which of the following statements about the western Highlands and lowlands of Ethiopia is NOT correct? (UEE – 2004/2012)
    A. It is the single largest physiographic region in Ethiopia
    B. It makes Ethiopia the most mountainous and plateau land in Africa.
    C. It is the most densely settled and extensively cultivated part of Ethiopia.
    D. It stretches from Tigray in the North to the Ethio – Somalia border in the south.

Explanation: The Western highlands and associlated lowlands stretches from Tigrai (Ethiopia – Eritrean border in the north to Gamo Gofa (Ethiopia – Kenyan border) in the south

Answer: D

32. What Makes Ethiopia the "roof of North – Eastern Africa"? (UEE – 2004/2012)
    A. Its compact shape
    B. Its proximity to the region
    C. Its large proportion of highlands
    D. Its large proportion of a rable land

Explanation: 56 percent of Ethiopia's land is highland. This fact has given Ethiopia the name "Roof of North eastern Africa"

Answer: C

33. Which of the following land form serves as a natural bridge connecting the highlands of Gonder and Gojam with the highland of Wello? (EUEE- 2005-2013)
    A. The Tigrean Plateau
    B. The yeju – Wadla Delanta Ridge
    C. The North central Massifs
    D. The shean plateau

Explanation: The highlands of Gonder and Gojjam with the highlands of Wello connected by a land bridge called yeju – Wadla Delanta.

Answer: B

34. One of the following statements correctly describes the terrain characteristics of Ethiopia. (EUEE – 2005/2013)

A. The land form is dominated by mountains having elevation of over 4000m.
B. The highest peak is located in the south eastern highlands
C. Ethiopia has the most elevated land mass compared to its neighbors.
D. Lowlands are confined to the western margins of the

Explanation: Ethiopian is the only country in the region or compared to its neighbors with such a high proportion of elevated surface.

Answer: C

35. Which one of the following branches of Geography would you study if you were interested in understanding the major forces that are shaping the land forms of Ethiopia?

A. Climatology
B. Geomorphology
C. Biogeography
D. cultural Geography

Explanation: Geomorphology is the branch of Geography study about the major internal and external forces that are shaping the land form of the earth.

Answer: B

7.3.3. Drainage systems and water Resources of Ethiopia

- Drainage pattern – refers to the fabric or surface arrangement of the main rivers and their tributaries.
- Drainage systems refers to the direction and destination of the rivers.
- The drainage pattern of the Ethiopian rivers and streams is a result of geological events of the tertiary period which created the present surface configuration of the country.
- Ethiopian highlands serve as a watershed separating the western, eastern and the rift valley drainage systems.
- Ethiopia is among a few African countries with enormous water resources. It is a country of large number of rivers and lakes.

General Characteristics of Ethiopian Rivers

- Ethiopian rivers are energetic
- They flow from the highlands of the interior to the peripheral lowlands and then to seas and lakes bouncingly.
- These flow conditions have made Ethiopia known as the water tower of north eastern Africa" and as the watershed

between the Mediterranean sea and the Indian clean drainage systems.

- Ethiopian rivers are characterized by
  - → Most of the major rivers descend from great heights and flow through deep gorges.
  - → They push through rapids and waterfalls along their courses.
  - → They have international characteristics since they flow out of the national territory.
  - → They have rapids, falls, gorges and cataracts that hinder – transportation along with the fluctuating volume of water of the rivers.

⇒ A few of Ethiopian rivers have different names when they cross the border of Ethiopia, E.g River Abbay becomes Blue Nile in the Sudan, River Tekeze becomes River Athbara in the Sudan River Ghenalle becomes River Juba in Somalia and River Baro becomes River sobat in the Sudan.

Rivers	Catchment area (km^2)	Length in km			Major Tributaries
		Inside	Outside	Total	
Wabishebelle	205,407	1340	660	2000	Ramis, erer, Dakafa, Fafen, Yerer, Gobelle, Gallleti, Mojo
Abbay	198,508	800	560	1360	Dabus, Didessa Fincha, Guder, Muger, jemma, Beshillo, shinta, Dinder
Ghenalle	168,141	480	570	1050	Dawa, Weyb. Welmel, Mena
Awash	113,709	1200	-	1200	Akaki, Kessem. Borkena, Mille

Tekezze	87,733	608	560	1,168	Tirai, Anghereb, Ghiba, Guang
Ghibe/omo	77,205	760	-	760	Gojeb, Gelgel Ghibe
Baro	75,718	227	280	507	Akobo, Gilo

Table. 7.4. Major rivers of Ethiopia and their tributaries

Drainage systems of Ethiopia

The drainage systems of Ethiopia are broadly classified in to three major groups.

i) The Western (Mediterranean) drainage system
ii) The south eastern (Indian ocean) drainage system,
iii) The Inland (Rift valley) drainage system.

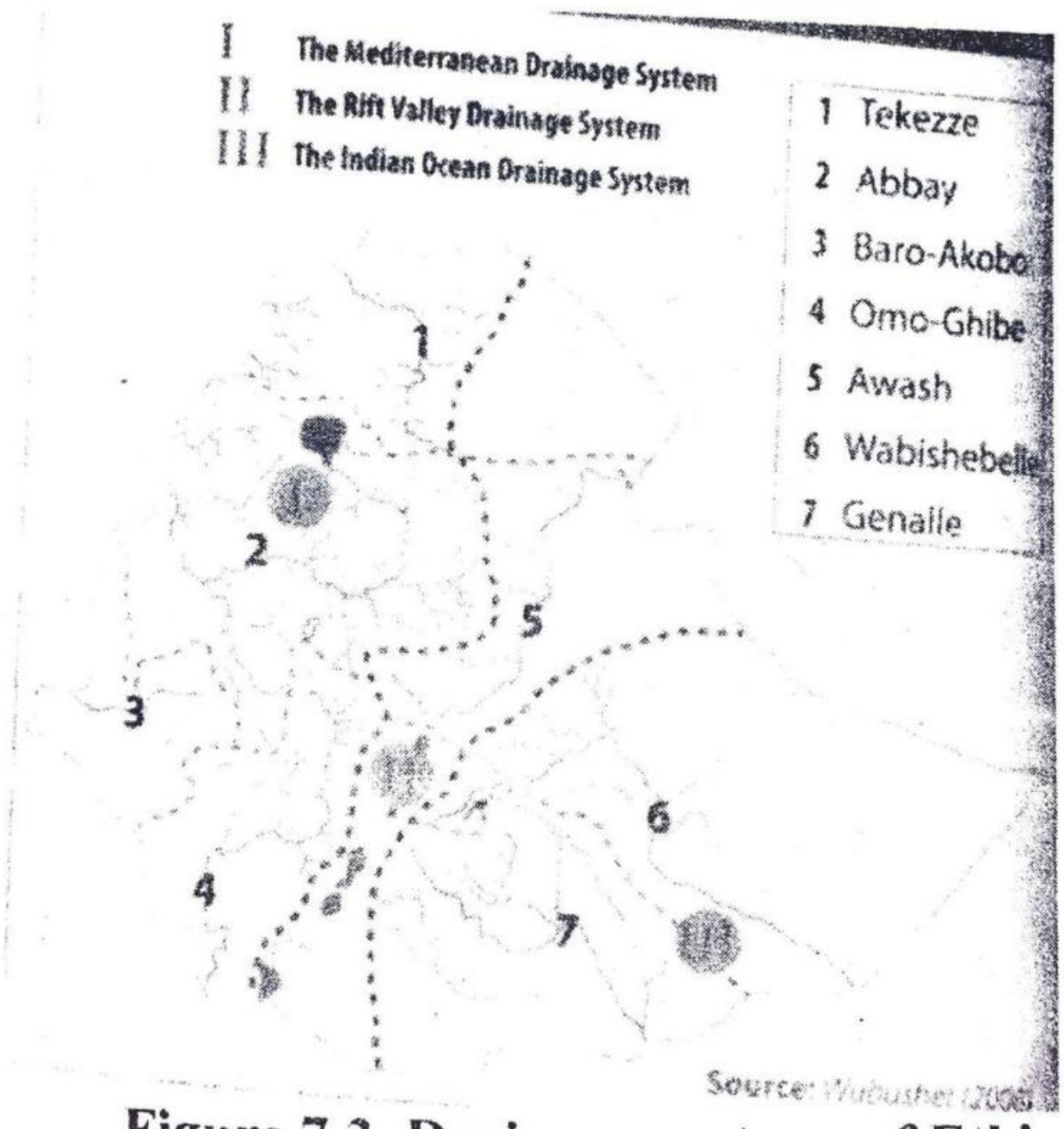

Figure 7.2. Drainage systems of Ethiopia

i) The Western (Mediterranean) Drainage system

- It is the largest of all, both in aerial extent and volume of water out flow.
- It contributes sixty percent of the country's total annual water discharge.
- It drains forty percent of the total area

- It is made up of major river basins namely.

→ River Tekezze – drains the massifs of western Lasta, northern Gondar/ semen and south western, western and central Tigray.

→ River Abbay – The source of river Abbay is Gelgel or Gishe Abbay from choqe massifs. A large number of streams seen the river from the plateaus of western shoa, south western Wello, northern Wollega, and northern Illubarbor

- River Baro – akobo – drain the wettest highlands in the south west and the south west and crosses the border to gain the Nile.

ii) The South eastern Drainage System

- It is the second largest drainage system.
- In includes all rivers that drain towards the Indian ocean were they discharge.
- It consists of the Genalle and Wabishebelle.
- It flows southeast, across the Somalia arid and semi-arid areas.
- It contributes about 32% of the country's total annual water flow.
- It is only the Ghenalle that reaches the Indian ocean while river wabishebelle ends up in the sands of Benadir desert in Somalia.

The Inland (Rift Valley) Drainage System

- It is the smallest of the three systems in terms of catchment area, discharge of water and volume of water
- It is characterized by

→ No outlet to the sea

→ Small, areal discharge of water

→ Relatively lesser volume of water when compared with others.

→ A number of lakes and small streams.

→ The lakes occupy fault depressions in to which smaller streams flow. For example, Meki and katar flow in to Lake zway, Bilate and Gidabo flow in to

Lake Abbaya while sagan and Woito flow in to Chew Bahir.

- The major river that drains the rift valley is river awash. It discharges in to lake Abbe on the border between Ethiopia and the Republic of Djibouti.

(=) The awash River basin;

- It is the most utilized in the Rift valley.
- The basin covers a total area of 110 thousand square kilometers and serves as home to 10.5 million inhabitants.
- It rises from the Shewan plateau near Ginchi town, a town at about 100 kilometrs west of Addis Ababa, and flows along the Rift Valley
- The middle and lower courses are part of the great rift valley system (the upper course is not part of the system).
- The lower Awash River basin comprises the defaicalluvial plains of the Tendaho, Assaita, and Dif Behri areas, and of the terminal lakes area.

(=) The Omo – Ghibe basin;

- It is originated in the central plateau (western Oromiya) and flows in to lake turkana.
- In the upper course it is known as Gilgel Ghibe, in the middle course Ghibe and in the lower course the name of the river is Omo.

Lakes of Ethiopia

- Ethiopia is rich in lakes.
- They are mostly located in drier areas. This implies that the lakes are of structural origin rather than climatic effect.
- The natural lakes found in Ethiopia can be classified in to highland and rift valley lakes.

Highland Lakes. They are found outside the rift valley.

- They are found in mountains and plateaus.
- They are either watershed or crater lakes
- Acrater lake is formed after an explosive volcano breaks a mountain open, leaving a deep mouth.
- Watershed lake is created when the rain water files the valley or the lower place.

- Tana is one of the highland lakes. It was formed when the basalt lava (Aden series of Quaternary period) dammed the area.
- Crater lakes includes many smaller but deeper lakes in and around Debre Zeit (Bishoftu), eg. Guda, Bishoftu, Babo Gaya, Hora, kuriftu, Arsedi, Green lake are the known ones.
- Wonchi (30km) and end to the south east of Ambo), Hyke (near Dessie), Ashenge (Tigray) and Ziquala (shoa over Mt. Ziquala) are the other crater lakes.

Rift Valley Lakes

- They are found in a linear pattern along the floor of the rift valley.
- They are formed in the depressions and basins caused by tectonic activities along the Rift valley.
- They include lake ziway, Langano, Abiyata, Shalla, Awassa, Abaya, Chamo, Galila, Koka (man made lake).
- Lake shalla is the deepest lake and lake Ziway is the shallowest lake in Ethiopia.

Artificial Lakes

- They are formed when people dam rivers to generate hydroelectric power. For example, koka, finicha, Melkawakana, Abasamuel, Tekezze, etc.
- They can also be formed when drinking water is accumulated in reservoirs, eg. Gefersa near Addis.

Table 7.5 Depth area and location of Ethiopia lakes

Lakes	Area (mk^2)	Maximum depth (meters)	Location
Tana	3,600	9	Highland
Abaya	1,160	13	Rift valley
Chawo	551	10	Rift valley
Ziway	434	4	Rift valley
Shalla	409	266	Rift valley
Langano	230	-46	Rift valley
Hawassa	229	-10	Rift valley
Abijalla	205	14	Rift valley
Haik	35	23	Highland
Hashenge	20	25	Highland

Significance of Ethiopia Lakes and Rivers

- Lakes and rivers are Ethiopia's main water resources.
- Ethiopia's lakes and rivers are useful for:
  - → Main source of hydroelectric power (H. E. P) supplies for the country. For example, River Ghibe – Gilgel Gibbe – 1,2 and 3 H.E.P. plants, river Awash – Awash 1, 2 and 3 H.E.P Project. And River Fincha – Fincha H.E.P. project.
  - → Potential for fishing. For example, Lake chamo, Lake Abbay, Lake Tana, Rivers Baro, etc.
  - → They have also good natural scenery.
  - → They are important tourist attractions.
  - → They are again very important water sources for irrigation. For instance, River Awash is the most utilized because of the flat plains it crosses for hundreds of kilometers.
- They are important for transportation services. For example, rivers Baro.

Illustrative Questions

36. Which landform group (physiographic region) of Ethiopia forms an internal drainage system? (EHEECE – 2001/2009)
   A. The Rift Valley Region
   B. The Associated Lowlands of North Western Highlands.
   C. The associated Lowlands of south Eastern Highlands.
   D. The shewan plateau and south western Highlands.

Explanation: The riftvalley region is landform group (physiographic region) of Ethiopia form all internal drainage system, eg. Awash river

Answer: A

37. Which of the following rivers of Ethiopia belongs to the rift valley drainage system? (EHEECE – 2002/2010)
   A. Awash
   B. Abay
   C. Gehenal
   D. Beshillo

Explanation: Awash river is belong to the rift valley drainage system.

Answer: A

38. Which of the following river basins of Ethiopia forms an inland drainage system? (EHEECE – 2003/2011)
    A. The Abay river basin
    B. The Baro and Akobo river baseis
    C. The Ghenale and dawwa river basins
    D. The omo river basin

Explanation: The Omo river basin of Ethiopia from an inland drainage system in the south western part of Ethiopia.

Answer: D

39. Which of the following drainage systems of Ethiopia does NOT have a fixed source region and a fixed flow direction? (UEE – 2004/2012)
    A. The awash Drainage system
    B. The Tekezze Drainage system
    C. The Gibe – Omo drainage system
    D. The rift valley Drainage system

Explanation: In Ethiopia the rift valley drainage systems has not fixed source region and flow direction.

Answer: D

40. Which of the following drainage systems of Ethiopia is the largest in terms of catchment area? (UEE – 2004/2012)
    A. The Abay drainage system
    B. The Ghibe – Omo drainage system
    C. The Baro – Akobo drainage system
    D. The wabshebelle drainage system

Explanation: In terms of catchment area the wabeshebelle drainage system is the largest which covers 205, 407 sq. km. Abay, Gi Omo, Baro – Akobo – 198,508 sq. km, 77, 205 sq. km and 75, 718sq. km respectively.

Answer: D

41. Which of the following rivers of the south – eastern Ethiopia Drainage system carries more water than the rest? (EUEE 2005/2013)
    A. Dawa
    B. Wabishebelle
    C. Weyb
    D. Ghenale

Explanation: Asfar as its large length of 1340kms is concerned wabishebelle river carries more water than the rest.

Answer: B

42. The third largest lake of Ethiopia is
A. Lake Tana
B. Lake Ziway
C. Lake Abaya
D. Lake Abijata

Explanation: The third largest lake of Ethiopia is Lake Ziway (443km).

Answer: B

43. An Ethiopian river which is relatively Less suitable for the generation of hydroelectric power is (EHEECE 2002/2010)
A. Baro
B. Ghibe
C. Abay
D. Tekezze

Explanation: Most of the rivers of Ethiopia has great potential for the generation of hydroelectric power this river include Ghibe, Abbay and Tekezze.

Answer: A

44. Which of the following lakes is located on an international boundary?
A. Tana
B. Ashenge
C. Abe
D. Abaya

Explanation: Lake Abe located on an international boundary between Ethiopia and Djibouti.

Answer: C

45. Controlling the increasing pollution of rivers in big cities and towns of Ethiopia requires all of the following measures except
A. Intensification of environmental education
B. Creating a system whereby polluters rectify their wrong actions.
C. Taking legal actions to discoverage continuation of similar practices.
D. Banning further establishment of chemical industries in such cities and towns.

Explanation: Controlling the increasing pollution of rivers in big cities and towns of Ethiopia requires intensification of environmental education, creation of a system where by polluters rectify their wrong actions, and taking legal actions to discourage continuation of similar practices.

Answer: D

46. River Ghibe/Omo forms a closed drainage system and eventually drains in to _____ (UEE. 2003/2011)

A. Lake Chamo
B. Lake Abe
C. Lake Turkana
D. Lake Fincha

Explanation: River Gibe/ Omo forms a closed drainage system and virtually drains in to lake Turkana.

Answer: C

47. Which one of the following rivers flows through the western most point of Ethiopia?

A. Tekezze
B. Ghibe
C. Akobo
D. Abay

Explanation: River Tekeze flows through the western most point of Ethiopia.

Answer: A

48. One of the following rivers of Ethiopia is NOT presently used for the generation of hydroelectric power?

A. Ghenale
B. Tekeze
C. Wabishebelle
D. Abay

Explanation: Presently Tekeze, wabishebele, Gebe and Abay used for the generation of hydro – electric power in Ethiopia.

Answer: A

49. Which of the following lakes of Ethiopia are known to have areasonably high potential for the production of soda ash?

A. Abijata and shalla
B. Abaya and Chamo
C. Tana and Ziway
D. Haik and Ashenge

Explanation: In Ethiopia Abaya and Chamo lakes of Ethiopia are known to have areasonably high potential for the production of soda ash.

Answer: B

50. Which of the following Ethiopian rivers is known for its large variety of fish species?

A. Baro
B. Awash
C. Abay
D. Omo

Explanation: Baro river is known for its large variety of fish species.

Answer: A

51. Which of the following lakes of Ethiopia is found in a region known for its production of salt?

A. Lake Afdera
B. Lake Zuway
C. Lake Chamo
D. Lake Abaya

Explanation: In regional location lake Afdera is found in the Afar region which is a source of salt.

52. The deepest and shallowest lakes of Ethiopia are ____ and ____ respectively. (EUEE – 2005/203) **Answer: A**

A. Shalla and Ziway
B. Tana and Ashenge
C. Tana and Ziway
D. Shalla and ashenge

Explanation: The deepest and shallowest lakes of Ethiopia are shalla and Ashenge.

53. Which part of the Awash river valley has the highest potential for the development of irrigated agriculture? (EUEE – 2005/2013) **Answer: D**

A. Upper Awash
B. Middle Awash
C. Lower Awash
D. Awash Delta

Explanation: The Lower Awash valley has the highest potential for the development of irrigated agriculture.

7.3.4. Water – Resource conservation and management in Ethiopia **Answer: C**

- Studies indicate that all of the Ethiopian rivers, put together, have the potential to produce about 63 billion kwh of electricity per annum and can support the estimated three million hectares of irrigable land in the country.
- Ethiopia has been known as the "Water Tower of North eastern Africa" for the last fifty to sixty years.
- Ethiopia is the second richest African country in terms of water resource potential, following the Democratic Republic of Congo.
- Despite all these potentials, the water resource available and the actual use to the country's social, economic and environmental development show marked gap. This, paradox areas from the prevailing lack of proper planning and harnessing of the water resource. In resources to this need, the Federal Democratic Republic of Ethiopia has adopted a national conservation strategy for natural resources under the Ethiopian Environmental protection Authority. The

Authority has developed policy, goals, objectives and principles

Overall policy Goals

The overall policy goals are

- Improve and enhance the health and quality of life of the people
- Promote sustainable social and economic development through management and use of
  - Natural human made and cultural resources
  - The environment
- Perform these takes in a manner that meets the needs of the present generation without compromising the ability of future generations to meet their own needs (sustainable development

Policy Goals Related to water Resources

In relation to water resources the goals of the Environmental protection Authority are

- Ensure that the control of environmental health hazards
- Recognize that natural ecosystems
- Ensure that any proposed introduction of exotic species to water ecosystems as subject to detailed studies and environmental impact assessment
- Promote the protection of the interface between water bodies and land (for example lake shores, river banks and wetlands
- Involve water resource users particularly Women and animal herders in the local planning, designing and follow up of water policies programs and projects
- Recycle waste water when it is found to be safe for health and the environment.
- Promote to the extent possible viable measures to artificially recharge ground and surface water resources.
- Promote effective water management techniques at the level for improved performance of medium to large irrigation schemes.
- Provide technical and credit support to the private sector water resource development activities

Illustrative Examples

54. Ethiopian rivers are known to have an international significance due mainly to their (EHEECE – 2001/2009)
    A. Seasonal fluctuation
    B. Flow on rapids and falls
    C. Narrow and winding course
    D. Contribution to arid regions in the Horn and North east Africa.

Explanation: Ethiopian rivers are known to have an international significance due to their contribution to and regions in the Horn and North East Africa.

Answer: D

55. What is the prominent development activity based on the Awash river? (EHEECE – 2001/2009)
    A. Hydro-electric power generation
    B. Eco – tourism
    C. Irrigation
    D. A and C are correct

Explanation: Irrigation and hydro – electric power generation are the prominent development activities based on the Awash rivers

Answer: D

56. An Ethiopian river which has the lowest flow variability is (EUEE 2005/2013)
    A. Tekeze
    B. Awash
    C. Baro
    D. Ghenale

Explanation: An Ethiopian river which has the lowest flow variability is Baro.

Answer: C

7.4. Climate of Ethiopia and The Horn

- Climate is along – term data about the average weather conditions of a place or a region over along period of time; such periods is typically as long as 30 years or more.
- Climatic elements include, precipitation, temperature, humidity, air pressure, Winds, etc.
- Climate influences all human endeavor.

7.4.1. Factors Influencing the Spatial and Temporal Distributi of Climatic Elements in Ethiopia and the Horn.

- The distribution of elements over the surface of the earth is unev in terms of magnitude and time.
- The spatial and temporal distribution of climatic elements governed by the climate control factors.
- They include latitude, altitude, revolution of the earth and t inclination of the earth's axis, distance from the sea, mounta barriers, weather systems and cloud cover.

Latitude

- It is the angular location of a place or point with reference t the direct rays of the sun.
- It indicates the location of Ethiopia with in the tropics.
- Ethiopia's and the Horn's location within the tropical Zon results in.
  - → High temperatures during most of the year
  - → High daily (diurnal) ranges of temperature
  - → Relatively small annual ranges of temperature
  - → Little difference between summer and winter in th ratios of daylight to night.

Altitude

- It modifying temperatures from one place to another.
- Temperature and altitude are conversely proportional when a general decrease in temperature for increases in elevation
- The average rate at which temperature changes per unit of altitudinal difference is known as lapse rate.
- The rate of change is 6°c per 1000 meters. This change is called environmental lapse rate or atmospheric lapse rate.
- It determines the spatial distribution of temperature in Ethiopia.
- Because of altitude difference temperatures decrease toward the interior (where the highlands dominate) from the peripheral lowlands (where mean annual temperatures often reach more than 30°c)

evolution of the Earth and Inclination of the Earth's Axis

- The axis of the earth inclines 23.5° to the normal of the ellipti

- The earth revolves around the sun, this inclination produces a change in the direction of the sun's rays, there by affecting the length of time that the sun shines on the earth every other day
- Changes in the length of the day and in the directness of the sun's rays cause seasons. These result in the temporal-variation of temperature in a year in Ethiopia and in the Horn.

Distance from water bodies

- Land masses absorb and release heat energy more quickly than water bodies do.
- It affects the Horn's climate only in coastal areas that are adjacent to the Red sea and the northwestern Indian ocean.

Mountain Barriers

- It can affect climate in that they exert influence on the spatial distribution of rainfall.
- Places located on the leeward side of mountains (also called rain shadow) receive little rain. This effect occurs in the north western and north eastern lowlands of Ethiopia and the Horn.

Weather systems

- They are the apparent shift of the overhead sun between the Tropic of Capricorn and the tropic of cancer results in the oscillation of Inter Tropical convergence zone (ITCZ) south and north of the equator. This in turn brings about the invasion of different weather systems. These invading weather systems bring the variations of rainfall and temperature over Ethiopia
- The most important weather systems that have greater impacts over Ethiopia are the.

a) The North east Trade winds (prevalent in December, January and February)
   - → It is also known as sub Tropical Jet air streams
   - → They are generated in the sub - tropics due to temperature contrasts and prevail over Ethiopia in winter and bring no moisture to the interior parts.

b) The Equatorial Westerlies (prevalent in June, July and August)

→ It is also known as Equatorial Jet streams.

→ They originate from the south Atlantic due to the attraction of low pressure cells over south Western Asia and prevail over Ethiopia in summers providing the big rains ('kiremt' rains).

c) The equatorial easterlies (prevalent in September and October, March and April.

→ It is also known as tropical Easterlies.

→ They are air streams that originate from the northern Indian ocean and Arabian sea.

→ They blow over Ethiopia in spring and autumn bringing rains to the south eastern lowlands and to the adjacent highlands.

7.4.2. Spatial and Temporal variation of Temperature in Ethiopia.

A. Spatial variation of Temperature in Ethiopia

- It is mainly due to variation in altitudes. Even places of the same latitudinal positions experience different amounts of mean annual temperatures due to the altitudinal effects.
- The two factors that most affect the spatial variation of temperature in Ethiopia:

→ Cloud cover are

→ Altitude

Cloud cover

- It is the covering of parts of the earth by clouds.
- It can prevent the heat absorbed by the earth from escaping in to the atmosphere.
- The presence or absence of cloud cover produces spatial temperature variations in Ethiopia and elsewhere in the Horn.

Altitude

- It is a major determining factor of Ethiopia's agro – climatic Zones.

- In Ethiopia, the impact, the impact made by altitude has resulted in the formation of five agro climatic zones. traditionally expressed as temperature zones of Ethiopia.

Table 2.6: **Ethiopia's Agro – climatic zones**

Altitude in Meter	Mean annual temperature in ^0C	Traditional agro – ecological name	Global – equivalence
3,300 and above	< 10	Wurch/Kur	Alpine or Afro Alpine
2,300 – 3,300	10 – 15	Dega	Temperate
1,500 – 2,300	15 – 20	Woina Dega	Subtropical
500 - 1,500	20 – 30	Qolla	Tropical
Below 500	>30	Bereha	Desert

Wurch/Kur Zone Areas

- They have the highest altitudes and lowest temperatures.
- They have temperatures of less than 10^0C.
- These areas exist only in the very high mountains of south-Gonder. Wollo. Shoa. Arsi and Bale.
- Example. Mt. Ras Dashen in semine Gondar. Mt. Guna in South Gondar. Mt. Megezez in North Shoa and Mt. Batu in Bale. etc.

Dega – **Zone Areas**

- They are highland areas with lower altitudes and higher temperatures than Wurch – Zone areas
- They were the home concentrated human settlement.
- They were chosen because of the presence of favorable conditions such as secure location (from which people could defend themselves from threats), reliable rainfall, absence of diseases such as malaria, etc.
- Due to high concentration of human population, it has been intensively cultivated and has a high rate of soil erosion, overgrazing and deforestation.
- Example; Dinsho in Bale, chillallo in Aris, Hulla insidama and Debresena in North Shewa.

Woina – Dega – Zone Areas

- They contain most of Ethiopia's agricultural land
- They are the country's main areas producing such as surplus grain and Inset and its derivatives.
- In the woina – Dega Zone, as in the Dega Zone, there can be two growing seasons when rainfall reliability is high.

Qolla Zone and Bereha – Zone Areas

- They are largely confined to lowland areas with altitudes of 1500 meters and below.
- They are sparsely populated and their populations are primarily engaged in pastoralist activities.
- They occupy the peripheral areas of Ethiopia and Eritrea, most of Somalia and Djibouti.

B. Temporal Variation of temperature in Ethiopia

- Temperatures vary from season to season.
- In most of Ethiopia, high temperatures are recorded from march to June and low temperatures are recorded from November to February.
- These variations are primarily due to the tilting of the earth at $23\frac{1}{2}^0$ to the normal elliptic and the distance of the overhead sun and its apparent north – South movement across the equator as the earth revolves around the sun.

7.4.3. Spatial and Temporal variation of Rainfall in Ethiopia and the Horn

- Rainfall, like temperature is a major element of climate.
- Several geographic factors influence the formation and spatiotemporal distribution of rainfall in Ethiopia.

A. Spatial Variation of Rainfall in Ethiopia and the Horn

- It is the result of the invading strength of the moisture laden winds and the presence of barrier that checks the moisture arrival in to the area.

B. Temporal variation of Rainfall in Ethiopia

- It is the result of the apparent shift of the over head sun combined with the shift of the In tropical convergence zone and the nature of the prevailing weather systems following the oscillation of the ITCZ.

Tropical Convergence Zone (ITCZ)

- a low pressure zone
- formed by the convergence of North east Trade winds and the Equatorial Westerlies
- it shifts north and south of the equator following the position of overhead sun
- in July ITCZ found at the Tropic of cancer During this time ethiopia and the Horn under the influence of the Equatorial westerlies and Easteries winds bring moisture to the highland
- in January ITCZ shifts to the Tropic of Capricorn leaving the region for the prevalence of the Northeast Trade winds that are moisture laden - During this time only the southern coastal lands and the afar region receive rain
- March and September. ITCZ is around the equator The Equatorial westerlies provide rain to the highlands of Ethiopia and to the central and south eastern lowlands and

4.4. Rainfall Regions in Ethiopia and the Horn

- In the middle and high - latitudes, seasons are classified as winter, spring, summer and autumn.
- Based on rainfall distribution, both in space and time, five types of rainfall regions can be identified in Ethiopia and the Horn These are.
  - Year - round rainfall region (wet in most months
  - Summer rainfall region
  - Autumn and spring rainfall region
  - Winter rainfall region
  - Merged spring, summer, and autumn rainfall region

Year - round Rainfall Region

- This region includes the south western plateau comprising the regions of Wellega, Keffa, Illubabor and Gamo Gofta
- It can be represented by stations Gore, Mizan, Metu, Bonga, Gambella, etc.
- The region has more rainy days than any other part of the country the average rainfall varies from 1400mm to 2200mm.

- It is represented by the letter B is on Figure 7.3.

Summer Rainfall Region

- It is also known as kiremt (June, July and August) in Ethiopia
- During this time the sun is overhead north of the equator.
- It is the largest in the country.
- It consists of the North west Highlands and Western lowlands.
- It can be represented by stations Debremarkos, Fitche, Gondar, Bahirdar, etc.
- Its moisture – bearing winds are the Equatorial Westerlies and Easterlies.
- It is represented by the letter A in figure 7.3.

Autumn – and spring Rainfall Region

- It covers the south eastern highlands and associated lowlands up to the Somalia coasts.
- They can be represented by stations such as Gode, Moyalle, Jigiga, Yebello, Baydhabo,
- The region's moisture – bearing winds are the Equatorial Easterlies
- The region's average rainfall varies from less than 500 to 1000mm.
- It is represented by the letter E in Figure 7.3.

Winter Rainfall region

- It consists of the eastern escarpment of the Western highlands, the middle Rift valley section, the Afar subdivision and Eritrea.
- They can be represented by stations such as Mitswa, Assaita and Djibouti.
- The region's moisture – bearing winds are, for the most part, the North Easterlies.
- The air mass is continental (dry) and has only a short sea trajectory (that is the Red sea)
- The regions total annual moisture is very low and of short duration.
- It is represented by letter Din Figure 7.3.

Merged spring, Summer and Autumn Rainfall Region

- It is the smallest in the country.

- It consists mainly of the western foothills of the southeastern Highlands
- The region's total annual rainfall varies from 1500mm to 1000mm
- It covers a corridor that stretches from the Sidama Highlands to the Hararge plateau
- They can be represented by stations such as Assaita, Awash, etc
- It is represented by the letter C in Figure 7.3

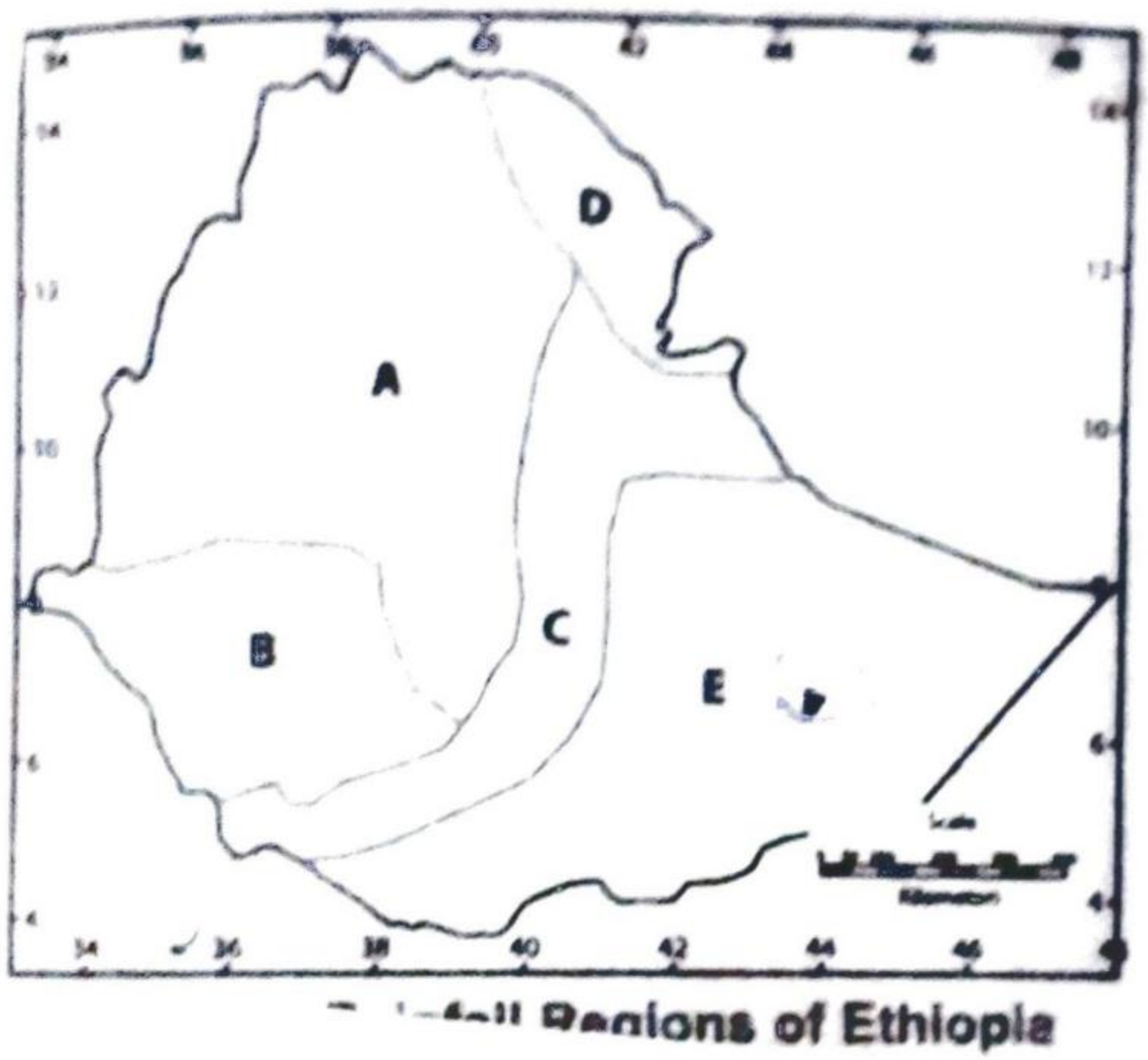

Figure 7.3. Rainfall Regions of Ethiopia

Variability and Intensity of Rainfall

i) **Variability of rainfall**: refers to the high probability of a given amount of rainfall to be repeated each year.

→ The effect of variability of rainfall is high in marginal rainfall areas (low lands) than on the highlands.

→ Variability is inversely related to total amount of rainfall. It is shorter where amounts of rainfall are low.

ii) **Intensity of rainfall**

- It refers to the amount that fall within a unit time.
- The rate of intensity determines the rate of run off, amount of infiltration in soil and flood occurrence.

- Rainfall intensity is high in lowlands and low on highlands in Ethiopia.
- The rain that falls on the ground can undergo the process.
    1. Get in to the soil by infiltration/seepage/
    2. Flow down – slope in to streams as run off /surface flow;
    3. Return to the atmosphere through evapo transpiration
- Factors affecting rainfall efficiency in a region are temperature, winds, Run off and seepage.

Rainfall regions of Ethiopia

There are four rainfall regions in Ethiopia. Namely Monomodal, Bimo – dal, Trimodal and diffuses pattern.

i. **Monomodal rainfall Region**

→ It is found in the Western half of Ethiopia and characterized by single maximum rainfall pattern daily a year.

ii. **Bimodal rainfall region**

→ It is located in southern and south eastern parts of Ethiopia getting double maxima rainfall within a year.

iii. **Trimodal Rainfall region**

→ It is located in the central and south eastern highlands and adjoining lowlands. This rainfall is characterized by three distinct seasons, namely Bega, Belg, and Kiremet.

iv. **Diffused rainfall region**

→ This rainfall, region is located in the extreme north eastern areas. It is characterized by no distinct rainfall pattern.

Illustrative Questions

57. In Ethiopia, temperature tends to decrease towards the inner part of the country mainly because of the effects of (EUEE – 2000/2008)

A. Ocean currents
B. Global warning
C. altitude
D. latitude

Explanation: Altitude is one of the factor that affect (increase or decrease) climate in Ethiopia. As move in to the interior the altitude of Ethiopia increase are temperature decreases.

Answer: C

58. Which of the following best explains the prevailing spatial and temporal pattern of rainfall in Ethiopia?
   A. Country's location within the tropics
   B. Higher elevation the country has in eastern Africa
   C. Country's location in relation to the Atlantic and Indian oceans
   D. B and C are correct answers

Explanation: Higher elevation and location in relation to Indian and Atlantic ocean the tropic is affect the prevailing spatial and temporal pattern of rainfall in Ethiopia.

Answer: D

59. Why are the Denakil plains receiving one of the lowest amounts of annual rainfall in Ethiopia? (EUEE – 2000/2008)
   A. They are far away from the sea
   B. The land has a very low altitude
   C. They are located mainly on the lee – ward side of the equatorial westerlies
   D. They are located mainly on the lee – ward side of the central and northern highlands of Ethiopia

Explanation: The Denakil plains receiving the lowest amount of annual rainfall in Ethiopia because the land has very low altitude.

Answer: B

60. Which factor of climate seems more important to the climatic of the North east African Region of which Ethiopia is apart? (EHEE 2001/2009)
   A. Altitude and latitude
   B. Latitude and distance from the sea
   C. Distance from the sea
   D. Altitude and ocean current

Explanation: The two most important factors that determine the climate of North east African Region is Altitude and latitude.

Answer: A

61. The coefficient of variation of rainfall in Ethiopia (EHEECE 2001/2009)
   A. [illegible] than Kiremt
   B. [illegible] for the entire country

C. Is higher for highlands than lowlands
D. Is the highest in the Horn of Africa

Explanation: The coefficient of variation of rainfall in Ethiopia in higher for the highlands than lowlands.

Answer: C

62. Which statement is not true about the climate of Ethiopia?
A. Lowlands are drier and warmer than the highlands.
B. Both temperature and rainfall amount decrease with increasing elevation.
C. Equatorial westerlies winds are important moisture bearers in keremt.
D. Rainfall is seasonal and the amount decreases when one moves away from the south western part of the country.

Explanation: The climate of Ethiopia characterized by altitude. Lowlands are driver and warmer than the highlands decrease and also temperature and rainfall increase with increasing elevation.

Answer: B

63. Wettest regions of Ethiopia are found in the
A. North eastern part
B. South eastern part
C. south Western part
D. central part

Explanation: In Ethiopia, the south western part such as wellega, keffa, Illubabor and Gamo Goffa referred as the wettest region.

Answer: B C

64. The temperature Zone that is traditionally referred to as "Dega" in Ethiopia is the area that lies above the elevation of (EHEECE 2002/2010)
A. 3300 meters
B. 300 meters
C. 2300 meters
D. 2000 meters

Explanation: The temperature zone that is traditionally referred to as Dega in Ethiopia is the area the lies above the elevation of 3000m.

Answer: B

65. Which of the following statements reflects the seasonal and annual distribution of rainfall in Ethiopia?
A. In the Bega season, rainfall exceeds 1000mm in most areas.
B. The amount of rain in Bega season is more than that in the Belg.

C. In most parts of the country, more than half of the annual rainfall is received in the Keremt season.
D. The country receives most of its rainfall in spring and autumn.

Explanation: In most parts of Ethiopia more than half of the annual rainfall is received in the keremt season.

Answer: C

66. In which of the following parts of Ethiopia do you expect to find the highest variability of annual rainfall.
A. The south western highlands
B. The Arsi – Bale massifs
C. The rift valley lakes region
D. The north eastern lowlands

Explanation: The north Eastern lowlands of Ethiopia expected to find the highest variability of annual rainfall.

Answer: D

67. Which of the following localities of Ethiopia would most probably receive the lowest annual rainfall? (EHEECE-2002 2010)
A. Assosa
B. Adama
C. Asaita
D. Gambela

Explanation: Asaita located in the arid Rift valley area would most probably received the lowest annual rainfall below 250mm.

Answer: C

68. What is the main criteria that is used to identify seasons as "Bega", "Belg" and "Kiremt" in Ethiopia?
A. Temperature distribution
B Distribution of annual rainfall
C. Weather systems
D B and C are correct

Explanation: The main criteria that is used to identify seasons as bega and kiremt in Ethiopia are distribution of annual rainfall and weather system.

Answer: D

69. A locality in Ethiopia has a mean annual temperature of 18°c. To which of the following traditional temperate zones does it most probably belong?

A. Wurch
B. Dega
C. WainaDega
D. Qolla

Explanation: If the region has a mean annual temperature of 18°c. this region is most probably belong to Dega.

Answer: B

70. According to Ethiopian traditional classification of temperature Zones, the term "Woina Dega" refers to areas that have climate (EHEECE 20003/2011

A. Temperature
B. Sub-tropical
C. tropical
D. desert

Explanation: According to Ethiopian traditional classification of temperature zones, the term "Woina Dega" refers to areas that have temperature climate.

Answer: A

71. Which part of Ethiopia receives the lowest amount of rainfall during the "Kiremt" season?

A. The rift valley lakes region
B. The north – eastern and south – eastern lowlands
C. The north – western lowlands
D. The lowlands of Gambella and Beni – shangul Gumiz

Explanation: In Ethiopia during the "Kiremt" season the lowest amount of rainfall receives in the North Eastern and South eastern lowlands.

Answer: B

72. The climate of Ethiopia is mainly influenced by _____

A. Latitude
B. Distance from the sea
C. altitude
D. ocean current

Explanation: The climate of Ethiopia is mainly influenced by altitude.

Answer: C

73. Which one of the following has the weakest impact on seasonal and annual climatic conditions of Ethiopia?

A. The Atlantic ocean
B. The Indian ocean
C. The southwest Asian Landmass
D. The south east Asian Landmass

Explanation: The weakest impact on seasonal and annual climate conditions of Ethiopia is the south East Asian landmass.

Answer: D

74. Which of the following lowlands of Ethiopia receive the lowest amount of annual rainfall?
   A. The Baro lowlands
   B. The rift valley lakes region
   C. The Tekeze – Angehrib lowlands
   D. The omo river basin

Explanation: The Tekeze – Angehrib low land in the North western lowland receive the lowest amount of annual rainfall.

Answer: C

75. Which of the following towns of Ethiopia has the longest rainy season and the lowest rainfall variability?
   A. Gondar
   B. Gore
   C. Assela
   D. Dire Dawa

Explanation: From the given alternatives gore town of Ethiopia has the longest rainy seasons and the lowest rainfall Variablity.

Answer: B.

76. Which of the following parts of Ethiopia lacks a distinct seasonal pattern in its receipt of rainfall?
   A. Extreme south – eastern areas
   B. Central highland and lowlands
   C. Extreme north – eastern areas
   D. Southern highland and lowlands

Explanation: The extreme south Eastern areas of Ethiopia lack a distinct seasonal pattern in its receipt of rainfall.

Answer: A

77. Which of the following statements about rainfall in Ethiopia is correct? (UEE – 2004/2012)
   A. In general lower altitudes have greater annual precipitation than higher altitudes.
   B. The amount of rainfall received by any two places located on the same altitude is more or less the same
   C. The amount of rainfall received by any two places located on the same latitude is more or less.
   D. In general the south western highlands tend to have greater annual precipitation than the north western highlands.

Explanation: In Ethiopia, the south western highlands have by virtue greater annual precipitation than the north western highlands.

Answer: B

78. What is the principal reason behind the relatively lower mean daily temperatures recorded over much of highland Ethiopia in July and August?
    A. The high angles of the sun
    B. The chilling effect of rain bearing winds
    C. The increase in soil moisture due to heavy rainfall
    D. The continuous presence of more or less thick cloud cover

Explanation: In the rainy summer season the air temperature over the highlands of Ethiopia is low because of the thick cloud cover which reduces the amount of solar radiation received.

Answer: D

79. Which of the following is NOT directly affected by the vertical distribution of climate in Ethiopia?
    A. The spatial distribution of vegetation
    B. The spatial distribution of population
    C. The spatial distribution of urban centres
    D. The spatial distribution of linguistic groups

Explanation: The vertical distribution of climate in Ethiopia is NOT directly affected by the spatial distribution of linguistic groups.

Answer: D

80. Two places in Ethiopia are located at the same latitude and receive the rays of the sun at the same angle. However, both places happen to have different temperature records which of the following is most likely responsible for this?
    A. Wind direction
    B. Altitudinal variation
    C. cloud cover
    D. length of day

Explanation: Because of inverse proportional of altitude and temperature, altitude determine the temperature condition of a given area.

Answer: B

81. Which of the following towns in Ethiopia has the longest rainy season? (EUEE 2005/2013
    A. Debre Tabor
    B. Weldiya
    C. Goba
    D. Mettu

Explanation: Year – round rainfall region has more rainy days than any other part of the country due to this fact Mettu belong to this region.

Answer: D

82. Which of the following areas of Ethiopia usually receives rainfall in autumn and spring? (EUEE – 2005/2013)
A. The south – Western lowlands
B. The north – western highlands
C. The south – Eastern lowlands
D. The central and south – eastern highlands

Explanation: The autumn – and – spring rainfall region covers the south eastern highlands and associated lowlands up to the Somalia coasts.

Answer: C

83. Which of the following statements in NOT correct about rainfall in Ethiopia?
A. It is unreliable in both amount and period of arrival
B. Highlands generally receive more rainfall than lowlands
C. Its amount and duration decrease progressively from the south west to the north east
D. The timing of its onset is more dependable in lowlands than on highlands.

Explanation: In general, reliability or dependability of rainfall is more on the highlands than on the lowlands of Ethiopia

Answer: D

84. Which of the following statements correctly describes the characteristics of rainfall in Ethiopia? (EUEE – 2005/2013)
A. Rainfall goes down to about 200mm in the western lowlands
B. Rainfall distribution is affected mainly by latitudinal location of a given area.
C. Rainfall in Ethiopia is the result of summer monsoon and tropical easterlies
D. Rainfall reliability decreases in the highlands

Explanation: Rainfall in Ethiopia is the result of summer monsoon and tropical easterlies.

Answer: C

85. Which of the following statements in NOT correct about the distributions of temperature in Ethiopia. (EUEE – 2005/2013).

A. There is generally a high daily range of temperature in the country.
B. The country experiences a regular receipt of solar heat throughout the year.
C. There is generally a very low annual range of temperature in the country.
D. Places that are on the same latitude generally experience the same temperature.

Explanation: the distribution of temperature in Ethiopia experienced high diurnal (daily) range of temperature.

Answer: C

86. In the rainy summer season, the air temperature over the highlands of Ethiopia gets low because of the (EUEE-2005/2013)
A. Thick cloud cover
B. Clear sky at nigh
C. effect on Elnino
D. low forest cover

Explanation: In the rainy summer season, the air temperature over the highlands of Ethiopia gets low because of the thick cloud cover.

Answer: A

7.5. Natural Vegetation and Wild Animals of Ethiopia

7.5.1. Types of Natural vegetation of Ethiopia

- Natural vegetation refers to any original plant grown in and covering an area.
- Ethiopia's climate, relief, soils and other geographic factors have resulted in the existence of different types of natural vegetation
- There are about 1,000 species of natural, vegetation in Ethiopia of which 1,000 plant species are endemic.
- The types of natural vegetation is an area are strongly determined by temperature and rainfall.

Natural vegetation's Relationship to Altitude and Rainfall

- In Ethiopia, the types of the natural vegetation of an are highly correlated with altitude and rainfall, as they are with temperature.
- The lowlands have harsh environments due to low rainfall and are characterized by xerophytic plants.

- The highlands are characterized by different types of tree stands and map forests.

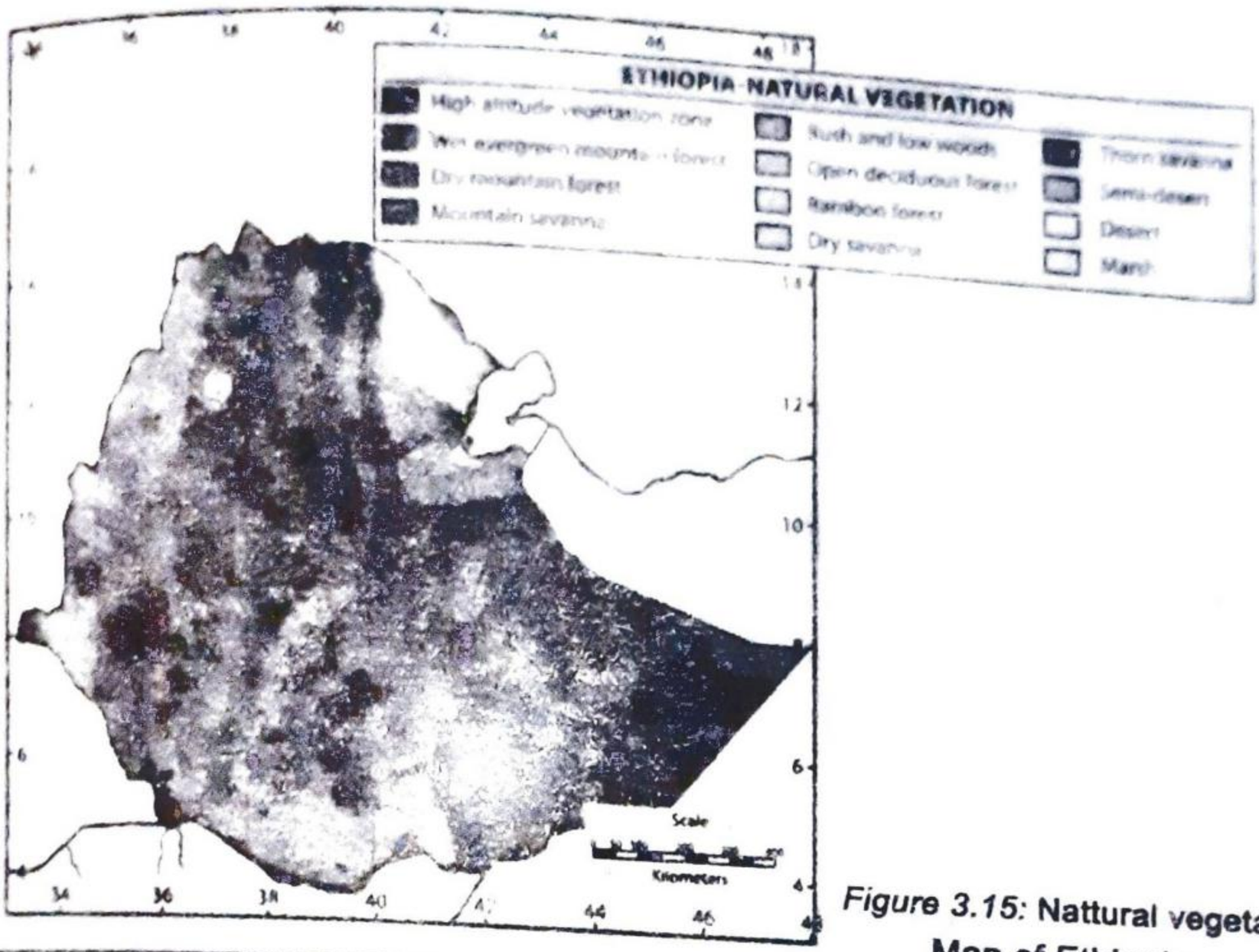

Figure 3.15: Nattural vegetal Map of Ethiopia

Figure 7.4. Natural vegetation map and Ethiopia

- The natural vegetation of Ethiopia is classified in to five major types depending largely on the influence of climate and altitude.
  - → Afro – alpine and sub – Afro alpine
  - → Forests
  - → Woodland savanna
  - → Steppe vegetation
  - → Desert and semi desert vegetation

Afro – Alpine and sub Afro – Alpine vegetation

- It is found at very high altitudes (above 3300 meters)
- It is often known as high mountain vegetation.
- The Afro Alpine type grows at an a 4620 mas/ while the sub – Afro Alpin elevation (mostly 3000 – 3300masl).
- They are Afro – alphine vegetation is dominated by scrubs and woodland.
- The Afro Alphine vegetation is dominated by Gibra (Lobelia rhynchopetalum) and Asta *Erica arborea)

Forests

- They are found in areas where the altitude varies from 450 – 3300m above sea level and where the total annual rainfall is between 200 – 2200mm
- In Ethiopia, forests are found in all places except in the peripheral arid lowlands and very high mountain peaks where the altitude extends in to the Wurch Zone.
- Based on altitudinal Zonation, we classify the forests of Ethiopia in to
  - → Highland forests (forests that grow over 1500meters above sea level)
  - → Lowland forests (forests that grow below 1500 meters above sea level)

Highland forests

- It grow between 1500 – 3300 mas (altitude)
- They inclued kerkha (arundinarial) 2800 – 3000 mas/ Tid (Juniperous procera) or coniferous trees 2200 – 2800 mas l, Ziga (podocarpus), 1800 – 2200 mas and woira (oliia Africana) and Kosso (Hagenia Abbyssinia (1500 - 1800).
- In Ethiopia these forests are found particularly in central and south eastern highlands.

Lowland forests

- It grow below 1500m altitude
- They are known as gallery / riverine forests, that grow along the river banks of Awash, Wabishebelle, Ghanale etc where moisture is available in the soil.
- The mean annual rainfall exceeds 500mm, Baphia forest predominates
- The dominant trees are sholla and Warka.

- Based on the dominant tree composition, Ethiopian forests can be classified in to Broad – leaf and coniferous forests.

Broad – leaf forests:

- They are hardwood forests.
- They are found in the wettest part of Ethiopia where the annual rainfall ranges from 1400 – 2000mm.

- These forests include the forests of wellega, Keffa, Illubabur and Gamo goffa Zones.
- The most dominant tree species are Wanza, Weira, Kerero, Tikur Inchet, Bisana etc.

Coniferous forests

- They are composed of softwood trees
- They are found in the isolated parts of the shoan plateau and the Arsi – Bale massifs.
- This type of forest is found in pocket areas of central, Southern and south eastern highlands. They are unlawfully removed from other places.

Woodland Savanna

- They are found in both highland and lowland areas
- The altitudinal range is 250 – 2300m.
- The mean annual rainfall range is between 200 – 1400mm.
- They are grass mixed with scattered trees, shrubs and bush lands.
- Ethiopia's savanna grasslands are found in the southern half of Zeway, Langano, Abiiyatta and Hawassa.
- They are made up of Juniperious woodlands – mountain grasslands with tide trees, Accacia woodlands – grasslands with acacia (Ghirar) trees and mixed deciduous woolands – grasslands with trees like sholla and warka.

Semi – desert and Desert Vegetations

- These plants grow in the harsher climate regions of up to 1400meters elevation with a total annual rainfall below 600mm, i.e. in the north western lowlands, Afar lowlands and the Ogaden lowlands.
- It consists of short acacia, thorn bushes, succulent plants and a few rough grasses.

Problems of Ethiopian Natural vegetation

- The most serious vegetation problems of the country are deforestation overgrazing and burning.

- Due to the presence of large scale deforestation in Ethiopia the forest land occupies less than 3% of the total area of the country.
- The grassland that listed for source of food in livestock production leads the savanna land of Ethiopia for destruction. Finally, man burns trees for charcoal and clear land for cultivation.

Conservation Measures

- Reforestation – planting trees after every cutting
- A forestation – planting an bare and unproductive lands.
- Controlling forest fires (burning practices)
- Practicing agro forestry
- Using alternative source of energy
- Controlling overgrafing
- Capacity building and institutional development.

Illustrative Questions

87. Which feature describes broad leaf forests of Ethiopia? EUE – 2000/2008)
    A. Consists of xerophytic plants
    B. Composed of soft wood trees
    C. The most wide spread forest types
    D. Found in the wettest parts of the country

Explanation: The equatorial Rain forest is made up of dense, tall and broad leafed evergreen trees because this area is the wettest south western part of Ethiopia.

Answer: D

88. Which one of the following is NOT a reason behind the rapid depletion of forest resources in Ethiopia? (EUE – 2000/2008)
    A. Rapid growth of towns and cities
    B. Un wise land clearing and cultivation methods
    C. The existence of a proclamation that recognizes private ownership of forests.
    D. The low consumption of electricity and fossil fuels for cooking purposes.

Explanation: Resources behind the rapid depletion of forest resources in Ethiopia includes ... rapid population

growth in towns and cities, unwise land clearing and cultivation methods, and low consumption of electricity and fosil fuels for cooking purpose.

89. Which types of vegetation cover relatively the most extensive region in Ethiopia? (EUE – 2001 2009)

Answer: C

A. Highland forests
B. Savanna and Horn bush
C. steppe and semi – desert
D. Afro – Alpine

Explanation: In Ethiopia steppe and semi – desert types of vegetation cover relatively the most extensive regions in Ethiopia.

90. One of the following is NOT true about forest resources of Ethiopia

Answer: C

A. Most natural forests of Ethiopia consists of tropical hardwoods
B. Timber from the natural forests of Ethiopia has low demand on the world market
C. Most of the country's forests are located in remote and in accessible areas.
D. Much of the timber that is needed for construction comes from natural forests.

Explanation: most natural forest consists of tropical hardwoods Timber from the natural forest of Ethiopia has low demand on world market and most of the Ethiopia forests are located in the remote and in accessible areas.

91. The remnants of the original forest covers of Ethiopia are commonly found (UEE 2002-2010)

Answer: D

A Inside and around the compounds of ancient churches and monasteries.
B On the banks of rivers, the south western plateau and on the steep slopes of mountains in less populated areas.
C On the Tigray plateau
D A and B are correct

Explanation: The remnants of the original forest covers of Ethiopia are commonly found inside and around the compound of

ancient churches and monasteries and on the banks of rivers
the south western plateau and on steep slopes of mountains
in less populated area.

Answer: D

92. Which of the following constitutes the largest share of Ethiopia's forest reserve? (UEE- 2003/2011)
   A. Coniferous forests
   B. Gallery (Reverine) forests
   C. Bradleat rainforest
   D. Mixed deciduous woodlands

Explanation: The largest share of Ethiopia's forest reserve is coniferous forests.

Answer: A

93. Which one of the following factors is NOT responsible for the degradation of forest resources in Ethiopia?
   A. Indiscriminate cutting of trees for full purpose
   B. Clearing of forest for agricultural purpose
   C. Appropriate forest policy and management
   D. Setting forest fire to improve the quality of the soil

Explanation: Factors for the degradation of forest resources in Ethiopia includes in discriminate cutting of trees for full purpose, clearing of forests for agricultural purpose and setting forest fire to improve the quality of soil

Answer: C

94. All things considered, the single most important factor responsible for the rapid depletion of forests in Ethiopia is _____
   A. Recurrent drought
   B. High industrial demand for forest resources
   C. Expansion of settlements and cultivation
   D. The high contribution of the sector to the country GDP

Explanation: The most important factor responsible for the rapid depletion of forests in Ethiopia is expansion of settlements and cultivation.

Answer: C

95. The type of natural vegetation that covers the largest area of the Ethiopian lowlands is _____
   A. Acacia savannah and thorn bush
   B. Podocarpus forests

C. Arundinoria forests
D. Juniperous Woodlands

Explanation: The type of natural vegetation that covers the largest area of the Ethiopian lowland is Acacia savannah and thorn bush.

Answer: A

96. Which one of the highland forests is found on the highest altitude? (UEE – 2004/2012)
A. Juniperous
B. Podocarpus
C. Arundinaria
D. Hagenia Abyssinia

Explanation: The highland forest kerkha (arundinaria) found the highest altitude of 2800 – 3000 masl.

Answer: C

97. Where are the most extensive forest reserves of Ethiopia found?
A. The lowlands of Gambella
B. The Arsi – Bale massifs
C. The Jemjem plateau
D. The south – western highlands

Explanation: The most extensive forest reserves of Ethiopia found in the Arsi – Bale massifs.

Answer: B

98. Which of the following is a correct match between types of vegetation and their geographic location in Ethiopia? (EUEE-2005/2013)
A. Afro – alphine – Bale mountains
B. Montanue grassland – Afar depression
C. Deciduous wood land – semien mountains
D. Reverine forests – Danakil depression

Explanation: The afro – alphine natural vegetation is found in Bale mountains.

Answer: A

99. Which of the following types of the natural vegetation of Ethiopia is typically found on both the highlands and the lowlands? (EUEE – 2005/2013)
A. Gallery (riverine) forests
B. Coniferous forests
C. Wood land savannahs
D. Hagenia forests

Explanation: A type of natural vegetation in Ethiopia that found both the highlands and the lowlands as woodland savannahs.

Answer: C

100. The broad leaf forests of Ethiopia include all of the following species EXCEPT
A. Kosso
B. Wanza
C. Kerero
D. Tikur Inchet

Explanation: All are belong to broad leaf forests where as kosso is under highland forests.

Answer: A

7.5.2. Wild Animals of Ethiopia

- The existence of different climatic – vegetation regions has enabled Ethiopia to host awide veriety of wild animals.
- Ethiopia has about 277 large mammalian species 862 bird species, 201 reptile species and 63 amphibians species.
- 7 species of mammals and 25 species of birds are found only in Ethiopia (i,e. Endemic).
- These wild animals are found in many parts of the country, but largely concentrated in the south and western parts.

Types of Wild Animals in Ethiopia

The wild animals found in Ethiopia can be groped in to the following broad categories

Common wild animals

- They are animals commonly found in many places of the world.
- It including hyna and the jackal

Game animals:

- It is also known as lowland animals.
- They include herbirores and carnivores.
- Examples of Herbivories, such as giraffes, wild asses, zebra. Gazelles, elephants, oryx, etc.
- Examples of carnivores which includes lions, leopards, cheetahs etc.

Arboreals:

- Altemative we call it tree climbing animals.

- They are moistly found in the rainforest regions of Ethiopia.
- Ethiopia's arboreal animals, such as monkeys, colobus monkey, apes, baboons, etc.

Aquatic animals:

- They are live in water
- Example, Fish, crocodiles and hippopotamus.

Birds:

- They are found in the rift vally sanctuaries and forest lands.
- Ethiopia has different kinds of both endemic and migratory birds; for example, flamingoes, white pelicans, fish eagles, Geese and Abyssinian Ground Horn bill.

Rare/ Endemic animals

- They are found only in Ethiopia.
- Today these animals exist in very small number on highlands at great risk of extinction. Eg. Walia Ibex, Semeen fox etc.

Endemic mammals of Ethiopia and their location

- Walia Ibex (wild goat – on the semein mountains.
- Mountain Nyala (Dega Agazen) – on the Bale mountains – in heather and woodland areas.
- Gelada (chelada) Baboon – on the semen mountains.
- Menilik's Bush Buck (Dukula) – on the central [illegible]ale Highlands.
- Swayne's Harte beest (Korkay) – in Netchsar National park and sankalle sanctury.
- The Ethiopia wolf previously known as semen fox (Key kebero) – on the semen and Bale highlands.
- Wild Ass (Yedur Ahiya) – in Afar and S.E low lands.

Photography 7.5. Wildlife of Ethiopia

Causes of the extinction of wild – animals

The main reasons for such increasing destruction of wild animal in Ethiopia include.

- The destruction of their habitats (deforestation) due to the rapid growth rate of farmlands, urbanization and industrialization:
- Expar f grazing land
- The wide practices of illegal hunting by the local people looking for meat, skin, fur, horn and ivory:
- Frequent wild fires
- The shrinking of their habitat for grafing purposes;
- Shortage of food and water because of recurring droughts.

Conservation Measures

The following possible conservation actions are recommended to protect and achieve sustainable use of this resource.

- Establish national parks, game reserves and sanctuaries.

National parks: are conservation areas for wild animals in which legal hunting is allowed, with some restrictions.

Sanctuaries: are wild – animal conservation areas where hunting is strictly. Example = Afar Gewane.

Game Reserves: are wild – animal conservation areas where tourists are allowed to practice licensed hunting.

- Wild animals protection through national legislation.
- Monitor and administer existing conservation areas properly.
- Establishing strong laws that prohibit illegal hunting.
- Educate the public about environment protection.
- Protect habitats.
- Educate and encourage local communities to protect their animals' habitats and resources.

Uses of wild life in Ethiopia

Wild animals are used for

- Educational value
- Aesthetic value/ recreation and refreshment
- Source of revenue via tourism
- Desire for controlled hunting; and
- Potential for domestication

Illustrative Questions

1. Which one of the following indicates a correct match between the endemic wild animals of Ethiopia and their specific geographic location in the country? (EUEE – 2000/2008)
   A. Key kebero - Mago park
   B. Walia Ibex – Ntchsar park
   C. Wild Ass – Semen mountains
   D. Menlik's Dukula – Bale highlands

Explanation: The correct match between the endemic wild animals in Ethiopia and their specific geographic location in the country is Menlik's Dukula – Bale

Answer: D

102. Where do you expect to find the largest population of the colobus monkeys (guerezas) in Ethiopia? (EHEECE – 2005/2013)
A. In green areas of urban centers
B. In the tropical rainforest areas of the country
C. In the mountain grasslands of northern Ethiopia
D. In the steppe and semi – desert vegetation areas

Explanation: The largest population of the colobus monkeys (guerezas) in Ethiopia found in the tropical rainforest areas of the country.

Answer: B

103. Which of the following alternatives show a correct match between wild animals and their geographical location in Ethiopia?
A. Giraffes – semen mountains
B. Walia Ibex – awash park
C. Wild ass – Afar plain
D. Menelike bush buck Dukulla – Nachsar park

Explanation: Among the given alternatives the correct match is wild ass found in the afar plain.

Answer: C

104. Endemic mammals in Ethiopia (EHEECE – 2001/2009)
A. Exist in very small numbers at risk of extinction
B. Occur in may parts of the country in great numbers.
C. Are found mainly in the lowland areas
D. Are confined to zoos established in big cities like Addis Ababa

Explanation: Endemic mammals in Ethiopia exist in very small numbers at risk of extinction.

Answer: A

105. The principal cause of the increasing reduction of biodiversity in Ethiopia over the past century is (EHEECE – 2002/2010)
A. Introduction of exotic species from other continents
B. Land pollution through chemical industries
C. Destruction of forest to get fire wood, construction materials or land for cultivation
D. Acid rain resulting from emission of harmful gases

Explanation: The principal causes of the increasing reduction of biodiversity in Ethiopia over the past century was the distraction of forest to get firewood, construction materials or lands for cultivation.

Answer: C

106. Most of the national parks and game reserves of Ethiopia are found in the (2002/2010)
   A. Northern half of the country
   B. Southern half of the country
   C. Southern and eastern parts of the country
   D. Western and northern parts of the country

Explanation: Most of the national parks and Games reserves of Ethiopia are found in the southern half of the country especially in the southern nations, nationalities and peoples regional states and oromiya regions.

Answer: B

107. Which one of the following strategies will not help to conserve wildlife resources? (2003/2011)
   A. The protection of natural vegetation
   B. Establishment of national parks and game reserves
   C. Controlling illegal hunting
   D. Expansion of crop cultivation

Explanation: The strategies help to conserve wildlife resources includes the protection of natural vegetation, establishment of natural parks and same reserves and controlling illegal hunting.

Answer: D

108. Which of the following endemic mammals of Ethiopia is confined to the eastern lowlands of the country? (UEE-2004 2012)
   A. Wild Ass
   B. Gelada Baboon
   C. Mountain Nyala
   D. Menelik's Bushbuck

Explanation: In Ethiopia rare (endemic) mammals like wild Ass (Yedur Ahiya) found in the Afar and south east lowlands.

Answer: A

7.6. Soils of Ethiopia

7.6.1. Formation of Soils in Ethiopia

- It is the uppermost loose or unconsolidated material overlying the earth's crustal rocks.
- Its major components are water, air, organic and inorganic minerals.
- It is a dynamic, natural and complex substance which can support animals and plants.
- A good soil for growing most plants should have about 45% minerals, 5% organic matter, 25% air and 25% water.
- Soil formation is known as pedogenesis.
- The formation and properties of soils are the results of parent material, climate, organisms, topography and time.

7.6.2. Types of soil in Ethiopia

1. Nitosols (Red – basaltic soils)

- They are dominant in the western highlands of wellega, kefa, Illubabor, the southern highlands of sidamo, central highlands of shoa and eastern highlands of Hararghe
- They are found in areas of heavy rainfall.
- They are leached soils.
- They are the most intensively cultivated soils. Because they are friable, have high moisture storage capacity and stable structure.
- They cover about 12 percent of the country.
- They are highly leached and lack soluble minerals like sodium, calcium, etc., but they are rich in iron and aluminum.
- They are potentially good for farming and other agricultural practices since they are friable, and have astable structure
- They are the best soils for coffee, inset and cereals.

2. Vertisols (Black Basaltic soils)

- They are extensively in many parts of north western central and southeastern highlands of Ethiopia where there are pronounced wet and dry seasons.
- Cover about 10 percent of the total land of Ethiopia

- They have high clay content.
- They are sticky and difficult for farming.
- They are left for grafing where the drainage in poor.

3. Acrisols
- They are basically found in association with coarse Nitosols in south western humid areas of Ethiopia.
- They cover about 4.5 percent of the country.
- They are characterized by low productivity capacity due to low possession of nutrients.
- They are extremely leached soils.
- They are more acidic than the Nitosols.

4. Cambisoils
- They are the results of most recent lava and quaternary period sediments.
- They are young and shallow soils with coarse texture.
- They are mostly found on the rugged and slopy terrain of the country
- Cover about 11.6% of the country and used for growing wheat and barely.

5. Regosols
- Are young and shallow soils with course texture.
- They have low agricultural value.
- They are found in the Danakil and eastern ogaden region.
- They cover about 4 percent of the country.
- They occur on steep terraced or eroded slopes along the north eastern escarpment and at the eastern end of the chercher highlands where that grows.

6. Xerosols:
- They are young and shallow soils found in arid and semi – arid regions, with a weekly developed profile.
- They are extensively found in the north eastern escarpment, northwestern and south eastern lowlands, et. ogaden.
- Are characterized by high salt content and humus deficiencies.
- Have little significance for agriculture except places where they could be irrigated.

- They cover about 8 % percent of the country

7. **Vermosols**
   - They are found in desert and semi desert areas
   - They are salty, acidic and have a weakly developed profile
   - They are not suitable for cultivation even if they are irrigated

8. **Livisols**
   - are found in association with nitosols, especially in the lake Tana plains, part of north central and eastern highlands as well as in the southern lowlands.
   - They have good chemical nutrients.
   - They develop in areas where there are clearly marked dry and wet seasons and leaching is not very high.
   - They are not as suitable as vertisol for farming.
   - They are intensively cultivated in areas where they are not water logged.
   - They cover about 6 percent of the country.

9. **Lithosols**
   - They are young soils developed on steep slope and areas of very low precipitation.
   - The north – eastern and south – eastern lowlands are partly covered with lithosols.
   - They are poor maturity and their location on steep slopes.

10. **Fluvisols**
    - They are found in the lower river valleys of Awash, Wabishebelle, Ghenale, Omo, Baro, Abay and in the Rift valley lakes.
    - They cover about 10 percent of the country's total areas and are found on the plains.
    - Are soils that rivers have transported from highlands to lowlands.
    - They have very good agricultural potential.

7.6.3. Soil problems and conservation in Ethiopia, soil problems in Ethiopia

- Soil erosion is the most serious problem of land resource in Ethiopia

Soil Erosion by Running Water

- Too much soils is carried away by the rivers to the neighboring countries. For example, River Abbay carries 3000 4000 million cubic metric tons of soils annually. Generally the highlands experience a loss of about 2000 tons of soil per/km^2/ year.

Factors that Accelerate Soil Erosion in Ethiopia

The main causes for such heavy loss of soil are

- **Natural/ physical factors**: the rugged topography and torrential summer rains.
- **Human made factors**: includes deforestation, Bad cultivation practices (over cropping, over cultivation, slash and burn), keeping too many livestock (over grazing), unscientific tilling and using in adequately maintained tracks and trails.

Conservation Measures to Combat Soil Erosion

Possible conservation measures that have been recommended for minimizing soil loss by erosion in Ethiopia.

- Reforestation
- Afforestation
- Terracing mountain hills
- Avoidance of tiling land not suitable for cultivation
- Controlling grazing/ limiting grazing to carrying capacity
- Shelter – belt and wind break in the arid and semi arid areas.
- Extended irrigation schemes along major rivers and their tributaries
- Constructing check dams.
- Controlled growth of population
- Intercropping
- Developing improved grass types that can feed more livestock.
- Proper construction of tracks and routes so that drainage could not wash them easily.

Illustrative Questions

109. A region in Ethiopia which is known for its exemplary use of traditional soil and water conservation schemes is (EUEE – 2001/2009)

A. Kibre Mengist
B. Kulito
C. Konso
D. kobbo

Explanation: In Ethiopia the Konso people is an exemplary use of traditional soil and water conservation schemes.

Answer: C

110. Which of the following soils types of Ethiopia is most suitable for the production of coffee. cereals and inset (EUEE – 2002/2010)

A. Xersols
B. Nitosols
C. Lithosols
D. Yermosols

Explanation: The type of soil in Ethiopia which is the most suitable for the production of coffee, cereals and inset is Lithosols or Red Basaltic soil.

Answer:

111. Which of the following conservation practices is most effective to minimize soil loss and ensure a permanent supply of water or degraded mountain sides of Ethiopia?

A. Controlled grzting
B. Terracing on mountain sides
C. Building more national parks and game reserves
D. Reforestation

Explanation: Reforestation is the most effective to minimize soil loss and ensure a perennial supply of water on degraded mountain – soils of Ethiopia.

Answer: D

112. Which of the following of Ethiopia is characterized by deep profiles and high moisture storage capacity?

A. Xerosols
B. Nitosols
C. Yermosols
D. Regosols

Explanation: Nitosols soil of Ethiopia characterized by deep profiles and high moisture storage capacity.

Answer: B

113. Which of the following types of traditional soil fertility improvement methods do you recommend for a farm community that has plenty of land and a very small herd of livestock?

A. Fallowing
B. Croprotation
C. Application of manures
D. inter cropping

Answer: C

Explanation: Application of manures is the traditional soil fertility improvement methods that recommended for a farm community that has plenty of land and a very small herd of livestock.

114. Which one of the following soils are salty, acidic desert semi desert in character and are unsuitable for cultivation even if they are irrigated?

A. Vertisols
B. Yermosols
C. xerosols
D. Nitosols

Answer: C

Explanation: xerosols soils are salty, acidic desert semi desert in characteristics and are unsuitable for cultivation even if they are irrigated.

115. Which of the following is NOT among the main reasons behind high rates of soil erosion in Ethiopia? (UEE – 2004/2012)

A. High intensity of rainfall
B. Overgrazing by wild animals
C. Overgrazing by domestic animals
D. Expansion of agriculture and settlements

Answer: C

Explanation: Among the main reasons overgrazing by wild animals is not a cause for high rates of soil erosion in Ethiopia

116. Which one of the following soil types is covering large areas of Ethiopia and mostly associated with densely populated areas?

A. Regosols
B. Nithosols
C. lithosols
D. Fluvisols

Answer: B

Explanation: Nithosols soils is covering large areas of Ethiopia and mostly associated with densely populated areas.

117. Which of the following attributes of fluvosols best explains why they are among the little used soils in Ethiopia?

Answer: B

A. They are young soils
B. They occur in semi-arid and thinly populated areas
C. They usually occur in the lower valleys of major rivers
D. They are associated with alluvial and lacustrine deposits

Explanation: Because, they usually occur in the lower valleys of major rivers.

Answer: C

118. ___ is a soils conservation method that involve leaving a cultivated land idle for a certain period of time. (EUEE 2005/2013)

A. Manuring
B. Mulching
C. fallowing
D. gulley rehabilitation

Explanation: Fallowing is a soil conservation method that involves leaving a cultivated land idle for certain period of time.

Answer: C

119. In what main way does afforestation help to control soil erosion?

A. By regulating the flow of water
B. By adding humus to the soil
C. By adding humus to the soil
D. By reducing the soil trampling effects of herds of hooted animals

Explanation: Afforestation help to control soil erosion by regulating the flow of water

Answer: A

120. Which of the following **NOT** strongly related to the geographical location of Ethiopia?

A. The fact the Ethiopia is a member of IGAD
B. The socio – cultural behavior of the people of Ethiopia
C. The fact that Ethiopia is a member of the UN
D. The time zone to which Ethiopia belongs

Explanation:

Geographical location of Ethiopia is strongly related with

- Ethiopia's Time Zone i.e. +, 3 GMT
- People socio – culture behavior
- Ethiopia being a member of IGAD/ Inter Governmental Authority on. Development

Answer: C

121. In which part of Ethiopia can you find the oldest layers of the upper sandstone?
   A. north western parts of the country
   B. southern most tips of country
   C. south – eastern parts of the country
   D. central parts of the country

Explanation:
Upper sandstone is the **oldest and thickest** in the north – western part of Ethiopia, because of **Regression and Deposition**

Answer: A

122. Which of the following is an example of an inland drainage system?
   A. The Abay Basin
   B. The Ghenale Basin
   C. The Rift Valley Lakes Region
   D. The Baro River Valley

Explanation:
The drainage system of Ethiopia are:
- The western/ Mediteranean sea, Abbay, Tekezze, Baro
- The south eastern/ Indian ocean, Ghenale
- The Inland/ Rift – Valley lakes region

Answer: C

123. Which one of the following lakes has shown the greatest tendency of drying up over the past two to three decades?
   A. Lake chamo
   B. Lake Asdhengie
   C. Lake Kuriftu
   D. Lake Haromaya

Explanation:
Due to unwise use of water resources **Lake Haromaya** exposed for drying up

Answer: D

124. Which of the following law lands has the wettest climate?
   A. The Denakil Plains
   B. The Chew Bahir Rift Valley
   C. The Baro Lowlands
   D. The Lower Awash

Explanation:
The Baro – Akobo lowlands are the **wettest** lowlands because of their **wind ward site of** equatorial westerlies resulted to get rain in most of the months

Answer: C

125. If all places with elevations of 1000 or more meters above means sea level are defined as highlands, which of the following is a lowland town?

A. Asosa
B. Gode
C. Jigjiga
D. Ziway

Explanation:

Important towns of western lowlands are:

- Gambella
- Metema
- Assosa

Answer: A

126. Which of the following statements is **NOT** correct about the characteristics of the main rivers of Ethiopia?

A. They have mostly stable year round flows
B. They have narrow and winding courses
C. They tend to flow through deep gorges or canyon
D. They have many rapids and waterfalls

Explanation:

Ethiopian rivers are characterized by:

- Steep profile
- form rapid and water falls
- Seasonal Fluctuation in water volume.

Answer: A

127. Which of the following parts of Ethiopia has the greatest known potential for the production of geothermal electricity?

A. The North Central Massifs
B. The South – Western Lowlands
C. The Arsi – Bale Massifs
D. The Ethiopian Rift Valley

Explanation:

Economic potential of the Ethiopian Rift – Valley include

- Geothermal energy
- Irrigable agriculture
- Extraction of salt
- Archeological site

Answer: D

Explanation:
Evidences for unstable physiographic division of the Ethiopian rift Valley
- Intense volcanic activity
- Minor faulting
- Thermal springs and Fumaroles

Answer: C

129. Which of the following forms the bedrock of the most intensively cultivated soils of Ethiopia?
A. The Trap series rocks
B. Quaternary lava
C. Mesozoic marine sediments
D. The basement complex

Explanation:
The soil of Ethiopia were basically derived from the rocks of **Basement complex, Mesozoic cold sediments** and rocks of **Trapean and Aden volcanic series laves** with quaternary sediments

Answer: D

130. Which of the following statements about the Wabishebelle River is **NOT** correct?
A Its waters flow largely through semi – arid and arid lands
B. It is the longest river in the country
C. It forms a large delta when it enters the Indian Ocean
D. It has the largest catchment area in the country

Explanation:
Wabishebelle
- The longest river in Ethiopia
- Has large catchment area
- Flow through arid and semi – arid lands
- Does not reach the Indian ocean

Answer: C

131. Which of the following non – metallic minerals of Ethiopia used in the production of cement?
A. Limestone
B. Clay
C. Soda Ash
D. Silica

Explanation:
As tree of sedimentary rock – made from the remains of animal shell provide as source in cement production

Answer: A

132. Koka Dam is to Awash river as Melka Wakena Dam is to
River

A. Omo
B. Ghenale
C. Wabeshebelle
D. Dawa

Explanation:
River Ghibe - **Gilgel Gibbe 1, 2, 3 H.E.P plants**
River Awash, **Awash 1,2,3 H.E.P Plants**
River Fincha - Abbay tributary - **Fincha H.E.P plant**
River Ghenale: **Melka wakena**

Answer: B

133. **Since ancient times, Ethiopians have been extracting gold mainly from?**

A. **Volcanic veins in Precambrian rocks**
B. **Alluvial deposits**
C. **Quaternary lava deposits**
D. **Mesozoic marine deposits**

Explanation:
Most of the Ethiopian's extracting Gold mainly from Alluvial deposits

Answer: B

134. **During which of the following groups of months are the lowest mean monthly temperatures of most towns of highland Ethiopia recorded?**

A. November, December and January
B. September, October and November
C. February, March and April
D. June, July and August

Explanation:
Most of the Ethiopian high lands experience **mean- Monthly temperature** as low as 0^0c between November and January

Answer: A

135. What is the basis for classifying seasons into Bega, Belg and Kiremt in Ethiopia?

A. Annual rainfall distribution and weather systems
B. Annual distribution of temperature
C. Altitudinal distribution of temperature
D. Altitudinal distribution of rainfall

Explanation:
classification of Bega, Belf and Kiremt seasons in Ethiopia basis on **Annual rainfall, distribution** of **weather system**

Answer: A

136. Which of the following areas of Ethiopia is located on the leeward side of rain – bearing "Kiremt" winds?
A. The north – western highlands
B. The south – western highlands
C. The Dalol Depression
D. The Gambela lowlands

Explanation:
North – western and North – eastern lowlands of Ethiopia are located on the **leeward site of** "Kiremt" rain bearing winds that is rain shadow

Answer: A

137. Which of the following statements about rainfall variability in Ethiopia is correct?
A. Rainfall variability decreases with decreasing altitude
B. The highlands have lesser rainfall variability than the lowlands
C. Rainfall variability is lowest in areas that have a bimodal rainfall pattern.
D. Rainfall variability increase with increase in the amount of annual rainfall

Explanation:
Rainfall variability (deviation from the expected amount and time decrease **from the south western highlands in all directions**

Answer: B

138. Which of the following river courses of Ethiopia marks part of the international boundary that Ethiopia shares with Southern Sudan?
A. Angereb
B. Gojeb
C. Akobo
D. Beles

Explanation:
Akobo as major tributary of Baro serves international boundary that Ethiopia shares with southern sisian

Answer: D

139. Which of the following tree species of Ethiopia is typically found at higher altitudes?
A. Sholla
B. Ghirar
C. Warka
D. Kosso

Explanation:
Ethiopia's highland tree species are • Kerkha, • Tid, • Zigba, • woira and • kosso

Answer: D

140. What are the major controls of weather and climate in Ethiopia?
A. Longitude and pressure belts
B. Latitude, altitude and weather systems
C. Winds and pressure belts
D. The north – south movement of the overhead sun

Explanation:
Major factors influence weather and climate in Ethiopia are
- Latitude • altitude • Distance from the sea
- Weather system like /Equatorial Easter lies and west erlies/

Answer: B

141. **To** which temperature zone do the higher grounds of the Arsi – Bale Massifs belong?
A. Tropical
B. Temperate
C. Subtropical
D. Alpine

Explanation:
WURCH/ Alphine climate exist at latitudes equal to or more than 3,300m ams, found on the highest areas id
- Semen mountain in north Gonder • Arsi – Bale massifs
- South Gondar – Mount Guna • Wollo and Shoa

Answer: D

142. Which of the following statement is correct about general distribution of temperature and rainfall in Ethiopia?
A. Both temperature and rainfall increase with decreasing altitude
B. Both temperature and rainfall increase with increasing altitude
C. With decreasing altitude temperature decreases while rainfall increases

D With decreasing altitude temperature increases while rainfall decreases

Explanation:

There is a General decrease in **Temperature** for increase in elevation and **Rainfall** decreases for decreasing altitude

Answer: D

43. Where do you expect to find the largest population of arboreal animals in Ethiopia?

A. In mountain grasslands C. In the tropical rainforest areas

B. In woodland savannahs D. In the semi arid and arid lowlands

Explanation:

Ethiopia's arbonreas/ animals that climb up threes like **Colobus monkey** and **baboons**, are mostly found in the **Tropical rainforest regions** **Answer: C**

44. Which of the following mammals of Ethiopia is presently facing the greatest danger o extinction?

A. Wild ass C. Jackals

B. Baboons D. Monkeys

Explanation:

In relation with expansion of Grazing land '**Gelada**' O**chelad** ' **baboon**. Fund in the Semein highlands currently facing great risk of extinction/ News to disappear

Answer: B

45. Which of the following soils of Ethiopia is most intensively cultivated?

A. Nitosols C. Vertisols

B. Cambisols D. Luvisols

Explanation:

Cambisols Ra[illegible] potentially good for [illegible] cultivated soil type for [illegible]

Answer: A

[illegible] ble for land

Explanation:

Soil Erosion in Ethiopia accelerated by:-

- Deforestation
- Bad cultivation practices
- Keeping to many livestock/ over grazing

Answer: D

147. Which of the following can be a good reason for the limited agricultural use of fluvisols in Ethiopia?
   A. Their low nutrient content
   B. Lack of awareness by farmers
   C. Their occurrence in semi – arid and thinly populated areas
   D. Their occurrence on flat areas

Explanation:

Fluvisols are associated with **river, sea** and **lake deposition** transported from highlands to lowland are found extensively in the lower regions

Answer: A

148. What is the dominant type of natural vegetation found in the southern most parts of Ethiopia?
   A. Semi – dessert steppe
   B. Acacia savannah and thorn bush
   C. Deciduous woodland
   D. High savannah

Explanation:

Short acacia savannah, Thom bush, few rough – grasses are dominant type of southern most part of Ethiopia natural vegetation

Answer: B

149. To which category of forests do the broadleaf rainforests of Ethiopia belong?
   A. Coniferous forests
   B. Highland forests
   C. Equatorial rainforests
   D. Tropical rainforests

Explanation:

Broad leaf ever green forest belongs the **Tropical rainforests.**

Answer: D

150. Which of the following is **NOT** a method used to combat the depletion of natural vegetation?
   A. Controlling forest fires
   B. Increased reliance on biomass energy

C. Reforestation
D. Proper management of existing forests

Explanation:

Approaches of fighting natural vegetation rapid depletion:-

- Reforestation
- proper management
- controlling forest fire
- Afforestation

Answer: B

151. Which of the following parts of Ethiopia is most suitable for the large scale production of tea?

A. The Arsi – Bale highlands
B. The South – western Highlands
C. The upper Awash Valley
D. The Gambella lowlands

Explanation:

The south – western highlands of Ethiopia is best suited – agro – climatic zones for the large scale production of tea.

Answer: B

152. Which of the following crops covers the largest cultivated area in the main growing season of Ethiopia?

A. Sorghum
B. Wheat
C. Barley
D. Maize

Explanation:

Barely covers the largest cultivated area in the main Growing season of Ethiopia.

Answer: C

153. Outcrops of old crystalline basement complex rocks are found in all of the following parts of Ethiopia **EXCEPT** in

A. Eastern Somali
B. Northern Tigray
C. Central Sidama
D. Southern Omo

Explanation:

In Ethiopia out – crops of old cry staline basement complex rocks, due to continuous delutation fount in

- Northern and central Tigray
- Southern Omo
- central sidama
- southern Bale and Borena

Answer: A

154. The Ethiopian Rift Valley is part of the world's greatest rift Valley system which extends from _____.

A. Eritrea to Kenya
B. Israel to Eritrea
C. Syria to Mozambique
D. Djibouti to South Africa

Explanation:

The world great Rift Valley system stretches from Syria in the middle – fast to Mozambique in SE Africa

Answer: C

155. Which of the following areas of Ethiopia is known for its rich potash deposit?

A. The Abay Gorge
B. The Rift Valley ales Region
C. The Danakil Depression
D. The Chew Bahir Rift

Explanation:

The Denakil Depression is rich in potash – deposit

Answer: C

156. The formation of inselbergs in parts of Ethiopia and the Horn is one of the geologic events that took place during the

A. Precambrian Era
B. Paleozoic Era
C. Mesozoic Era
D. Cenozoic Era

Explanation:

The massive denutational activities in the Paleozoic – era resulted in the formation of inselburgs

Answer: B

Review Questions on Unit Seven

Part I. True or False questions

Write "A" if the statement is true and "B" if it is false

A 1. Most of Ethiopia's medieval and later settlements are concentrated in Dega – Zone agro – climatic areas.

B 2. The inland (Rift valley) drainage system is the second largest drainage system.

B 3. Altitude is the angular location of a place or point with reference to the direct rays of the sun.

A 4. In Ethiopia highland rainfall is more dependable than lowland rainfall.

B 5. Slash – and – burn practices is the major conservation measures of natural vegetation.

Part II. Matching Question
Match items in column "A" with that of column "B"

A	B
6. ITCZ G	A. Debre Markos
7. Year – round rainfall Region L	B. North east Trade winds
8. Summer-Rainfall Region A	C. Environmental lapse rate
9. Autumn and spring Rainfall Region N	D. Mitswa
10. Winter Rainfall region D	E. Tropical
11. Merged spring, summer and Autumn Rainfall Region J	F. Leeward side and rain shadow
12. Wurch/kur R	G. Low pressure zone
13. Dega I	H. High pressure zone
14. Woina Dega O	I. Temperate
15. Qolla E	J. Assaita, Awash
16. Bereha K	K. Desert
17. Latitude M	L. Metu
18. Weatehr systems B	M. Angular location of a place
19. Mountain Barriers F	N. Moyalle
20. Altitude C	O. Subtropical
	P. Rainfall
	Q. Temperature
	R. Alphine or Afro – Alpine

Part III. Completion questions
Fill in the blank space with appropriate words or phrases

21. _____ lake is formed when a sheet of lava dams up a shallow surface depression.
22. ___ is a dome shaped plateau that serves as a watershed between the Awash and Abbay River basins.
23. The oldest and longest geological era is called _____
24. The shape of Ethiopia is _____
25. The main geological events of the Paleozoic era were _____
26. ____ are conservation areas for wild animals in which legal hunting is allowed, with same restrictions.
27. The practice of placing too many livestock on a given place of land is _____
28. River Baro become _____ in the Sudan
29. ___ refers to the fabric or surface arrangement of the main rivers and their tributaries.

30. The southern subdivision of the Ethiopia Rift system, the Chew Bahir rift, is also known as

Answer Key to Review Questions on Unit Seven

Part I.

1.A 2.B 3.B 4.A 5.B

Part II

6.G 7.L 8.A 9.N 10.D 11.J 12.R 13.I 14.O 15.E
16.K 17.M 18.B 19.F 20.C

Part III

21. Watershed Lake
22. The shoan plateau
23. Precambrian Era
24. Compact shape
25. Denudation and penplanation
26. National parks
27. Overgrazing
28. sobat
29. Drainage pattern
30. The Omo – Ghibe trough

Population of Ethiopia and the Horn

8.1. Population Theories

- Modern population theory is generally thought to have evolved in the late eighteenth century, in the writings of Malthus.
- Malthus's work focused on population and in the economic and social issues associated with it.
- Various scholars have attempted to explain the relationships that exist between population and factors such as resource, economy. Politics and other social issues.
- The main purpose of population theory is to seek appropriate answers to the following population – related questions.
  - → How many human beings can be the planet feed and the environment sustain?
  - → How does population growth determine the socio – economic issues of countries?
  - → How do biological, economic, social and political factors determine population growth?

8.1.1. Malthusian population theory

- Thomas Robert Malthus (1766 – 1834) was an influential English economist who founded the science of modern demography.
- He warned of a constant tendency for human population growth to exceed food production.
- He agreed that population increase led to increase demand for food less food per person, – Increased Mortality /Death/ - Decreased Fertility/ Birth → Decrease in population Growth.
- Malthus took an exceedingly pessimistic views
  - Passion between sexes is inevitable and universal;

can only increase by a constant amount, in arithmetical progression (1 – 2 -3 -4 -5), but that the human population has tendency to multiply in geometric progression, (1-2-4-816), multiplying itself by a constant amount each time.

- Population would outstrip food supply until catastrophe occurred
- Malthus referred to the catastrophes as positive checks that control fast population growth and in the long term maintain a balance between population and resources.
- In order to avoid the inevitable occurrence of the positive checks, human beings should adopt preventive checks.
- Malthus was totally against family planning methods.
- In general, Malthus believed that the "hot passion or sexual urge" of people could lead to over population, which in turn, would result in poverty and other catastrophes (positive checks). He conclude that "the poor are to be blamed for their own poverty. Neither wages nor providence, nor society is to be blamed."

8.1.2. Anti – Malthusian population theories

- Anti – Malthusians are optimists who argue against Malthus.
- They argued that population increase led to increase demand for food – Improvement in technology – population Growth continues.
- Anti – Malthusian theories are optimistic.
- There are two anti – Malthusian theories – one was developed by Karl Marx, and the other by ester Boserup.

A. Marxian population Theory

- He was one of the leading opponents of Malthus.

- He believed that no population problem under a socialist made of production rather population becomes problem of surplus people only under capitalist modes of production.
- While Malthus focused on individual actions and considered "moral restraint (preventive checks) to be the solution to population problems, Marx focused on the economic structure of society and believed the solution would be found in a new social order.

B. Boserupian Population Theory

- Esther Boserup (1910 – 1999) was a Danish economist of the 20th century. In 1965, she developed a "theory on population and Agriculture".
- She argued against Malthus.
- She argued that technological advancement would ensure that food supply would keep up with population growth.
- She formulated an optimistic theory about the influence of population growth on agricultural development.
- She presented the positive effect of population growth an agricultural transformations in the following ways.
  - → Forest – fallow cultivation
  - → Bush – Fallow cultivation
  - → Short – Fallow cultivation
  - → Annual cropping and
  - → Multiple cropping
- Boserup concluded that population would continue to grow because that increased demand for food and other necessities would lead to new innovations and technological advancement.

Illustrative Questions

1. Which one of the following is a possible conclusion of the Malthusian population theory? (UEE – 2004/2012)
   A. Population increase can motivate technological innovations
   B. Disease and ware should be allowed to avoid overpopulation
   C. Population increase can intensify agriculture to maintain food security

D. Economy grows in geometric progression as population growth arithmetically.

Explanation: Malthusian population theory stated that economy grows in geometric progression as population growth arithmetically.

Answer: D

2. Boserup said that "Necessity is the mother of invention" what does it mean? Explain briefly?

Explanation: Boserup stated that As the size of population increase it results in technological innovations and advancement and also increased demand for food and other necessities would lead to new innovations and technological advancement.

3. How did the Malthus and Marx views differ about rapid population growth?

Explanation: Malthus believed that the "hot passion or sexual urge" of people could lead to over population result in poverty and other catastrophes, on the other hand Marx stated that no population problem under asocialist mode of production rather under capitalist system.

4. Explain briefly the concept or view of malthesia preventive checks?

Explanation: It is moral restraint such as late marriage, avoiding sexual conduct before marriage, etc proposed by Malthus as controls of fast population growth.

5. What is the main purpose of population theory?

Explanation: The main purpose of population theory is to sect appropriate answers such as population size, and environment sustain population growth and its socio economic impact and biological, social and political factors determine population growth.

8.2. Trends of Population Growth and Structure in Ethiopia.

8.2.1. Trends of population growth in Ethiopia

- One difficult condition in the study of population in countries like Ethiopia is the absence of reliable information.

- The most important source of population information (demographic data) that enables us to understand population growth rate and its trends in a country is census.
- Census is defined as the periodic counting of the entire population of a country.
- All available source indicate that the current population of Ethiopia is growing rapidly.
- Ethiopia is the third most populous country in Africa, next to Nigeria and Egypt.
- According to the 2007 census, the population of Ethiopia is 73.8 million.
- The 2008 estimate of the annual growth rate of Ethiopia population is 2.23 percent.
- Due to the result of the government's efforts to regulate fertility rates through the delivery of family planning services the population growth rate changing from 3.1. percent in 1984 and to 2.0 percent in 2007.

Table 8.1. population estimates and growth rates in Ethiopia (1900 - 2007)

Year	1900	1920	1940	1960	1970	1984	1994	2007
Population (million)	11.8	12.9	16.3	23.5	29.5	42.6	53.5	73.8
Growth Rate (%)	0.2	1.0	1.5	2.2	2.8	3.1	2.9	2.0

8.2.2. Age and sex structure of the Ethiopia population

A. Age structure

- It refers to the distribution of population by age groups.
- It is the pattern that results from the distribution of members of a population in to different age categories.
- Two important stastical tools for understanding age structure are; Age groups and population pyramids.

Age Groups:

- The age – structure of the Ethiopia's population was typical of a developing country.

- The most widely used age groups are the five – year age groups 0-4, 5-9, 10-14, 15 – 64 and 65^{+}.
- It is characterized by high level of young and low level of old population.
- Age groups 0 – 14 (young) and 65^{+} old (elderly) age are economically dependent and age group 15 – 64 we call it economically active population.
- The high percentage for the young age group is the result of a high birth rate and natural increase. Whereas low percentage of old age group is the reflection of a high mortality rate which result in low life expectancy.

Population pyramid:

- It is the graphic representation of the age distribution of a given population by sex.
- The population pyramid of Ethiopia has broad base and this implies that very large properties of population belongs to younger age group.
- Ethiopia's old – age (65^{+} years) dependent population was very small – only 3.2% in the 1994 and 2007 census.

Age Dependency Ratio

- It is the relationship between the working or economically active population and the non – working population.
- It is imply that people in the young and old ages are dependent on the working – age population.
- The formula for calculating the age dependency ratio (ADR) is = ADR =

$$\frac{\text{Economical ly dependent popualtion(young+old)}}{\text{Economical ly active popualtion(Adult)}} \times 100$$

- Age dependency percentage in Ethiopia have ratios between 85 and 105.
- For example, the value of 82 for the age dependency ratio of Ethiopia in 2007 shows that for every 100 persons in the working age group, there are about 82 dependents.

Negative implication of heavy youth dependency

- Moving of more women in to child bearing reproductive age.

- The capacity to save is highly constrained at both the household and the national level.
- Diversion of limited national investible resources to social service.
- The demand for the nation's employment opportunities will increase as the youth group reaches working age; this results in a further high levels of un employment.

B. Sex structure

- The sex structure of a population is shown by its sex ratio, which is the ratio of male population to female population.
- Sex ratio is usually expressed as the number of males per 100 females in a population.
- A ratio greater than 100 shows a greater number – called on excess of males than females.
- The formula for calculating a sex ratio is:

$$\text{Sex Ratio} = \frac{\text{Male population}}{\text{Female population}} \times 100$$

Table 8.2. Sex ratios of Ethiopia population, by area

Census Year	Rural	Urban	National
1984	100.9	86.8	99.4
1994	102.6	93.3	101.3
2007	102.5	98.8	101.9

- Ethiopia's sex ratios vary between rural and urban areas.
- Rural ratios are higher than urban.
- Sex ratios in Ethiopia are generally lower for urban areas and higher for rural areas. This is primarily due to large number of females migrating to urban areas.
- Sex ratio in Ethiopia also varies with age.

Illustrative Questions

6. One of the features that charities the Ethiopian population is a shorter doubling period. Which of the following could best explain this feature. (EUE -2000/2008)
   A. Low rate of migration
   B. High rate of migration
   C. high fertility rate
   D. low rate of mortality

Explanation: High fertility (birth rate) is one of the characteristics of the Ethiopian population that contributed for the shorter period of doubling period.

7. Life expectancy refers to (EHEECE – 2001/2009)

Answer: C

A. The number of years one is expected to live at any age.
B. The number of years one is expected to live at birth.
C. The number of years that rural people are expected to live at mid – age.
D. The number of years that urban people are expected to live at mid-age.

Explanation: Life expectancy refers to the number of years one expected to live at any age.

Answer: A

8. Which one of the following statements is correct about the Ethiopian population? (EHEECE – 2002/2010)

A. It is the second largest in sub – Saharan Africa
B. It accounts for five percent of the world population.
C. It accounts for 18% percent of the African population.
D. It has the fastest growth rate in Africa.

Explanation: Ethiopia is the second largest country in population in the sub-sahara African region next to Nigeria.

Answer: A

9. The difference in the size of the Ethiopian population between the 1984 and 1994 censuses is about (EHEECE – 2002/2010)

A. 32million
B. 22 million
C. 12 million
D. 2 million

Explanation: According to central stastical authority the difference in the size of Ethiopia population between the 1984 and 1994 census is about 12 million because in 1984 the population of Ethiopia was about 40,100,000 and in 1994 the total population is about 53,500,000 peoples.

Answer: C

10. The major cause of rapid population growth in Ethiopia is ____ (EHEECE 2003/2011)

A. High rate of natural increase
B. High rate of net in – migration

C. Low infant mortality rate
D. High crude death rate

Explanation: The major cause of rapid population growth in Ethiopia is high rate of natural increase.

11. What is the most possible demographic implication of high proportion of young age population? **Answer: A**
A. Short doubling time
B. Short life expectancy
C. low dependency ratio
D. High dependency ratio

Explanation: In any country high proportion of young age population imply that high dependency ratio.

12. The present rate of population growth in Ethiopia is (EUEE – 2005/2013) **Answer: D**
A. 1.6%
B. 2.6%
C. 3.6%
D. 4.6%

Explanation: The present rate of population growth in Ethiopia is 2.6 percent

13. The population pyramid for Ethiopia shows that there is a (EUEE – 2005/2013) **Answer: B**
A. Low proportion of old age population
B. Moderate dependency ratio
C. Low proportion of young population
D. Two – thirds of the population fall under working age population.

Explanation: The population pyramid of Ethiopian population indicates a high dependency ratio, a very low proportion old people, more young population and low economically active population.

14. A youth dependency ratio of 82 shows that for every (EUEE - 2005/2013) **Answer: A**
A. 100 persons in the working age there are 82 persons in the young age
B. 82 persons in the working age there are 100 persons in the young age

C. 18 persons in the working age there are 82 persons in the young age.
D. 82 persons in the working age there are 12 persons in the young age

Explanation: A youth dependency ratio of 82 shows that for every 100 persons in the working age there are 82 persons in the young age.

Answer: A

8.3. The Spatial Distribution of Population in Ethiopia

- Geographically, the spatial distribution of population is a concept related to the areal spread of people on the earth.
- Population distribution refers to the way how population spreads out over a given area.
- Population density in the average number of people per `square kilometer in a given area.
- As of 2000, Ethiopia had one percent of the world's total population and eight percent of Africans total population.
- UN projections indicate that Ethiopia will have about 2.6% and 9.1%share from the worlds and Africa's totals, respectively by the year 2030.
- In Ethiopia, the population is UN evenly distributed for ratios hygiene and human related reasons.

Densely and sparsely populated Areas Ethiopia

- Spatial distribution of population by region very significant
- The largest population of population is found in the oromiya region (26.093 933.)
- The smallest population of person is in the Harari region (183,415).
- The Amhara region (17,221,976)have the second largest population in the country.
- The SNNP region (14,929,548) have the third largest population in the country.
- According to the 2007 Census, at the national level, 83,9% of the total population lives in rural places, while the urban population accounts for only 16.1%.

- In terms of density of population, Addis Ababa, Dire Dawa, and Hareri regions have exceptionally different and much higher population densities than the rest of the nation, because they are Urban-dominated regions.
- The most densely populated region in Ethiopia is the southern nations, nationalities and peoples region (SNNP)(133 persons 1km2),followed by the Amhara (108.2 persons/Km^2) and Tigray (86.2 persons/Km^2) regions.
- The most sparsely populate region in Ethiopia is the Somalia, Afar and Gambela region.
- Some of the zones with very high population densities are Ghedeio (424 persons/Km^2) Sidama (299 persons/Km^2) Kembata Alaba-Tembaro (299 persons/Km^2) and Hadiya (264 persons/Km^2)
- There are zones with population densities of less than 10 persons per square kilometer are Kamashi (5.7 persons $1km^2$) and Metekel (7.7 persons/km^2)

Table 8.3. Regional Distribution of Population in Ethiopia (2007)

Region	**Total Population**	**Total Area (km^2)**
Tigray	4,316,988	50,078,64
Afar	1,390,273	96,707
Amhara	17,221,976	159,201.66
Oromiya	26,993,933	353,006.81
Somalia	4,445,219	209,252
Benishangul	784,345	49,281.46
SNNP	14,929,548	112,343.19
Gambela	307,096	25,802.01
Harari	183,415	311.25
Addis Ababa	2,739,551	530.14
Dire Dawa	341,834	1,213.20

Source = CSA, stastical Report of the 2007 Census

8.4. Factors Affecting Population Distribution in Ethiopia

The un even (un equal) population spread on the earth's surface result from two different forms of factors which often operate jointly i.e. physical and human factors.

8.4.1. Physical Factors

The physical factors affecting population distribution include

- Climate (mainly rainfall and temperature)
- Soil fertility
- Natural water supply
- Vegetation cover
- The existence of epidemic disease/ environmental diseases.

- Altitude is the most significant physical factor that affects the distribution of population in Ethiopia.
- Due to the above fact, 77.5 percent of the population of Ethiopia lives in areas with altitudes above 1800 meters, and that these areas constitute only 37.6 percent of the total area of the country.
- Areas below 1800 meters, which constitute about 63 percent of the total area of Ethiopia, supports only 22.5 percent of the total of the country.

8.4.2. Human Factors

Human factors refer to human related causes influencing population distribution in Ethiopia. These factors are divided in to two groups. They include:

i) **Types of economic activity**

- Types of economic activities practiced in different parts of the country affect population distribution. For instance in areas where nomadism and shifting cultivation are practiced, there will be low population density, on the other hand, highlands which grow 'enset' and 'coffee' have higher population density than areas known for cereal crop production.

ii) **Historical pattern of population Movement**

- The pattern of movements of the two major groups of people (semetic and cushetic has directly affected the distribution of population in the country)
- The concentration of population was connected with defence and security purpose.

Illustrative Questions

15. In which area of Ethiopia do your expect to find the lowest crude density of population? (UEE – 2000/2008)
    A. Sidama
    B. Desse Zuria
    C. Elkere
    D. Debre Tabor

population

A. Enset
B. Sorghum
C. Pepper
D. Millet

Explanation: Enset is the crop associated with higher density population in Ethiopia in southern part.

Answer: A

17. A land carrying capacity refers to (EHEECE – 2001/2009)
    A. The minimum population that can live on a given place of land.
    B. The capacity of inhabitants in a given area to maintain land productivity.
    C. The capacity of a fertile land over and above a less fertile one.
    D. The maximum number of people that can live on land without damaging the resource base.

Explanation: A land carrying capacity refers to the maximum number of people that can live on a land without damaging the resource base.

Answer: D

18. The three most populations regional states of Ethiopia put in their rank order are (EHEECE 2002/2010)
    A. Oromia, Amhara and Tigray
    B. Oromia, SNNPR and Tigray
    C. Oromia, SNNPR and Amhara
    D. Oromia, Amhara and SNNPR

Explanation: Oromiya, Amhara, and the southern Nations, Nationalities and peoples regions are the most populous regional states of Ethiopia first to third ranks. **Answer: D**

19. The regional state (kilil) in Ethiopia that has a population density lower than the national average density is
    A. SNNPR
    B. Tigray
    C. Benishangul - Gumuz
    D. Harari

its total area

B. The eastern half of Ethiopia is as densely populated as its western half

C. There is an even distribution of population in most parts of southern Ethiopia

D. The Hararghe plateau is as sparsely settled as the south eastern lowlands.

Explanation: The distribution of the population of Ethiopia shows that about 80% of Ethiopia's population lives on about 50% of its total area.

Answer: A

8.5. Settlement Patterns of Ethiopian Population

- The physical factors influencing settlement types are primarily related to the magnitude of rainfall and availability of water.
- The most important human factors that influence settlement types include the level of development, the kind of land ownership and the need for community defense.
- Ethiopian settlements are broadly categorized as rural and urban. The primary bases for this dichotomy are the dominant economic activities and the degree of population density.

8.5.1. Rural Settlements

- It simply to describe any place as country side or more vaguely as non-urban place.
- It is a settlement whose entire settlers depend on agriculture for their livelihood.
- Ethiopia is characterized by predominantly rural settlements (85 percent).

- Settlements in the rural areas of Ethiopia can be broadly grouped in to two permanent and temporary/ mobile settlements.

Permanent settlements

- They are settlements which do not show frequent changes in their location.
- In Ethiopia highland settlements which depend on crop cultivation are areas of permanent settlement.
- The permanent rural settlements of Ethiopia can also be broadly divided in to two the scattered (diffused or dispersed) settlements and the grouped (nucleated) settlements.
- In areas of scattered settlements, homesteads are separated by relatively long distances.
- The grouped settlements are characterized by a large number of homesteads/ households concentrated in one place.

Temporary settlements

- They are simply mobile settlements in habited by nomadic people of low land Ethiopia. This because the harsh climatic condition forces people living in these areas to seasonally move from one area to another with their herds.
- It is difficult to provide social services to mobile settlements.

8.5.2. Urban settlements

- It include residence in cities and towns.
- The process of a development of urban settlements is known as urbanization.
- Urbanization in Ethiopia is not influenced by European colonialism. It is an unplanned, natural phenomenon of the late 19^{th} and early 20^{th} centuries.
- In Ethiopian the criteria used to distinguish urban from rural area are population size of 2000 and above, the presence of chartered municipality and two – thirds of the population in the settlement are engaged in non – agricultural activities.
- The interconnection of the different parts of the country by all weather roads, which radiate form Addis Ababa and the five – year Italian occupation are the major factors contributed to the development of urban centers in Ethiopia.

and Hararghe plateaus.

- The main causes for a high concentration are
  - → Concentration of industries
  - → Concentration of service – giving constitution such as transport, schools, health, water supply, and electricity services.
- The annual growth rate of urban centers is 4%

Illustrative Questions

21. Which of the following regions has the largest proportion of the urban population of Ethiopia? (UEE – 2002/2010)
    A. The Amhara National Regional state
    B. The Oromiya National Regional state
    C. The Tigray National Regional state
    D. The southern Nations, Nationalities and people Region.

Explanation: The largest proportion of the urban population in Ethiopia has found in the Tigray National Regional state compared to others.

Answer: C

22. Which region of Ethiopia is characterized by the largest number of urban settlement in Ethiopia? (UEE – 2004/2012)
    A. Tigray
    B. Oromiya
    C. Amhara
    D. SNNPR

Explanation: Oromiya region of Ethiopia is characterized by the largest number of urban settlement.

Answer: B

23. Define the term urban and urbanization?

Explanation: Urban refers to cities and towns where as urbanization means a processes of population concentration in areas specialized in non – agricultural activities.

24. What are the unique features of rural settlements?

Explanation: Like other settlement system rural settlements are settlement type which are non – urban place and settlers depend on agriculture for their livelihood.

25. List down the criteria that distinguish urban from rural area;

https://t.me/ethiolibrary1

- It refer
- Population change in any country is affected by fertility (births), Mortality (deaths), and migration (movement in to or out of the nation) which change the size and structure of the population of an area.
- Population change in any country is the result of
  - → The difference between fertility (births) and mortality (deaths) which is called natural increase.
  - → The balance between in migration (in – migration) and emigration (out – migration), which is known as act migration.
- Ethiopia is characterized by a high rate of population growth.

8.6.1. Fertility in Ethiopia

- It determine the size and structure of the population of a country.
- It refers to the occurrence of birth in a given country or region.
- It measures the rate by which a population adds to itself through live birth.

Common measures of fertility

- **Crude Birth Rate (CBR):** this is the total number of births occurring in a given year, per 1000 population.
- **General Fertility Rate (GFR):** this one is the number of birth s occurring in a given year per 1000 women in the reproductive age (i.e.) women aged 15 - 49
- **Age Specific Fertility Rate (ASFR):** This is the number of births that occur in a given year per women in the reproductive ages, presented in five – year age groups.
- **Total Fertility Rate (TFR):** This is the number of children a woman may have produced by the end of her reproductive period, given the current ASFR.

measured by TFR (5.4), GHR (179) and CBR (35.7), is the highest in the world.

- Fertility is considerably higher in the rural areas than the urban areas. For instance, EDHS 2005 showed that the TFR in rural areas was 6.0, which was almost two and half times higher than the 2.4 TFR in urban areas.
- The total fertility rate (TFR) for Ethiopia is 5.4 children per woman
- In regional level Addis Ababa has allow TFR of 1.4 and Oromiya has a high 76.2.

Factors that aggravate high fertility rate

- Low levels of family planning practices, due to lack of awareness and religious beliefs;
- Early marriage
- Altitude of population towards children, considering them as assets
- Low social status of woman
- High infant and child mortality (death) ratio.

8.6.2. Mortality in Ethiopia

- It is the occurrence of deaths in a given population.
- The main measure of mortality are.
  - **Crude Death Rate (CDR)**: this is the total number of deaths occurring in a given year, per 1000 people.
  - **Infant mortality Rate (IMR)**: This is the number of deaths in a year among infants under one year of age, per 1000 live births.
  - **Child mortality Rate (CMR)**: this is the number of deaths in a year among children between one and five years of age, per 1000 children between the same ages.

- **Under five Mortality Rate**: this is the number of deaths in a year among infants and children between birth and five years of age, per 1000 live births.
- **Maternal Mortality Rate (MMR)**: this is the annual number of maternal deaths occurring during pregnancy, child birth, or within two months after the birth or termination of pregnancy, per 1000 women between 15 and 49 years of age (reproductive age),
- **Adult Mortality Rate (AMR)**: this is the number o deaths in a year among Adults between 15 and 49 years of age, per 1000 adults in the same age group.
- **Age – specific Death Rate (ASDR)**: This is the number of deaths that occur in a given year per 1000 of the population in five – year age group.
- **Life Expectancy:** this is the average number of years a newborn baby is expected to live if he/she is exposed throughout life to the prevailing pattern of age – specific death rates.

Levels and Differentials of Mortality in Ethiopia

- Ethiopia is still characterized by a high level of mortality
- The mortality level of Ethiopia's population, as measured by IMR (80), CMR(56), MMR (1.34) and AMR (5.99), is the highest in the world.
- IMR and CMR rates are used to indicate a countries' health status and socio economic development status.
- The IMR and CMR in Ethiopia in 2005 were 80 and 56 respectively
- The IMR of Ethiopia's population much higher than average in the less developed countries (57/1000).
- Mortality is urban areas is consistently lower than in rural areas. In urban areas was 66 deaths per 1000 live births, compared to 81 deaths per 1000 live births in rural areas.
- In regional level deferential in infant, child and under – five mortality levels in Ethiopia. The lowest level is that of Addis Ababa (72/1000),on the other hand, the highest levels are in

- The average [illegible] mortality rate for the period 1994 – 2005 was 1.34.

Causes of high mortality rate

- Poverty
- Low standard of living
- Civil war
- Recurrent drought
- Famine and
- Prevalence of HIV/AIDS.

Migration

- It is the movement of people across a specified boundary for the purpose of establishing a new made of life. i.e. internal and international .
- In Ethiopia migration has very little effect in the process of population change.
- Internal migration of Ethiopia has the main effort of redistributing the population size between rural and urban areas. This rural – urban migration causes urban growth in Ethiopia. But international migration has limited impact on the processes of population change in Ethiopia.

Illustrative Questions

26. Which of the following NOT Among the major factors that contribute to the prevalence of high fertility rates in Ethiopia? (EUEE- 2003/2011)
    A. Early marriage
    B. The low status of women in the Ethiopian society
    C. Increased availability of health services.
    D. Increased availability of family planning services.

Explanation: Increased availability of family planning services are the major controlling or reducing method of fertility rates in Ethiopia.

Answer: D

27. Which of the following is most strongly associated with the fertility level of the population of any country? (UEE – 2004/2012)
    A. The geographical location of the country
    B. The degree of the natural resources endowment of the country
    C. The stage of the economic development attained by the country.
    D. The degree of political stability enjoyed by the population.

Explanation: In any country the stage or level of economic development strongly associated with fertility because it determine women status and level of education.

Answer: C

28. Which of the following can result in the decline of the Total Fertility Rate (TFR) of the population of a country like Ethiopia. (UEE – 2004/2012)
    A. Increase in age at marriage
    B. Declining child morality rate
    C. Declining infant mortality rate
    D. Increase in maternal mortality rate

Explanation: Because of increase in age at marriage, there is decline of the total Fertility Rate (TFR) of the population of Ethiopia.

Answer: A

29. What are the major factors that affect population change in any country?

Explanation: Population change in any country is affected by fertility (births), mortality (deaths), and migration (movement in to or out of the nation).

30. Write measures of fertility?

Explanation: Measures of fertility includes crude Birth Rate, General Fertility Rate, Age specific Fertility Rate and Total Fertility Rate.

31. What is the difference between infant and child Mortality?

Explanation: Child mortality is the number of death in a year among children between one and five years of age where as

births.

32. Why in rural area has highest rates of mortality than urban area?

Explanation: This is due to shortage of sufficient clinic, hospitals and physicians.

8.7. Impacts of Rapid Population Growth in Ethiopia

- The population growth rate is grows rapidly, the demand for food supply, health facilities, energy, water and transport increases.
- The negative results of rapid population growth have caused many environmental and socio – economic problems.
- Population growth causes serious environmental degradation in the area where it occurs, including deforestation, population, soil erosion, depletion of resources, etc.

8.7.1. Deforestation

- It refers to the removal of forest cover of an area without adequate replacement.
- It is the process of the indiscriminate destruction of the natural vegetation cover of a forest area.
- Due to rapid population growth and the increasing population's the forest cover resource of Ethiopia has been declining significantly.
- Increasing demand for agricultural and settlement lands is a major cause of forest destruction in the country.
- In Ethiopia, there is a direct correlation between population density and deforestation.
- Rapid population growth leads to deforestation mainly because of peoples increasing needs for wood, agricultural land, settlement land and grazing land.
- Its direct consequences include accelerates soil erosion, destroys biodiversity, affects rainfall by decreasing evapotranspiration, shortages of wood supply and affects the natural beauty of the affected areas.

8.7.2. Pollution

- It refers to any undesirable change in natural conditions of water, air and other components of the natural environment that has negative effects on the health and activities of human beings and other living creatures.

in household, industries, farm lands, means of transportation, and so on mainly caused by human activities

- In Ethiopia, population is a major problem in over populated urban centers.
- In Ethiopia, rapid population growth leads to environmental pollution by increasing emission of the amounts of pollutants such as.
  - → Sewage, solid wastes, and poliutant gases generated by households.
  - → Pollutant gases, liquids, and solid chemicals generated by expanded industries.
  - → Pollutant gases generated by the increasing number of automobiles
  - → Agricultural pollutants, such as fertilizers, pesticides, animal wastes, etc.

Illustrative Questions

33. Rapid rural population growth in Ethiopia has little contribution to

A. Deforestation
B. Desertification
C. pollution
D. soil degradation

Explanation: As far as the fact that pollution related to development of urban growth so Rapid rural population has little contribution to pollution.

Answer: C

34. Explain briefly the impact of rapid population growth on the development processes?

Explanation: Rapid population growth result is lack of food, growth demand for education, low access to modern health service, shortage of housing and lack of job opportunities and effects on resources.

35. Describe the impact of rapid population growth in Ethiopia on Education?

Explanation: Due to the high fertility and young age structure of the population the school age population has been growing

rapidly. This indicates that the alarming population growth rate highly affects the availability of better education in the country.

8.8. Population Policy of Ethiopia

- It is a policy that is formulated and implemented by a government in order to plan and control population growth, based on the economic, social, cultural, political, and demographic conditions of the country.
- It is needed mainly to address population related problems in a country.
- The population policies of countries can be broadly categorized in to two groups as = anti – natalist and pro – natalist policies.
  - → Anti – natalist population policy seeks to lower fertility rates, in particular, and population growth rates, in general.
  - → Pro – natalist population policy seeks to increase fertility rates, in particular, and population growth rates, in generals

History of Population Policy in Ethiopia

- Beofre adecade ago, there was no explicit population policy in Ethiopia.
- In April 1993, the transitional government of Ethiopia approved the national population policy. The goal of this policy is maximizing the welfare of the country by harmonizing the rates of population and economic growth.
- The type of population policy used in Ethiopia is basically anti – natalist such policies promote lowered fertility rates, in particular, and lowered population growth rates, in general.

Goals and strategies of Ethiopia's population policy

Ethiopia's Population =- Policy Goals

- Ethiopia's population policy has
  - Broad
  - General and
  - Specific types of goals

The broad goal of Ethiopia's population policy

- It is to promote social welfare by harmonizing the rate of population growth and the country's capacity for socio –

- Expediting socio-economic development processes through holistically integrated development programs;
- Reducing the rate of rural – to urban migration
- Ensuring environmental protections
- Reducing morbidity and mortality
- Raising the economic and social status of women; and
- Improving the social and economic status of vulnerable groups, such as adolescents, children, and the elderly.

Specific Objectives

- Reducing the total fertility rate (TFR);
- Increasing the prevalence of contraceptive use;
- Reducing maternal, infant and child morbidity and mortality rates, as well as promoting the level of general welfare of the population;
- Increasing female participation at all elves of the educational system;
- Removing all legal and customary practices that prevent women from the full enjoyment of economic and social rights;
- Insuring spatially balanced population distribution patterns:
- Improving productivity in agricultural activities and introducing off- farm and non – agricultural activities for the purpose of employment diversification; and
- Mounting effective country – wide population information and programs.

Ethiopia's Population – Policy strategies

- Expanding contraceptive distribution;
- Raising the minimum age of marriage for girls from 15 years to at least 18 years;
- Promoting breast – feeding as a means of birth – spacing;
- Implementing career counseling services in schools;
- Integrating women in to the modern sector of the economy;

- Affording all laws "impeding", in any way, the access of women to all social, economic, and cultural resources,"
- Establishing teenage and youth reproductive health counseling centers;
- Increasing research in reproductive health; and
- Promoting the involvement of males in family planning
- In general, the population policy covers all major ground that needs to be covered in providing directives on the management of population growth in the interest of sustainable development.

Illustrative Questions

36. The current population policy of Ethiopia has set clear targets to be achieved by 2015. Which one of the following is correct about such targets? (EHEECE – 2000/2008)
    A. Increasing birth rate by 25%
    B. Reducing population growth rate to 0.5%
    C. Reducing total fertility to six children per woman
    D. Increasing the prevalence of contraceptive use to 40%

Explanation: The current population policy of Ethiopia has set clear targets, to be achieved by increasing the prevalence of the contraceptive (family planning) use to 40%.

Answer: D

37. Which of the following is NOT among the general objectives of the national population policy of Ethiopia? (EHEECE – 2002/2010)
    A. Improving the social and economic status of women, youth, children and the elderly.
    B. Harmonizing the rates of economic and population growth
    C. Bringing an end to rural urban migration through speedy rural development programs.
    D. Narrowing the wide urban – rural gap in people's access to basic public services.

Explanation: From the given alternatives bringing an end to rural urban migration through speedy rural development programs is not the general objectives of the national

38. What does population policy mean?

Explanation: Population policy is a policy that is formulated and implemented by a government in order to plan and control population growth, based on the economic, social, cultural, political, and demographic conditions of the country.

39. Mention some of the rationales of Ethiopia's population policy?

Explanation: The rationales of Ethiopia's population policy is closing the gap between high population growth and low economic development and also improve the quality of life.

40. What is the main reasons Ethiopia used anti – natalist population policy?

Explanation: This is due to high population growth and low economic productivity.

8.9. Urbanization in Ethiopia

8.9.1. The concept of Urbanization

- It is an increase in proportion of population living in urban areas.
- It is the process whereby large numbers of people leave country side/ rural places and small towns in order to settle in cities and surrounding metropolitan/ urban areas.
- Items of degree (state) of urbanization, Ethiopia is one of the least urbanized countries in the world.
- In Ethiopia most rural migrants to the cities have bettered themselves in comparison to their former standard of living in rural areas.
- The rapid growth of population in urban centers has been causing serious problems such as overcrowding, substandard housing, homelessness, in adequate municipal services, crime, poverty, and pollution.

8.9.2. Trends of Urbanization in Ethiopia

- Urbanization is a recent phenomena in Ethiopia. It was introduced mainly after the late 19^{th} and early 20^{th} centuries.
- Factors contributing to urbanization in the nation include peoples' need for better living conditions and people's desire to come close to clustered settlements.
- In Ethiopia's major factors contributed to the origin and development of most urban centers.

all weather roads, and ... parts of the country by
→ The five - year Italian occupation.

- Most urban ceners of Ethiopia have developed a long major transport routes, which have attracted people to these areas.
- The shewan and the Harerghe plateaus have relatively large concentrations of urban centers and urban population.
- The major reasons for high urban population concentration are.
  - Concentration of industries that results in relatively higher opportunities for employment.
  - Concentrations of social services and facilities, such as schools, health institutions, water supplies, electricity, transportation, etc that result in, relatively, better living conditions.

Levels and Distribution of urbanization in Ethiopia

- In Ethiopia, the level of urbanization has been very low in the country.
- Less than 3 percent of Ethiopia's population lived in urban areas in the 1940s, but the three censuses of the country show that urbanization is increasing in Ethiopia.
- According to the 2007 census, the urban population rate ranges from 10 percent in SNNP to 100 percent in Addis Ababa.
- Addis Ababa city Administration is entirely urban, its percentage share of Ethiopia's total urban population is only 23.1 percent.
- The largest share, about 28 percent, is living in oromiya Region
- Gambela region has the smallest share of the country's urban population.

Illustrative Questions

41. Which of the following statements about urbanization in Ethiopia is correct? (UEE – 2004/2012)
    A. Ethiopia is the least urbanized country of the world
    B. Ethiopia's rate of urban growth is among the slowest in the world

C. Roughly one third of the GDP of Ethiopia comes from urban centers
D. More than 25 percent of the Ethiopian population

Explanation: In terms of degree (stage) of urbanization, Ethiopia is one of the least urbanized countries in the world.

42. What is the main factor that causes the accelerated growth of the largest cities of Ethiopia? (EUEE- 2005/2013) **Answer: A**
A. The rapid growth of the urban economy
B. The rapid growth of industrialization in the country
C. The advances made in the country especially in trade and tourism
D. The high rates of natural increase and net in migration

Explanation: The main factor that causes the accelerated growth of the largest cities of Ethiopia are high rates of national increase and net in migration.

43. The establishment of new urban centers in late 19th century and 20th century Ethiopia was triggered mainly by **Answer: D**
A. Construction of all weather roads
B. Discovery of key economic minerals
C. Military garrisons established during Menelik's government
D. A and C are correct answer.

Explanation: The establishment of new urban centers in late 19th century and 20th century Ethiopia was triggered mainly by construction of all weather roads and military garrisons established during Menelik's government.

Answer: D

44. What does urbanization mean?

Explanation: Urbanization means the process whereby large numbers of people leave country side. Rural places and small towns in order to settle in cities and surrounding metropolitan /urban areas.

45. Which term refers to the average number of children that would be born alive to a woman during her life time if she were to pass through all child bearing years?
A. Age specific fertility rate C. Total fertility rate

C. Crude birth rate
D. General fertility rate

Explanation:

$$\text{General Fertility Rate /GFR} = \frac{\textit{Total No of live births}}{\textit{No of womon age } 15-49_{yers}} \times 1000$$

GFR:- is the number of live births per 1000 women in the reproductive ages i.e 15 – 49years

Answer: D

46. Which of the following is **NOT** a major factor behind the very high fertility rates among rural women in Ethiopia?
   A. Traditional values attached to children
   B. Early marriage
   C. The social status of women
   D. The ethnic background rural women

Explanation:

Fertility rates are high in Ethiopia due to

- High social and economic value of children
- Early marriage
- social status of women
- Low level of family planning practices

47. In which altitudinal range do most of the people of Ethiopia live?
   A. More than 2600 meters above sea level
   B. 1800 – 2600 meters above the sea level
   C. 1400 – 1800 meters above the sea level
   D. Less than 1400 meters above the seal level

Explanation:

67% percent of the population of Ethiopia lives in areas with altitude above/800 meters, and that these areas constitute 31.6% of the total area of the country

Answer: B

48. One the following is **NOT** among the main factors that explain why most of the urban centres of Ethiopia are located on highlands
   A. The high density of population on the highlands
   B. The types of economic activities practiced on highlands
   C. The high risks of vector born diseases in the lowland

D. The limited supply of water and building materials in the law lands

Explanation:

Most urban centers of Ethiopia are located on the highlands because they are free from the risk of **Tropical diseases like** malaria and yellow fever

49. Which type of economic activity supports the largest population per unit area of land? **Answer: C**

A. Sedentary agriculture
B. Hunting and gathering
C. Nomadic herding
D. Nomadic herding

Explanation:

Crop farming areas have **Greater carrying capacity** and **higher densities of population**

50. Which of the following statements is correct about the Ethiopian population? **Answer: A**

A. Children under the age of 15 comprise one – third of the population of Ethiopia
B. Ethiopia is among the countries with almost a balanced sex ratio
C. Ethiopia has a large and fast growing proportion of old age population
D. The Ethiopian population accounts for 15% of the population of Africa

Explanation:

Ethiopia is among the countries with almost a balanced sex ratio

51. During the last three decades the mortality rate of the population of Ethiopia has **Answer: B**

A. largely remained unchanged
B. been rising and falling following a regular pattern
C. been showing a declining trend
D. been rising fast

people's access to health care services, improving women's status as possible mechanisms play a significant role in **reducing mortality rate** in the last three decades

Answer: C

52. The population policy of Ethiopia can be rightly categorized as
    A. anti – Mathusian
    B. Anti - natalist
    C. pro – natalist
    D. Anti – migration

Explanation:

As most developing nations Ethiopia's national population policy is **Antinatalist reducing birds**

Answer: B

53. The number of deaths in a year among under one year of age per one thousand live births is
    A. Infant Mortality Rate
    B. Crude Death Rate
    C. Child Mortality Rate
    D. Sibling Death Rate

Explanation:

the number of infant, who dies less than the age if one year per 1000 total live – birth, that is Infant mortality Rate.

$$IMR = \frac{No.\ of\ deaths}{Total\ live\ births} \times 1000$$

Answer: A

54. The most important physical factor influencing patterns of population distribution and settlement in Ethiopia is
    A. Latitude
    B. Altitude
    C. air pressure
    D. soil type

Explanation:

The most crucial physical factor influencing patterns of population distribution and settlement in Ethiopia is altitude

Answer: B

55. Which of the following factors is a leading cause for the prevalence of large family sizes in the Ethiopian society?
    A. The stage of development of the welfare system
    B. Increasing standard of living
    C. Improved social status of women
    D. Better health care and improved diets

The absence of ensuring social security leads to the existence of large family size in Ethiopia

1. Which of the following statements is correct about urbanization in Ethiopia?

Answer: A

A. Ethiopia has a long history of indigenous urbanization
B. Most of the pre 20th century Ethiopian urban centers are still standing
C. The rate of urban growth in Ethiopia is one of the lowest in the world
D. Approximately one – third of the Ethiopian population lives in urban areas

Explanation:

Even if the level and rate of urbanization has been increasing due to **massive rural – urban migration,** the rate in urban advancement is one of the lowest in the world.

Answer: C

2. One of the following is **NOT** among the most densely populated areas in Ethiopia

A. Borena
B. Sidama
C. Kembata
D. Hadiya

Explanation:

Borena as the nomaticherdint lowland areas of Ethiopia, it is spercely populated due to mobile rural settlement

Answer: A

3. Which of the following factors is responsible for the widespread development of squatter settlements in the urban centres of Ethiopia?

A. The wide gap between housing demand and supply
B. The shortage of building materials
C. The shortage of skilled labour
D. The scarcity of building land

Squatter settlement

came of unbalance between **Housing demand and supply**

Answer: A

Review Question on Unit 8

Part I. True or False questions

Write "A" if the statement is true and "B" if it is false

1. The age dependency ratio for more developing countries is usually between 50 and 75. F
2. Ethiopia is the third most populous country in Africa. T
3. Mortality refers to the occurrence of birth in a given country or region. F
4. A population pyramid is usually employed to show the age distribution of a given population by sex. T
5. In Ethiopia, most of the physical factors are influence by altitude. T
6. The arid and semi and lowland pastoralim is considered as "greed by space". T
7. Urban settlements include residence in cities and town. T

Part II. Matching questions

Match items in column "A" with that of column "B"

A	B
8. Rural settlement F	A. Diffused or dispersed
9. Urban settlement	B. Deaths under one year
10. Scattered settlement	C. Absence of frequent change
11. Nucleated settlement	D. number of males per 100 females
12. Permanent settlement C	E. During pregnancy
13. Temporary settlements T	F. Crop – farming
14. Young age group O	G. Metropolis
15. Working age group I	H. Grouped
16. Sex ratio D	I. Economically active population
17. Infant Mortality B	J. Lowland parts of Ethiopia
18. Child Mortality M	K. Deaths between 15 and 49
19. Maternal mortality E	L. cities and towns
20. Adult Mortality Rate K	M. Deaths of child between one and five
	N. Deaths between 12-30
	O. Economically dependent

Part III Completion Questions

Fill in the blank space with appropriate words or phrases.

21. CDR is the total number of births occurring in a given year per 1000 population
22. _____ is the graphic representation of the age distribution of a given population by sex.
23. _____ is the average number of people per square kilometer in a given area.
24. The relationship between the working or economically active population and the non – working population is called _____
25. The difference between fertility (birth) and mortality (death(is called _____
26. The process of development of urban settlements is known as _____
27. _____ was a Danish economist of the 20th century, developed a "theory on populations and Agriculture."
28. The processes of the in discriminate destruction of the natural vegetation cover of a forest area are called. _____
29. _____ refers to any undesirable change in natural conditions of water, air, and other components of the natural environment that has negative effects on the health and activities of human beings and other living creatures.
30. _____ refers to the way how population spreads out over a given area.

Answer Key to Review Question on Unit – 8

Part I True or False Item

1.B 2.A 3.B 4.A 5.A 6.A 7.A

Part II.

8.F 9.L 10.A 11.H 12.C 13.J 14.O 15.I 16.D 17.B 18.M 19.E 20.K

Part III

21. Crude Birth Rate
23. Population density
25. Natural increase
27. Ester Boserup
22. Population pyramid
24. Age Dependency Ratio
26. Urbanization
28. Deforestation
30. Population distribution

UNIT NINE

Economic Growth and Development Trend in Ethiopia

9.1. An overview of Growth and Development Trend in Ethiopia

- Economic growth and development are not the same. They have two different meanings.
- Growth – is the quantitative accumulation of the national capital.
  - → It is a change output regardless of cultural, political, economic and other improvements.
  - → It involves great input in order to produce more out put and change in the overall aggregates.
- Economic – development – is a process by which members, of a society increase their personal and institutional, capacities to mobilize and manage resources to sustainable improvements in their quality of life.
  - → It is a process involving economic as well as social and environmental changes.
  - → It is also the qualitative transformation of a whole society.
- Sustainable development – is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.
- According to official sources, the economy of Ethiopia shows a forward tendency with a growing rate of 11% per anum and is projected has been growing at a rate of 11% and is projected to increase in the coming 5 years.
- Economic growth and development trend in Ethiopia, there has been an encouraging trend since 1996. This fact has been released by welfare – Monitoring survey as follows.
  - → In rural areas, the literacy rate increased from 18% in 1995/96 to 31% in 2004/05.
  - → Gross enrollment ratio at primary grades level has increased from 37.4% in 1996 to 79.8 in 2004/05. Although still relatively high, primary dropouts in rural areas declined from 18.5% in 1996 to 13.6% in 2004 at secondary level from 29.3% to 16.5%.

years.

Illustrative Questions

1. All EXCEPT one of the following belong to factors that hinder development in Ethiopia (EUE. 2002/2010)
   A. Lack of modern technologies
   B. Rapid population growth
   C. Little saving and low investment capital.
   D. Poor natural resource base.

Explanation: The major factors for that hinders development in Ethiopia are lack of modern technologies, rapid population, growth than economic growth and little saving and low investment capital.

Answer: D

2. Which one of the following is not a serious factor of socio – economic development in Ethiopia?
   A. General scarcity of natural resources
   B. Shortage of trained manpower and lack of efficient management system.
   C. Systems of management of agricultural lands.
   D. Insignificant decline in the rate of population growth

Explanation: In Ethiopia scarcity (shortage) of national resources is not a series factor `of socio economic development rather shortage of trained manpower and lack of efficient management and traditional management of agricultural lands are a serious problems.

Answer: A

3. Which one of the following has been and continues to be a development – planning problem in Ethiopia?
   A. Lack of educated people and expertise.
   B. Failure to listen to international financial organizations

C. Formulation of ambitions plans for the betement of the lives of the poor
D. Failure to encourage genuine participation of the poor and insufficient understanding of their real needs.

Explanation: one of the basic development – planning problem in Ethiopia is lack of educated people and expertise.

Answer: A

4. One of the following is not recommends a solution for achieving sustainable development in Ethiopia.
   A. Raising investments on heavy industry
   B. Appropriate utilization on heavy industry
   C. Encouraging population growth
   D. Raising the levels of domestic saving and investment.

Explanation: Alarming rate of population growth is not a right solution for achieving sustainable development in Ethiopia.

Answer: C

5. One of the following does not characterize the current development situation of Ethiopia.
   A. Only few Ethiopia's live below the poverty line.
   B. The country's per capital income is the lowest in the world.
   C. Ethiopia is not able to produce enough food for its rapidly growing population.
   D. Ethiopians health and educational status is lowest even by the African standard.

Explanation: Figurative datas indicated that in Ethiopia 31 million people live below the poverty line, and between 6 and 13 million people are at risk of starvation every year.

Answer: A

9.2. Major Features of Ethiopian Economy

The major sectors of the Ethiopian economy are:

- Agricultural
- Industrial
- Distributed (service)

Agricultural sector

- Agriculture is one of the primary economic activities which includes the production of crops and rearing of livestock for either local consumption or commercial purpose.
- In Ethiopia, agriculture is an old economic activity, which has been practiced since 4000 BR. But still agriculture is largely traditional and subsistence in nature.
- The productivity of both crop and animal farming is very low.

- Among the various services sub-sectors, trade, hotels and restaurants had remarkable overall growth spanning.
- Banking and insurance follows showing an average growth rate performance of 18.5 percent from 2006/07 through 2009/10.
- Education ranks third with an average growth rate of 16.5 percent.

Industrial sector

- It is widely recognized that successful development of the industrial sector plays a key role in the economic and social progress of a nation.
- Ethiopia has had along traditional in the development of handicrafts and cottage industries, the introduction of modern manufacturing industries is only a 20^{th} century phenomenon.
- It ranked third in its contribution to Ethiopia's GDP.
- Its growth was small (13.0%) in comparison to that of the agriculture and service sectors.
- The industrial sector consists mainly of large and medium – size industries, small – scale and handicraft manufacturing, and construction sub – sector.
- It employs about 10 percent of the country's active labor force.
- The main products include textiles, food stuffs, beverages, cement, leather and leather products, metallic and non – metallic products, wood, and paper and pulp.
- Among the industrial sub-sectors, mining and quarrying has shown a tremendous growth performance accounting 44.2 percent of the industrial sector's growth performance.
- Due to recent development programs ADIL (agricultural development led Industrialization) there is a forward and back ward linking's between agriculture and industry.
- ADLI'S top – level goals are to ensure.

- Economic growth through a rural centre development strategy
- Improved living standards for rural and urban dwellers
- Effective education services
- Expansion of preventive and primary health care facilities.

- Sustainable development (promotion) of manufacturing industries depends on a number of requirements. These are also known as basic factors for the development of modern manufacturing industries.
- They include minerals and fuels, hydro electricity, agricultural raw materials and markets.

Illustrative Questions

6. Which one of the following is an indicator of the low level of performance of the agricultural sector in Ethiopia? (EUE, 2000/2008)
   A. High level of food – self sufficiency gap
   B. The low level of land and labour productivity
   C. Heavy dependence on making and use of compost
   D. A and B are correct indicators

Explanation: High level of food – self sufficiency gap and low level of land and labor productivity are the indicators of the low level of performance of the agricultural sector in Ethiopia.

Answer: D

7. The Ethiopian industrial sector is characterized by all of the following features EXCEPT one
   A. Scarcity of capital
   B. Lack of qualified manpower
   C. Dominance of high industries
   D. Production of high – technological equipment

Explanation: The Ethiopian industrial sector is characterized by – scarcity of qualified manpower and dominance of light industries.

Answer: D

8. People often describe agriculture as the 'backbone' of the Ethiopia economy, thereby implying, among others, that it
   A. Consumes the lion's share of the land
   B. Uses cutting age technology for production
   C. Depends more on direct rainfall irrigation
   D. Creates employment for most of the workforce

Explanation: People often describe agriculture as the back bone of the Ethiopian economy there by implying among other it creates employment for most of the work force.

Answer: D

9. Which one of the following is the most important factor, behind the infantile nature of the development of manufacturing industries in Ethiopia?
   A. Lack of raw materials
   B. Shortage of energy resources
   C. Absence of transport infrastructure
   D. The low purchasing power of the people

Answer: D

Explanation: The most important factor behind the infantile nature of the development of manufacturing industries in Ethiopia is the low purchasing power of the people

10. The Ethiopian agricultural sector experienced some new developments.
    A. Declining dependence on rain
    B. Declining contribution to the export mix
    C. Declining dependence on improved fertilizer
    D. Shift from traditional farming mode to amore mechanized one.

Answer: A

Explanation: The Ethiopian agricultural sector experienced declining dependence on rain is the new development over the last years.

11. A form of agriculture in Ethiopia in which farmers practice the rearing of livestock side by side with crop production is
    A. Pastoral nomadism
    B. Ranching
    C. mixed farming
    D. poultry

Answer: C

Explanation: Mixed farming is the form of agriculture in Ethiopia in which farmers practice the rearing of livestock side by side with crop production especially in the highland area.

12. Which of the following modern manufacturing industries in Ethiopia currently employs the largest labour force? (UEE - 2003/2011)
    A. Assembly of motor vehicles
    B. Textile industries
    C. Chemical industries
    D. Iron and steel manufacture

Answer: B

Explanation: Textile industries in Ethiopia currently employs the largest labour force.

13. The fact that contributed the most to the under development of the traditional (cottage) industries of Ethiopia is _____.
    A. Shortage of the necessary raw materials
    B. The negative attitude that the society has always had towards artisans or craftsmen.
    C. Lack of domestic demand for their products
    D. Shortage of capital and labor

manufacturing industries.

C. The country has a balanced mix of both light and heavy industries
D. The majority of the industries produce consumer goods and first stage agricultural products.

Explanation: The manufacturing industries in Ethiopia is produce consumer goods and first stage agricultural products.

Answer: D

15. Which of the following is the best estimate of the contribution of the industrial sector to the GDP of Ethiopia at present? (UEE. 2004/2012)

A. About 50 percent
B. Not less than 12 percent
C. Roughly 6 percent
D. Roughly 30 percent

Explanation: The contribution of the industrial sector to the GDP of Ethiopia at present not less than 12 percent or nearly 13 percent.

Answer: B

16. Compare and contrast the difference between GDP and GNP?

Explanation: GNP is normally measured by totaling all personal spending, all government spending and all investment spending by a nation's industry both domestically and all over the world where as GDP is the total value of currently produced final goods and services within a country's territory, usually in a year.

17. Describe why Ethiopia is at lower stage of development?

Explanation: Ethiopia's development has serious problems and challenges; the leading factors include lack of trained human power and efficient management systems, lack of adequate modern technology, Lack of adequate savings fort investment its these is the reason why Ethiopia is at lower stage of development.

9.3. Present Features of Ethiopian Socio – Economic Development.

Basically, when countries are categorized or grouped either as developed or developing countries, globally accepted measures or indices are used.

- The following are widely/ frequently used in the Ethiopian context.
  - GDP or GNP (Gross Domestic product/ Gross National product) per capita.
  - Illiteracy rate
  - Infant mortality rate
  - Expected life expectancy
  - Population growth rate
  - Percentage of population who have access to clean water
  - Level of infrastructural development, etc.
- Ethiopia's [resent socio – economic features are treated as follows

GDP Per Capita

- It is one of the lowest in the world.
- PASDEP'S target has achieved a growth rate of Ethiopia's GDP per capita status will join the middle in group countries with in the coming five years (2010 – 2015).
- The current status of Ethiopia's GDP per capita is estimated to be us $ 344 per annum.

Literacy rate

- The illiteracy rate in Ethiopia has been high
- The current literacy rate of Ethiopia is said to de 42.7% in the year 2009.
- Welfare Monitoring surveys undertaken so far exhibits a sharp rise in gross school enrollment that considerably confirm the decline of illiteracy rate in the country.

Infant Mortality Rate

- IMR in Ethiopia has been high.
- The UNDP 1997 census reveal that Ethiopia's infant mortality rate was 111/000 which even above the average of sub – Saharan countries (97/1000).

Expected life Expectancy

- It is the reflection of varied social, economic, psychological and other factors.
- In Ethiopia, it was 43 about ten years ago due to the prevalence of HIV impacts.
- The recent strategies adopted both in the primary health care services and supply of HIV/AIDS medicines extensively, the expected life expectancy in Ethiopia has reached to 55.8 years in 2009/10.

growth rate:

- Rapid population growth with an average rate per annum has been one of the many threats to Ethiopia's development pace.
- Addressing this issue GTP the continuation of PASDEP has designed to slow the rapid population growth by applying spacing of births through health service institutions and extension package services.

Percentage of Population with access to clean water

- For many decades lack of access to clean water has been out peoples serious problem.
- Most illness caused both in urban and rural areas are related to unsafe water.

Level of infrastructural development

- It includes avaguely of elements that promote the living standard of people.
- This development such as the road networks, electric power supply, rehabilitation/resettlement of pastoralists, access to health services and strengthen the urban – rural link.
- In the last ten years under PRP ((Poverty Reduction Programmed), SDPRP (sustainable Development for poverty Reduction programmed)and PASDEP (plan of Action for sustained development to End Poverty), significant measures have been made to change the picture and attain the millennium goals.

Illustrative Questions

18. In which of the following indicators does Ethiopia have a substantially higher figure than an average west European country? (UEE-2000/2008)

A. Level of urbanization
B. Rate of urban growth
C. per capita income
D. Energy consumption

Explanation: High Rate of urban growth is one of the indicators does Ethiopia have substantially higher figure than average west European country.

Answer: B

19. Which one of the following indicators of development is a physical factor for backwardness of Ethiopia?

A. Low GNP per capita
B. Rapid population growth
C. Level of investment
D. Drought

Explanation: Among the given alternatives drought belong to a physical indicators of development because most of the time it occurs by natural phenomena.

Answer: D

20. List the frequently used indicators of development in Ethiopia?

Explanation: Development indicators of Ethiopia includes GDP or GNP, Illiteracy rate, Infant mortality rate, expected life expectancy etc.

21. What is the need for knowing and identifying indicators of development?

Explanation: it is important for categorized or grouped either as developed or developing countries.

22. What is the reason large scale industries in Ethiopia are not an indicators of development?

Explanation: This is due to the fact that Ethiopia's modern manufacturing sector is backward.

9.4. Challenges and prospects of Socio – Economic Development For Ethiopia.

- The ultimate goal of all economic planning and execution is growth followed by development.
- Achieving economic especially has been a serious issue in many countries, especially in the developing counties.
- Different programs and strategies have been adopted and pursed, but often their goals have not been achieved or else have not been sustained. This has been the case in our country since the early 1960s.

9.4.1. Challenges to Ethiopia's socio – Economic Development

- Ethiopian governments have launched economic development programs to reduce, it not to end, poverty.
- Some 31 million people are at risk of starvation every year.
- The different factors hindering Ethiopia's progress (challenges) are interconnected and in combination constitute a "poverty trap" that prevents the country from breaking out of poverty. These factors are
  - Rapid population growth
  - Land – fragmentation
  - Environmental degradation has caused frequent droughts
  - Very low productivity
  - Low income
  - Low levels of investment
  - Dependence on un reliable rainfall
  - Structural bottlenecks
  - Lack of good governance and commitment to accomplishing tasks.
  - Low infrastructure coverage

Table 9.1. Distribution of main indicators of infrastructure availability

	Access to improved water (% of population)	Read density, 1999 (km/1000 people)	Electrical Power, 2001 (KW per- capita)	Telephone lines
Ethiopia	24%	0.48	8	5
Average for developing countries	79%	0.14	272	100
Sub – sahara Africa	58%	0.07	105	15
Low – income countries	76%	0.18	NA	28

Illustrative Questions

23. Which of the following is true about current development – status of Ethiopia?
    A. Decline population growth
    B. High income
    C. Land – fragmentation
    D. A and C

Explanation: currently, because of high rate of population growth Ethiopia facing land – fragmentation.

Answer: C

24. Identify the wrong combination
    A. Environmental degradation – drought
    B. Good governance – low infrastructural coverage
    C. Low income – low levels of investment
    D. Decline of population growth Less land fragmentation

Explanation: Good governance results commitment to accomplishing tasks in the right time and achieving high infrastructural coverage.

Answer: C

25. Which of the following challenges of Ethiopia's socio – economic development is directly or indirectly affects the other
    A. Low productivity
    B. Population growth
    C. dependence on rainfall
    D. structural bottlenecks

Explanation: In one way or the other high population growth is the root cause of Ethiopia socio economic development problems.

Answer: B

26. Describe how environmental degradation has caused frequent droughts?

Explanation: Let us take soil, because of high population pressure and high deforestation it will be deteriorate as a result decrease productivity leads to drought.

9.4.2. Prospects for Ethiopia's socio Economic Development

- Since Ethiopia has joined the world economy, a lot has been said and attempted to take out the country from backwardness. But, it is still defined as a country of mass poverty, frequently drought and famine, etc.
- Ethiopia's development problems are diverse and highly complicated economic, demographic, physical and social problems are deep rooted. Hence, an all – rounded, integrated and comprehensive remedies should be suggested to overcome or minimize the scenario
- The following are some of the attainable measures suggested.
    - Development of energy sources;
    - Promotion of industrialization
    - Modernization of agriculture
    - Conservation of natural resources
    - Proper utilization of agricultural potential;
    - Exploitation of niche markets and opportunities wherever they present themselves;
    - Expansion of exports in particular, diversifying exportable items to widen the economic base, reduce susceptibility to shocks and, in the long term, to reduce the prevailing dependency on foreign aid;
    - Promotion of better links between markets and produces to enable business to take place and to allow people easier access to essential services;
    - Promotion of capacity building regarding the work fore's education and skills.
    - Slowing the existing rapid population growth; and
    - Proper utilization of the potential of Ethiopia's women, who constitution about 48.9 percent of the population.

Illustrative Questions

27. Women empowerment and participation in the overall socio – economic development aspect can be promoted through (UEE-2004/2012)
    A. Gender sensitive policies
    B. Campaign against illiteracy
    C. Gender based division of labor
    D. Women focused family planning.

Explanation: women empowerment and participation in the overall socio – economic development aspects can be promoted through campaign against illiteracy.

Answer: B

28. Which one of the following is not true about a possible suggestion of Ethiopia's socio – economic development prospects
    A. Promotion of industrialization
    B. Conservation of natural resources
    C. Un fair women participation
    D. Expansion of exports

Explanation: participation of women in all parts of socio economic development aspect is bring societial change.

Answer: C

29. Explain briefly the duty and responsibility of each citizen in order to attain sustainable development?

Explanation: For most and important attention to bringing sustainable development in Ethiopia is each citizens should be focus on family planning and wise use of natural resources because of the fact that population growth and unwise use of natural resource is a serious problem of development.

9.5. Economic Relation

- Geographical and political factors drive, countries to establish economic relationship with other countries, including:
  - → Geographic factors, such as physiographic variation, that result in product variation.
  - → Political factors such as political affiliations due to temporary or permanent common interests.
  - → Economic factors such as comparative advantages.
- As a result of the above factors, the nation has economic relationship with high neighboring countries or other countries.

9.5.1. Ethiopia's Economic Relationships with Neighboring Countries.

- Ethiopia's export and import items and trade policy and strategy influence the economic relationships that the nation establishes with distant countries.
- Ethiopia's advantageous geographical proximity to countries of the middle east, the Gulf, and Europe also determine relationships with other counters.
- Ethiopia Export destinations have include western Europe, the Far- east, North American and central America.

- Europe was the leading export destination in the year 2006/07, taking 40.8% of the total value, and was followed by Asia, Africa and North and central America, accounting for 33.5%, 5.8%, respectively
- Germany was the single most important destination for Ethiopian exports between 2004 and 2006.
- Ethiopia's most important sources of imports in 2004/05 were the Middle east and Asia, accounted for 57.5%. Next was Europe, at 23.1%. the third and fourth places were taken by North /America and Africa.
- Saudi Arabia and china are the two most important sources of important for Ethiopian in recent years.

Duty Free and Quota – Free Market – Access opportunities

- Ethiopia, which is one of the developing countries, is expected to benefit from duty – free quota – free market – access opportunities.
- For example, the people's Republic of china has granted almost all Ethiopian export products duty – free access to its markets.
- This process will be facilitated by Ethiopia's advantageous geographical proximities to the middle East, Gulf countries and European countries

Illustrative Questions

30. Which of the following countries is least important as trade partner of Ethiopia? (UEE – 2000/2008)

A. Japan
B. Brazil
C. Sudan
D. Germany

Explanation: the least important trade partner of Ethiopian compared to Sudan and Germany is Brazil.

Answer: B

31. Which of the following statements about the international trade of Ethiopia is correct? (UEE – 2003/2011)

A. Ethiopia's balance of international trade has been persistently positive in recent decades.
B. The number of the major international trade partners of Ethiopia is quite small
C. Africa and Middle East are the major destinations of the export products of Ethiopia
D. As one of the main exporters to coffee, Ethiopia plays a key role in setting its market price.

Explanation: The number of major international trade partners of Ethiopia is quite small at the international trade.

Answer: B

32. Which one of the following countries receives the highest proportion of Ethiopian export? (UEE – 2005/2013)

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A. South Africa
B. Japan
C. Germany
D. United states of America

Explanation: Germany was the single most important destination for Ethiopian exports between 2004 and 2006.

Answer: C

33. What are the major factors that motivate countries to establish economic relationships?

Explanation: Factors includes geographic, political and economic which drive countries to establish economic relationships.

34. Which neighboring country of Ethiopian is its economic relationships are minor? Why?

Explanation: Because of political instability or unrest in Somalia economic relationships to Ethiopia are not strong.

35. Why Ethiopia establish good economic relationships with Middle East, the Gulf and Europe?

Explanation: This is because of geographical proximity

9.6. PASDEP (Plan for Accelerated and Sustained Development to End Poverty)

- Since the early 1960s, various development plans have been designed in Ethiopia to alleviate poverty.
- The EPRDF government initiated a five – year development program during its first year's rule (1992 – 1993), Later, it defined successive additional five – year development plans to achieve sustainable growth. These are:
    - Sustainable Development and poverty Reduction programme (SDPRP) 2000-2004.
    - Plan for Accelerated and sustainable Developments to End poverty 2005 - 2006 (PASDEP)
    - Growth and Transformation program – 2010 – 2015 (GTP)

PASDEP

- It is a document that describes the plan for accelerated and sustained.
- It was Ethiopia's guiding strategic frame work for the five – year period of 2005 – 2010.
- It represents the second phase of SDPRP, which covered 2001 – 2004.
- It is the link to GTP, which was designed for 2010 – 2015.
- It carries forward important SDPRP strategies related to human development, rural development, food security and capacity building.

- It was affected by a number of external and internal shocks- including; less support than expected offered by donors, poor rainfall and the global financial crises that began in 2008/2009 and swept the whole world.
- The strategy's bases included the MDGs Needs Assessment for Ethiopia

The PASDEP Strategy

The PASDEP program was based on these nine activities.

1. A massive push to accelerate growth. This strategy had two main thrusts:
   → Commercialization of agriculture, and
   → Promoting much moral rapid non – farm private – sector
2. A more differentiated approach to agriculture, focusing on Ethiopia's very different agro – ecological area.
3. A renewed look at several – urban linkages and the urban agenda.
4. Addressing the population challenge by implementing the existing national population strategy and making services available for spacing births.
5. Unleashing the potential of Ethiopia's women. The goals were:
   i. Increasing the number of girls who complete their schooling
   ii. Improving women's health
   iii. Liberating girl's and women's time from the unproductive hours spent fetching water supplies.
   iv. Improving several telephones services
6. Strengthening the infrastructure's backbone – the road network, water supply, electric power – supply, etc.
7. Managing risk and volatility. The goals were.
   i. Reducing repeated households cycles in and out of poverty that result from crop failure and major illness
   ii. Reducing the impact of shocks on public finances and government programs.
8. Intensive efforts to reach the MDGS. This goal included major effort between the government of Ethiopia and its foreign – aid partners.
9. Creating jobs. In Ethiopia, especially in urban areas, un employment is a serious challenge. The goals were:
   i. Managing the dynamics of population growth

(SDPRP)

D. Plan for Accelerated and sustainable development for End poverty (PASDEP)

Explanation: Ethiopia's current development strategy is growth and transformation program – 2010 – 2015 (GTP).

Answer: A

37. What is PASDEP?

Explanation: PASDEP is a document that describes that plan for accelerated and sustained Development to End poverty. It was Ethiopia's guiding strategic frame work for the five – year period of 2005 – 2010.

38. What is SDRRP?

Explanation: SDPRP is a document which describes sustainable Development and poverty Reduction program covers five – year period of 2000 – 2004.

39. What do you think that the objectives or goals of developed National development plan?

Explanation: The main objectives of establishing national development plan is bringing to sustainable development or improving socio economic aspects a country.

40. Explain the role of ADLI in Ethiopia sustainable development?

Explanation: ADIL (Agricultural Development Led Industrialization) made important enhancements to trade balance by decreasing imported raw materials and also it create a positive backward and forward linkage of agriculture and industry.

41. From where does the largest amount of the coffee that Ethiopia exports come from?

A. From large scale mechanized farms
B. From small holder of peasant farms
C. From the back yards of urban households

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D. From forests where wild coffee grows

Explanation:

Largest amount of the coffee that Ethiopia exports comes from **forest where wild coffee grows**

Answer: D

42. Which of the following is a widely accepted definition of the term "ethnic group"?

A. A Group of persons who have lived together in the same geographical unit

B. A group of persons that speak the same language and have the same religion

C. A group of persons who have the same religion regardless of where they live

D. A group of persons who are bound together by a common culture, language, customs, religion or race

D. Ethnic Group shares common cultural traditions: Customs, language, religion, race

Answer: D

43. Which of the following is the long – term national economic development strategy that Ethiopia adopted as of the early 1990S?

A. agricultural development Led – Industrialization (ADLI)

B. Climate Resilient Green Economy (CRGE)

C. Growth and Transformation plan (GTP)

D. The Millennium Development Goals (MDGs)

Explanation:

ADLI – as the long – term national economic development programme launched in 2002 – Followed by PASDEP/2005-2010/, which is the continuation of S, DPRP.

Growth and transformation programme/ GTP – 2010 – 2015 is the third phase of poverty eradication stratify

Answer: A

44. Which of the following statements about Ethiopia agriculture is **NOT** correct?

A. Only a small proportion of farm products reaches the market

B. The land held by farmers is getting smaller and smaller
C. Most farmers depend on irrigation to increase their food supply
D. A substantial proportion of farm products are consumed by pests

Explanation:

Most of the Ethiopian farmers rely on **Rain feed – agriculture** rather than on **Irrigation**

Answer: C

45. In which of the following countries does the mining sector make the lowest contribution to GDP

A. Botswana
B. Namibia
C. Ethiopia
D. Zambia

Explanation:

Agriculture is the dominant sector of Ethiopian GDP contribution followed by the service sector

Answer: C

Review Question on Unit 9

Part I. True or False questions

Write "A" if the statement of true and "B" if it is false

1. Growth is the qualitative transformation of a whole society.
2. The industrial sector ranked second in its contribution to Ethiopia's GDP.
3. One of the top – level goals of ADLI is effective education services.
4. An integrated and comprehensive national plan is required to fackle the diverse and complex causes of Ethiopia's under development.
5. Ethiopia's GDP per capita is estimated to be one of the least in the world.
6. Life expectancy of any country is the reflection of varied social, economic and psychological factors.
7. In Ethiopia, Agriculture had been the leading sector followed by the service sector.

Part II. Matching questions

Match items in column "A" with that of column "B"

A	B
8. Growth	A. cash and industrial crop
9. Development	B. Infant mortality rate
10. SDPRP	C. Weaving
11. PASDEP	D. Rapid population growth
12. GTP	E. cement and leather
13. Challenges of society Economic development in Ethiopia	F. Expansion of imports
	G. Quantitative accumulation of the national capital
14. Agricultural sector	H. 2006 - 2009
15. Industrial sector	I. 2000 - 2004
16. Distributed (service)	J. 2005 - 2015
17. Indicators of Development	K. 2010 - 2015
18. Traditional (cottage) industries	L. Qualitative transformation
19. Infrastructural development	M. Road networks, electric power supply.
20. Prospects for Ethiopia's socio Economic Development	N. Expansion of exports
	O. Hotels and restaurants

Part III Completion Questions

Fill in the blank space with appropriate words or phrases.

21. ___PAS___ was Ethiopia's guiding strategic framework for the five – years period of 2005 – 2010.
22. ___German___ was the single most important destination for Ethiopian exports between 2004 and 2006.
23. A change in output regardless of cultural, political, economic and other improvements are called _____
24. _____ is the total value of currently produced final goods and services within a country's territory, usually in a year.
25. _____ is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

26. The economy of Ethiopia shows a forward tendency with a growing rate of _____ per annum.
27. _____ and _____ are the two most important sources of imports for Ethiopia in recent years.
28. The Ethiopian economy is predominantly _____
29. _____ sector consists mainly of large and medium – size industries, small scale and handicraft manufacturing, and construction sub – sector.
30. The total gross national product of country divided by its total population is called _____.

Answer Key to Review Question on Unit – 9

Part I True or False Item

1.B 2.B 3.A 4.A 5.A 6.A 7.A

Part II.

8.G 9.L 10.I 11.J 12.K 13.D 14.A 15.E 16.O 17.B 18.C 19.M 20.N

Part III

21. PASDEP (plan for Accelerated and sustainable Developments to End poverty.
22. Germany
23. Growth
24. GDP (Grow Domestic Product)
25. Sustainable Development
26. 11 percent
27. Saudi Arabia and China
28. Agriculture
29. Industry sector
30. GNP (Gross National Product)

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